

Part III Lie Algebras and their Representations

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1 Motivation

[Lecture 1
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(This section is non-examinable)

Let G be a finite group. A representation of G is a group homomorphism $\rho: G \rightarrow \mathrm{GL}_n(\mathbb{C})$ for some $n \geq 1$. Two representations are isomorphic if they are equal after a change of basis of $\mathbb{C}^n$. Basic results in the representation theory of finite groups are:

- Every representation is a direct sum of irreducible representations.
- The number of (isomorphism classes of) irreducible representations of G equals the number of conjugacy classes of G .
- A representation is determined by its character $\chi = \mathrm{tr} \circ \rho$.

This leads to a very complete and beautiful theory (which I do not assume you are familiar with).

What about infinite groups? In mathematical nature, most infinite groups come equipped with some extra structure that is compatible with the group operations, like the structure of a manifold, leading to the theory of Lie groups. Some examples of Lie groups: $\mathrm{GL}_n(\mathbb{R})$ (invertible $n \times n$ matrices), $\mathrm{SL}_n(\mathbb{R})$ (determinant 1 matrices), $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SU}(n) = \{A \in \mathrm{SL}_n(\mathbb{C}) : A\bar{A}^t = I\}$, $\mathrm{SO}(n) = \{A \in \mathrm{SL}_n(\mathbb{R}) : AA^t = I\}$. A representation of a Lie group is a smooth group homomorphism $\rho: G \rightarrow \mathrm{GL}_n(\mathbb{C})$. Studying representations of Lie groups has applications to theoretical physics, differential geometry, harmonic analysis and number theory, to name a few.

An amazing insight, due to Lie and Killing, is that almost all of the information of a connected Lie group G can be read off from its behaviour ‘around the origin’. More formally, the tangent space of G at the origin e is a vector space $\mathfrak{g}$ over $\mathbb{R}$. The group structure on G can be used to define a pairing $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, called the Lie bracket, satisfying various properties, and the data $(\mathfrak{g}, [-, -])$ is called a Lie algebra.

Example 1.1. If $G = \mathrm{GL}_n(\mathbb{R})$, then $\mathfrak{g}$ is the vector space of all $n \times n$ matrices over $\mathbb{R}$, with Lie bracket $[A, B] = AB - BA$.

To give a precise instance of this passage of information, it is a deep theorem that the assignment $G \mapsto \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ induces a bijection on isomorphism classes of the following objects:

$$\left\{ \begin{array}{c} \text{Connected, simply connected} \\ \text{compact Lie groups} \end{array} \right\} \xrightarrow{\sim} \{ \text{Semisimple complex Lie algebras} \}.$$

Moreover this bijection identifies the representations on both sides. As an example, if $n \geq 2$ then representations of the compact Lie group $\mathrm{SU}(n)$ correspond bijectively to representations of $\mathfrak{sl}_n(\mathbb{C})$.

The upshot is that the study of Lie groups largely reduces to the study of pair $(\mathfrak{g}, [-, -])$ which is purely within the realm of linear algebra! In this course, we will develop the algebraic theory of Lie algebras, focusing on the case of semisimple complex Lie algebras.

Main goal of the course: state and prove the classification of semisimple Lie algebras and their representations. This is one of the most complete and beautiful theories of modern mathematics.

We can also consider groups that additionally have the structure of an algebraic variety; these are called algebraic groups. Such groups have Lie algebras too, and their study can again be largely reduced to that of their Lie algebra, so the study of Lie algebras is useful in algebraic geometry as well. For example, there is a bijection between isomorphism classes of the following objects:

$$\left\{ \begin{array}{c} \text{Algebraic representations} \\ \text{of } \mathrm{SL}_n(\mathbb{C}) \end{array} \right\} \xrightarrow{\sim} \{ \text{Representations of } \mathfrak{sl}_n(\mathbb{C}) \}.$$

Semisimple complex Lie algebras are classified by Dynkin diagrams, which are certain families of finite graphs. It turns out that such Dynkin diagrams (or more precisely, the simply laced diagrams of type ADE) classify a plethora of mathematical objects; these are called ADE classifications. See Wikipedia for more information about this beautiful topic.

2 Basic definitions

Let F be a field. Soon we will assume $F = \mathbb{C}$ but we can stay general for a few lectures.

Convention 2.1. Vector spaces will be finite-dimensional unless explicitly stated otherwise.

2.1 Definition of a Lie algebra

Definition 2.2. A *Lie algebra* over F is a vector space $\mathfrak{g}$ over F together with a bilinear pairing $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow F$ (called the Lie bracket) such that

1. $[-, -]$ is alternating: $[x, x] = 0$ for all $x \in \mathfrak{g}$.
2. $[-, -]$ satisfies the Jacobi identity: for all $x, y, z \in \mathfrak{g}$, $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

Remark 2.3. The first condition implies that $[x, y] = -[y, x]$ for all $x, y \in \mathfrak{g}$, and the converse holds when F does not have characteristic 2.

Lie algebras are typically denoted by fraktur letters $\mathfrak{g}, \mathfrak{h}, \mathfrak{p}, \dots$. When no confusion arises, we will simply write $[xy]$ instead of $[x, y]$.

Definition 2.4. A *morphism* of Lie algebras is a linear map $f: \mathfrak{g} \rightarrow \mathfrak{h}$ such that $f([x, y]) = [f(x), f(y)]$ for all $x, y \in \mathfrak{g}$. If such a morphism is bijective, we call it a Lie algebra *isomorphism*.

Definition 2.5. A *Lie subalgebra* of a Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h}$ such that $[xy] \in \mathfrak{h}$ for all $x, y \in \mathfrak{h}$.

2.2 First examples of Lie algebras

Quote 2.6 (Paul Halmos). *A good stack of examples, as large as possible, is indispensable for a thorough understanding of any concept, and when I want to learn something new, I make it my first job to build one.*

The most important example of a Lie algebra is:

Definition 2.7. For an integer $n \geq 1$, let

$$\mathfrak{gl}_n(F) := \text{Mat}_n(F) = \{n \times n \text{ matrices over } F\} \quad (2.2.1)$$

be the Lie algebra with Lie bracket $[A, B] = AB - BA$. We often simply write $\mathfrak{gl}_n$ when the field F is clear from the context.

It is easy to check that this is indeed a Lie algebra. Similarly, if V is a F -vector space, let $\mathfrak{gl}(V)$ be the Lie algebra whose underlying vector space is $\text{End}(V)$ and with Lie bracket $[f, g] = f \circ g - g \circ f$. If V is n -dimensional then choosing a basis of V determines a Lie algebra isomorphism $\mathfrak{gl}(V) \simeq \mathfrak{gl}_n(F)$.

The Lie algebra $\mathfrak{gl}_n(F)$ has subalgebras that we will also give a name:

$$\mathfrak{sl}_n(F) = \{X \in \mathfrak{gl}_n(F) : \text{tr}(X) = 0\}, \quad (2.2.2)$$

$$\mathfrak{b}_n = \{\text{upper triangular matrices}\} = \left\{ \begin{pmatrix} * & \cdots & * \\ 0 & \ddots & \vdots \\ 0 & 0 & * \end{pmatrix} \right\}, \quad (2.2.3)$$

$$\mathfrak{n}_n = \{\text{strictly upper triangular matrices}\} = \left\{ \begin{pmatrix} 0 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & * \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\}, \quad (2.2.4)$$

$$\mathfrak{d}_n = \{\text{diagonal matrices}\} = \left\{ \begin{pmatrix} * & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & * \end{pmatrix} \right\}. \quad (2.2.5)$$

One can explicitly check that these are Lie subalgebras. For the last one, the Lie bracket is even trivial; this is an example of an abelian Lie algebra.

Definition 2.8. A Lie algebra $\mathfrak{g}$ is *abelian* if $[xy] = 0$ for all $x, y \in \mathfrak{g}$.

Note that a subalgebra of $\mathfrak{gl}_n$ is abelian if and only if it consists of commuting matrices. Up to isomorphism, there exists a unique abelian Lie algebra of every dimension.

Similarly to defining the Lie groups SO_n and $\text{Sp}_{2\ell}$, we can define Lie algebras $\mathfrak{so}_n$ and $\mathfrak{sp}_{2\ell}$ using bilinear forms. Let V be a (finite-dimensional) vector space and $b: V \times V \rightarrow F$ a nondegenerate bilinear form. One can check that the subset

$$\{f \in \mathfrak{gl}(V) : b(fv, w) + b(v, fw) = 0 \text{ for all } v, w \in V\} \quad (2.2.6)$$

is a Lie subalgebra of $\mathfrak{gl}(V)$. If b is symmetric (i.e. $b(w, v) = b(v, w)$ for all $v, w \in V$), we denote the subalgebra (2.2.6) by $\mathfrak{so}(V, b)$. Concretely, choosing a basis of $V \simeq F^n$, there exists a symmetric matrix J such that $b(v, w) = v^t J w$ for all $v, w \in F^n$, and we have

$$\mathfrak{so}(V, b) = \{X \in \mathfrak{gl}_n(F) : X^t J + J X = 0\}. \quad (2.2.7)$$

If b is alternating (i.e. $b(v, v) = 0$ for all $v \in V$), we denote the subalgebra (2.2.6) by $\mathfrak{sp}(V, b)$.

We will usually make a specific choice of bilinear form. In the orthogonal case, let

$$J = \begin{cases} \begin{pmatrix} 0 & I_\ell \\ I_\ell & 0 \end{pmatrix} & \text{if } n = 2\ell, \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_\ell \\ 0 & I_\ell & 0 \end{pmatrix} & \text{if } n = 2\ell + 1. \end{cases} \quad (2.2.8)$$

and write $\mathfrak{so}_n(F) = \mathfrak{so}(F^n, J)$. In the symplectic case, we set $J = \begin{pmatrix} 0 & I_\ell \\ -I_\ell & 0 \end{pmatrix}$ and $\mathfrak{sp}_{2\ell}(F) = \mathfrak{sp}(F^{2\ell}, J)$.

Remark 2.9. If b is an alternating nondegenerate bilinear form on a vector space V , then the dimension of V is necessarily even, and there exists a basis of V for which b has Gram matrix J as above. So there is no ‘ $\mathfrak{sp}_{2\ell+1}$ ’.

Lecture 2
starts here

Exercise 1. Check that all of the above defined vector spaces are indeed Lie subalgebras of $\mathfrak{gl}_n$.

Remark 2.10 (non-examinable). The notation for the algebras $\mathfrak{gl}_n, \mathfrak{sl}_n, \mathfrak{so}_n, \mathfrak{sp}_{2\ell}$ is motivated by the connection with Lie groups, since they are the respective Lie algebras of the Lie groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), SO_n, Sp_{2n}$ when $F = \mathbb{R}$.

2.3 Representations of Lie algebras

Definition 2.11. Let $\mathfrak{g}$ be a Lie algebra. A **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ for some vector space V .

We also say that V is a $\mathfrak{g}$ -representation (or $\mathfrak{g}$ -rep for short), with the homomorphism being implicitly understood. There is a closely related notion of a $\mathfrak{g}$ -module:

Definition 2.12. A $\mathfrak{g}$ -module (or a $\mathfrak{g}$ -action) is a bilinear map $\mathfrak{g} \times V \rightarrow V, (x, v) \mapsto x \cdot v$ where V is a vector space, satisfying

$$[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$$

for all $x, y \in \mathfrak{g}$ and $v \in V$.

Given a representation ρ , we can define a $\mathfrak{g}$ -action on V , via $x \cdot v = \rho(x)(v)$; the fact that ρ is a Lie algebra homomorphism implies that this is indeed a $\mathfrak{g}$ -module. Conversely, given a $\mathfrak{g}$ -action on V , the association $\rho(x)(v) = x \cdot v$ defines a $\mathfrak{g}$ -representation.

Conclusion: the following notions are the same

$$\{\mathfrak{g}\text{-actions}\} = \{\mathfrak{g}\text{-representations}\}. \quad (2.3.1)$$

We will freely switch between $\mathfrak{g}$ -actions and representations without mention. Moreover, after choosing a basis of V , we may view a $\mathfrak{g}$ -representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ as a homomorphism $\mathfrak{g} \rightarrow \mathfrak{gl}_n(F)$.

2.4 First examples of representations

Let $\mathfrak{g}$ be a Lie algebra.

Definition 2.13. If V is a one-dimensional vector space, then the association $x \cdot v = 0$ defines a representation of $\mathfrak{g}$ on V , called the **trivial representation**. It corresponds to the zero homomorphism $\mathfrak{g} \rightarrow \mathfrak{gl}(V) = \mathfrak{gl}_1$.

The most important representation of any Lie algebra is the adjoint representation. If $x \in \mathfrak{g}$, let ad_x denote the linear map $\mathfrak{g} \rightarrow \mathfrak{g}$ given by $y \mapsto [xy]$. Since the bracket is bilinear, the association $x \mapsto \text{ad}_x$ is a linear map $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$.

Lemma 2.14. The map ad is a Lie algebra homomorphism.

Proof. We need to check that $\text{ad}_{[x,y]} = [\text{ad}_x, \text{ad}_y]$ for all $x, y \in \mathfrak{g}$. In other words, $\text{ad}_{[x,y]} = \text{ad}_x \circ \text{ad}_y - \text{ad}_y \circ \text{ad}_x$. Equivalently, $\text{ad}_{[x,y]}(z) = \text{ad}_x(\text{ad}_y(z)) - \text{ad}_y(\text{ad}_x(z))$ for all $z \in \mathfrak{g}$. Equivalently,

$$[[x, y], z] = [x, [y, z]] - [y, [x, z]]$$

for all $x, y, z \in \mathfrak{g}$. This identity is equivalent to the Jacobi identity! □

Definition 2.15. The representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is called the *adjoint representation*.

This is the crucial consequence of the Jacobi identity: that the adjoint representation is indeed a representation.

When a Lie algebra $\mathfrak{g}$ is defined as a subalgebra of $\mathfrak{gl}_n(F)$, then the tautological inclusion $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(F)$ is called the *defining representation*.

Example 2.16. Suppose $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$, which has basis $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ satisfying the relations

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

We have already seen 3 representation of $\mathfrak{sl}_2(\mathbb{C})$: the trivial one, the defining representation $\mathfrak{sl}_2 \rightarrow \mathfrak{gl}_2$, and the adjoint representation $\mathfrak{sl}_2 \rightarrow \mathfrak{gl}(\mathfrak{sl}_2) \simeq \mathfrak{gl}_3$, which in the basis $\{e, h, f\}$ has the form

$$\text{ad}_e = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{ad}_h = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad \text{ad}_f = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

2.5 Morphisms of representations

Definition 2.17. A *$\mathfrak{g}$ -homomorphism* (or simply morphism) between $\mathfrak{g}$ -representations V and W is a linear map $f: V \rightarrow W$ such that $f(x \cdot v) = x \cdot f(v)$ for all $x \in \mathfrak{g}$ and $v \in V$. This is also called a $\mathfrak{g}$ -linear map or $\mathfrak{g}$ -equivariant map. Write

$$\text{Hom}_{\mathfrak{g}}(V, W) \subset \text{Hom}(V, W) \tag{2.5.1}$$

for the subset of all $\mathfrak{g}$ -homomorphisms between V and W . This is a linear subspace of $\text{Hom}(V, W)$.

Definition 2.18. A $\mathfrak{g}$ -homomorphism between two representations is called a *$\mathfrak{g}$ -isomorphism* (or simply an isomorphism) if it is bijective. We say two representations are *isomorphic* or *equivalent* if there exists an isomorphism between them.

Proposition 2.19. Let $\rho, \rho': \mathfrak{g} \rightarrow \mathfrak{gl}_n(F)$ be two representations. Then ρ and ρ' are equivalent as representations if and only if there exists a matrix $g \in \text{GL}_n(F)$ such that $\rho'(x) = g\rho(x)g^{-1}$ for all $x \in \mathfrak{g}$.

Proof. An isomorphism $F^n \xrightarrow{\sim} F^n$ corresponds to an element $g \in \text{GL}_n(F)$, and the automorphism of $\mathfrak{gl}_n(F)$ is given by conjugation by g . $\square$

2.6 Properties of representations

Definition 2.20. Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a representation.

- The dimension of V is called the *degree* or *dimension* of the representation ρ .
- We say ρ is *faithful* if ρ is injective, equivalently every nonzero x has $x \cdot v \neq 0$ for some $v \in V$.
- We say a subspace $W \subset V$ is a *subrepresentation* of V if W is stable under the $\mathfrak{g}$ -action, i.e. $x \cdot w \in W$ for all $w \in W$ and $x \in \mathfrak{g}$. In that case W is itself a representation of $\mathfrak{g}$.

- We say V is *irreducible* if $V \neq \{0\}$ and the only subrepresentations of V are $\{0\}$ and V itself.

Example 2.21. The representations of $\mathfrak{sl}_2(\mathbb{C})$ described in Example 2.16 are all irreducible.

Lemma 2.22 (Schur's lemma). Let V, W be irreducible representations of a Lie algebra $\mathfrak{g}$ and $\varphi: V \rightarrow W$ a $\mathfrak{g}$ -equivariant map. Then:

1. The map φ is either zero or an isomorphism.
2. If F is algebraically closed (for example, $F = \mathbb{C}$), then φ is a scalar multiple of the identity map.

Proof. 1. If φ is nonzero, its kernel is a proper subrepresentation of V . Since V is irreducible, this kernel must be zero, so φ is injective. Similarly, by considering its image, φ is surjective.

2. Since F is algebraically closed, φ has an eigenvalue λ (corresponding to a root of the characteristic polynomial). But then the $\mathfrak{g}$ -equivariant map $\varphi - \lambda \cdot \text{Id}$ has an eigenvector in its kernel so is not an isomorphism. By the first part of the lemma, it must be zero.

□

3 Representation theory of $\mathfrak{sl}_2(\mathbb{C})$

Lecture 3
starts here

We assume $F = \mathbb{C}$ in this section and we will study representations of $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C}) = \mathfrak{sl}_2$. This concrete example will motivate much of the structure theory of general semisimple algebras later, and is also beautiful in its own right.

Recall that $\mathfrak{g}$ consists of 2×2 -matrices over $\mathbb{C}$ of trace zero, with Lie bracket $[X, Y] = XY - YX$. To study it, we will use the following basis:

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (3.0.1)$$

We can explicitly compute that

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h. \quad (3.0.2)$$

These will be the only matrix calculations we will do.

We have already seen some examples of representations of $\mathfrak{sl}_2$ in Example 2.16. In all of these examples, the image of h in $\mathfrak{gl}(V)$ is diagonalizable. This will be true in general, and the element h will play a crucial role in our classification of irreducible representations of $\mathfrak{g}$.

3.1 Irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$

Let V be a (finite-dimensional) irreducible representation of $\mathfrak{g}$, with corresponding homomorphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. Let's try to say as much as possible about V . For each $\lambda \in \mathbb{C}$, let

$$V_\lambda = \{v \in V \mid hv = \lambda v\} \quad (3.1.1)$$

be the λ -eigenspace of V . We know that $V_\lambda \neq 0$ for some $\lambda \in \mathbb{C}$ (since $\rho(h)$ has an eigenvalue), but we don't know yet that $V = \bigoplus V_\lambda$ since the h -action on V might not be diagonalizable. How do the elements e, f interact with these eigenspaces?

Lemma 3.1. For every $\lambda \in \mathbb{C}$, $e \cdot V_\lambda \subset V_{\lambda+2}$ and $f \cdot V_\lambda \subset V_{\lambda-2}$.

Proof. Let $v \in V_\lambda$, so $hv = \lambda v$. Since $he - eh = [h, e] = 2e$ and $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a Lie algebra homomorphism, we compute that

$$h(ev) = e \cdot (h \cdot v) + [h, e] \cdot v = e(hv) + 2ev \quad (3.1.2)$$

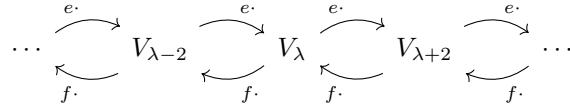
$$= e(\lambda v) + 2ev \quad (3.1.3)$$

$$= (\lambda + 2)(ev). \quad (3.1.4)$$

Therefore $ev \in V_{\lambda+2}$. The calculation for f is similar, using that $[h, f] = -2f$. $\square$

Lemma 3.1 implies that $W = \bigoplus_{\lambda \in \mathbb{C}} V_\lambda$ is a subspace of V that is stable under e, h, f so is $\mathfrak{g}$ -stable. It is also nonzero, since $\rho(h)$ has at least one eigenvalue. Since V is irreducible, this implies that $V = W$, so h is diagonalizable.

The following picture summarizes the situation:



Let $S = \{\lambda \in \mathbb{C} \mid V_\lambda \neq 0\}$. Elements of S are called the **weights** of V . Lemma 3.1 implies that if $\mu \in S$, then $\bigoplus_{\lambda \in \mu + 2\mathbb{Z}} V_\lambda$ is $\mathfrak{g}$ -stable. It follows that this subspace must equal V by irreducibility, so every two elements in S differ by an even integer.

How do we proceed further? By the finite dimensionality of V , in the above picture you can't keep hopping to the right. In other words, there exists a $\lambda \in S$ such that $V_\lambda \neq 0$ and $V_{\lambda+2} = 0$. Let $v \in V_\lambda$ be a nonzero element. This is called a **highest weight vector**.

Lemma 3.2. The subspace $W = \text{span}\{v, fv, f^2v, \dots\}$ is $\mathfrak{g}$ -stable.

Proof. It suffices to check that W is stable under e, h and f . We have $f \cdot W \subset W$ by construction. By Lemma 3.1, $h \cdot (f^n v) = (\lambda - 2n)(f^n v)$ so $h \cdot W \subset W$. It suffices to check that W is stable under e . Let's check that $e \cdot (f^n v) \in W$ for small values of n . If $n = 0$, then $e \cdot v = 0$ since $e \cdot v \in V_{\lambda+2} = 0$. If $n = 1$, then

$$e(fv) = (fe + [e, f])v \quad (3.1.5)$$

$$= f(ev) + hv \quad (3.1.6)$$

$$= \lambda v. \quad (3.1.7)$$

If $n = 2$, we compute

$$e(f^2v) = (ef)(fv) \quad (3.1.8)$$

$$= (fe + [e, f])(fv) \quad (3.1.9)$$

$$= f(e fv) + h f v. \quad (3.1.10)$$

Since $e fv = \lambda v$ and $f v \in V_{\lambda-2}$, we get $e(f^2v) = (2\lambda - 2)f v$. We can start to see a pattern here, and prove by induction (exercise) that for all $n \geq 0$ we have

$$e \cdot (f^n v) = n(\lambda - n + 1)(f^{n-1}v). \quad (3.1.11)$$

This proves that $e \cdot W \subset W$, concluding the lemma. $\square$

Corollary 3.3. *All the h -eigenspaces are one-dimensional: if $\lambda \in S$ then $\dim V_\lambda = 1$.*

Proof. Let $v \in V_\lambda$ be a highest weight vector. Then $W = \text{span}\{v, fv, f^2v, \dots\}$ is $\mathfrak{g}$ -stable so by irreducibility $V = W$. It follows that $S = \{\lambda, \lambda - 2, \dots\}$ and $V_{\lambda-2n}$ is spanned by $e^n v$ (which might be zero). $\square$

Corollary 3.4. *There exists a nonnegative integer $n \geq 0$ such that $S = \{-n, -n + 2, \dots, n - 2, n\}$ and V has dimension $n + 1$.*

Proof. Let $v \in V_\lambda$ be a highest weight vector. Let $n \geq 0$ be the largest nonnegative integer such that $f^n v \neq 0$. Then $W = \text{span}\{v, fv, f^2v, \dots, f^n v\}$. Formula (3.1.11) applied to $f^{n+1}v$ shows that

$$e(f^{n+1}v) = (n+1)(\lambda - n)f^n v. \quad (3.1.12)$$

Since $f^{n+1}v = 0$ and $f^n v \neq 0$, $(n+1)(\lambda - n) = 0$. Since $n+1 \geq 1$, this implies that $\lambda = n$. $\square$

Theorem 3.5. *Let $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ and $n \in \mathbb{Z}_{\geq 0}$. Then up to isomorphism $\mathfrak{g}$ has a unique irreducible representation of dimension $n + 1$, denoted by $V(n)$.*

Proof. We first show uniqueness. Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be an $(n+1)$ -dimensional irreducible representation of $\mathfrak{g}$. Let $v \in V$ be a highest weight vector. The above analysis shows that $hv = nv$ and $V = \text{span}\{v, fv, \dots, f^n v\}$. The proof of Lemma 3.2 explicitly describes the action of $\mathfrak{g}$ on V . Indeed, in the basis $\{v, fv, \dots, f^n v\}$ this action is given by

$$\rho(h) = \begin{pmatrix} n & & & \\ & n-2 & & \\ & & \ddots & \\ & & & -n \end{pmatrix} \quad (3.1.13)$$

$$\rho(e) = \begin{pmatrix} 0 & n & & \\ & 0 & 2n-2 & \\ & & \ddots & \ddots \\ & & & \ddots & n \\ & & & & 0 \end{pmatrix} \quad (3.1.14)$$

$$\rho(f) = \begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{pmatrix} \quad (3.1.15)$$

The matrix $\rho(e)$ has coefficient $i(n-i+1)$ at entry $(i, i+1)$, and zeroes elsewhere. This shows that every irreducible representation of $\mathfrak{g}$ of dimension $n+1$ must be isomorphic to one given by these matrices. To prove existence, we can merely check that the above big matrices indeed define a Lie algebra homomorphism $\mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}_{n+1}(\mathbb{C})$, by checking that the commutation relations $[\rho(h), \rho(e)] = 2\rho(e)$, $[\rho(h), \rho(f)] = -2\rho(f)$ and $[\rho(e), \rho(f)] = \rho(h)$. We still need to verify that this representation is irreducible. But if $W \subset \mathbb{C}^{n+1}$ is a $\mathfrak{g}$ -stable subspace, acting by e repeatedly shows that W contains a highest weight vector. Applying f repeatedly to this highest weight vector shows that W generates V . $\square$

Remark 3.6. *Another way to prove existence is to explicitly exhibit $V(n)$: it is given by $\text{Sym}^n V$, where $V = \mathbb{C}^2$ is the defining representation of $\mathfrak{sl}_2(\mathbb{C})$. (Example sheet 1)*

The h -eigenspaces on $V(n)$ are $-n, -n+2, \dots, n-2, n$ and n is the largest eigenvalue of h . That's why we denote this representation by $V(n)$ and call it the representation of highest weight n .

So we have completely classified all irreducible representations of $\mathfrak{sl}_2(\mathbb{C})$: there is exactly one of them in each dimension. But what if we start with a representation that is not irreducible? Is every representation a direct sum of irreducibles? For a general Lie algebra, this need not be true. For our specific example of $\mathfrak{sl}_2(\mathbb{C})$, complete reducibility still holds, as we will see in the next subsection.

3.2 Complete reducibility

Lecture 4
starts here

In this subsection only, let $\mathfrak{g}$ be a Lie algebra over any field F .

Definition 3.7. If V, W are $\mathfrak{g}$ -representations, then $V \oplus W = \{(v, w) : v \in V, w \in W\}$ has the structure of a $\mathfrak{g}$ -representation, via $x \cdot (v, w) = (x \cdot v, x \cdot w)$, called the direct sum of V and W .

Definition 3.8. We say a $\mathfrak{g}$ -representation is *completely reducible* if it is a direct sum of irreducible representations.

Lemma 3.9. A $\mathfrak{g}$ -representation V is completely reducible if and only if for every $\mathfrak{g}$ -subrepresentation $W \subset V$, there exists a subrepresentation $W' \subset V$ such that $V = W \oplus W'$.

Proof. (non-examinable) This is very formal but a little tricky; we only sketch the proof. For the backward direction, apply induction on the dimension of V . For the forward direction, let $W \subset V$ be a subrepresentation and let $\mathcal{F}$ be the set of all subrepresentations $U \subset V$ that satisfy $U \cap W = \{0\}$. Let $W' \in \mathcal{F}$ be a subspace that has maximal dimension among the elements of $\mathcal{F}$. We claim that $W \oplus W' = V$. Indeed, $W \cap W' = 0$ by definition of $\mathcal{F}$. Write $V = \oplus V_i$ where each V_i is irreducible. It suffices to prove that $V_i \subset W \oplus W'$ for each i . Fix such an i . If $V_i \subset W'$, then certainly $V_i \subset W \oplus W'$. Otherwise, $V_i \not\subset W'$, hence $V_i + W'$ has strictly bigger dimension than W' and so does not lie in $\mathcal{F}$. Therefore $W \cap (V_i + W') \neq 0$. Since $W \cap W' = 0$, it follows that $V_i \cap (W + W') \neq 0$. But $V_i \cap (W + W')$ is a nonzero $\mathfrak{g}$ -invariant subspace of the irreducible V_i , so it must equal V_i . So $V_i \subset W + W'$ for all i , concluding the proof. $\square$

Example 3.10. The defining representation of $\mathfrak{b}_2 \rightarrow \mathfrak{gl}_2$ (recall $\mathfrak{b}_2$ denotes upper triangular matrices) is not completely reducible: the first basis vector e_1 of F^2 spans a $\mathfrak{b}_2$ -invariant subspace which does not have a complement. (Check!)

3.3 More operations on representations

There are various ways in which we can make new representations from old ones on a Lie algebra $\mathfrak{g}$:

- (Quotient) If $W \leq V$ is a subrepresentation of a $\mathfrak{g}$ -representation V , then V/W has the structure of a $\mathfrak{g}$ -representation, via $x \cdot (v + W) = x \cdot v + W$.
- (Dual) If V is a $\mathfrak{g}$ -representation, then $V^* = \text{Hom}(V, F)$ has the structure of a $\mathfrak{g}$ -representation, via the formula $(x \cdot f)(v) = -f(x \cdot v)$. (Note the minus sign!)
- (Hom space) If V, W are $\mathfrak{g}$ -representations, then $\text{Hom}(V, W)$ is a $\mathfrak{g}$ -representation, via $(x \cdot \varphi)(v) = x \cdot \varphi(v) - \varphi(x \cdot v)$.
- (Tensor product) If V, W are $\mathfrak{g}$ -representations, then $V \otimes W$ is spanned by elements of the form $v \otimes w$ and the assignment $x \cdot (v \otimes w) = (x \cdot v) \otimes w + v \otimes (x \cdot w)$ extends by linearity to a $\mathfrak{g}$ -action on $V \otimes W$. This induces $\mathfrak{g}$ -actions on $\text{Sym}^n V$ and $\bigwedge^n V$. (See the hand-out on tensor products for more on this.)

You should check for yourself that these formulae indeed define $\mathfrak{g}$ -representations.

3.4 Complete reducibility for $\mathfrak{sl}_2(\mathbb{C})$

We now go back to the case $F = \mathbb{C}$ and $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$, and we will prove the following theorem:

Theorem 3.11 (Complete reducibility). *Every (finite-dimensional) representation of $\mathfrak{sl}_2(\mathbb{C})$ is a direct sum of irreducible representations.*

We will prove this later in this section. We first draw some important consequences. If V is a representation of $\mathfrak{sl}_2(\mathbb{C})$, the set of $\lambda \in \mathbb{Z}$ for which $V_\lambda = \{v \in V : h \cdot v = \lambda v\}$ is nonzero are called the **weights** of V . We view the set of weights as a multiset, where a weight λ has multiplicity $\dim V_\lambda$.

Corollary 3.12. *The isomorphism class of a representation of $\mathfrak{sl}_2(\mathbb{C})$ is determined by its weights (with multiplicity).*

Example 3.13. *Let V be a representation of $\mathfrak{sl}_2(\mathbb{C})$ with weights $5, 3, 3, 1, 1, 0, -1, -1, -3, -3, -5$. Then $V \simeq V(5) \oplus V(3) \oplus V(0)$.*

Example 3.14. *Suppose that V is a 5-dimensional representation and 3 is a weight of V . This means that $3, 1, -1, -3$ are all weights of V . By counting dimensions, we must have $V \simeq V(3) \oplus V(0)$.*

The proof of complete reducibility is a bit technical. The key tool will be the ‘Casimir element’.

Definition 3.15. *Let $\rho: \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a representation. The **Casimir element** associated with ρ is the linear map*

$$\Omega_\rho = \rho(e)\rho(f) + \rho(f)\rho(e) + \frac{1}{2}\rho(h)^2 \in \text{End}(V).$$

Note that the multiplications in the above expressions are understood to be composition of linear maps (i.e. ordinary matrix multiplications). Note that the element Ω_ρ is not necessarily of the form $\rho(x)$ for some $x \in \mathfrak{sl}_2(\mathbb{C})$. Also note that $\rho(e)\rho(f)$ does not equal $\rho(e f)$ (the latter does not even make sense, since it does not lie in $\mathfrak{sl}_2(\mathbb{C})$).

Lemma 3.16. *Let $\rho: \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(V)$ be a representation. Then the Casimir element Ω_ρ , seen as a linear map $V \rightarrow V$, is a homomorphism of $\mathfrak{sl}_2(\mathbb{C})$ -representations, i.e. it is $\mathfrak{sl}_2(\mathbb{C})$ -equivariant.*

Proof. We need to show that $\rho(x)\Omega_\rho = \Omega_\rho\rho(x)$ for $x = e, h, f$. This is an explicit calculation, which we do for $x = e$ only. For ease of notation, we will simply write e, h, f instead of $\rho(e), \rho(h), \rho(f)$, and we magically compute

$$\begin{aligned} e\Omega_\rho - \Omega_\rho e &= eef + efe + eh^2/2 - efe - fee - h^2e/2 \\ &= eef + eh^2/2 - fee - h^2e/2 \\ &= e([e, f] + fe) + ([e, h] + he)h/2 - ([f, e] + ef)e - h/2([h, e] + eh) \\ &= eh + efe - eh + heh/2 + he - efe - he - heh/2 \\ &= 0 \end{aligned}$$

This is the key property of the Casimir element, later we will see a more conceptual explanation of its construction. □

Corollary 3.17. *If ρ is irreducible of dimension $n + 1$, then $\Omega_\rho = \left(n + \frac{n^2}{2}\right) \text{Id}_V$.*

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Proof. By the previous lemma, Ω_ρ is an $\mathfrak{sl}_2(\mathbb{C})$ -equivariant map. By Schur's lemma, $\Omega_\rho = c \text{Id}_V$ for some scalar $c \in \mathbb{C}$. To compute the scalar, we determine the action of Ω_ρ on a highest weight vector v . The explicit description of $V(n)$ shows $(ef + fe + h^2/2)v = e(fv) + h^2/2(v) = nv + n^2v/2 = (n + n^2/2)v$. $\square$

Lemma 3.18. *Let $\mathfrak{h}$ be an abelian Lie algebra. Then every Lie algebra homomorphism $\mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{h}$ is the zero homomorphism.*

Proof. Let φ be such a homomorphism. Then $\varphi([x, y]) = 0$ for all $x, y \in \mathfrak{sl}_2(\mathbb{C})$. Applying this to x, y ranging over e, h, f shows that $\varphi(e) = \varphi(f) = \varphi(h) = 0$, as desired. $\square$

We will prove Theorem 3.11 by proving, using Lemma 3.9, that for every $\mathfrak{sl}_2(\mathbb{C})$ -representation V , every subrepresentation $W \subset V$ has an invariant complement, i.e., a subrepresentation $W' \subset V$ such that $V = W \oplus W'$. We will first treat the case where W has codimension 1.

Proposition 3.19. *Suppose that V is a $\mathfrak{sl}_2(\mathbb{C})$ -representation and $W \subset V$ a subrepresentation of codimension 1. Then there exists a trivial subrepresentation $W' \subset V$ such that $V = W \oplus W'$.*

Proof. Recall that the one-dimensional quotient space V/W has the structure of an $\mathfrak{sl}_2(\mathbb{C})$ -representation, via $x \cdot (v + W) = (x \cdot v) + W$ for all $v \in V, x \in \mathfrak{g}$. Since $\mathfrak{gl}_1$ is abelian, the action of $\mathfrak{sl}_2(\mathbb{C})$ is actually trivial, i.e., V/W is the trivial representation by Lemma 3.18. We will break up the proof into several cases.

Case 1: W is trivial Then V has a basis in which $\rho: \mathfrak{sl}_2(\mathbb{C}) \rightarrow \mathfrak{gl}(V) = \mathfrak{gl}_2(\mathbb{C})$ lands in the subalgebra $\left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\}$, i.e. lands in the subalgebra $\mathfrak{n}_2$. Since this subalgebra is abelian, ρ is the zero homomorphism by Lemma 3.18, so $V \simeq V(0) \oplus V(0)$ as representations and any vector space complement to W is also a complementary subrepresentation.

Case 2: W is irreducible and nontrivial Consider the Casimir element $\Omega_\rho \in \text{End}(V)$, which preserves the subrepresentation W . By the above Corollary 3.17, the restriction of Ω_ρ is a nonzero multiple of the identity map, and Ω_ρ induces the zero map on V/W . In other words, choosing a basis of W and extending it to a basis of V , Ω_ρ has the form

$$\begin{pmatrix} cI_{\dim W} & * \\ 0 & 0 \end{pmatrix}.$$

It follows that $V = W \oplus \ker(\Omega_\rho)$. Since $\ker(\Omega_\rho)$ is a subrepresentation of V , we have constructed a complementary subrepresentation to W .

Case 3: W general We now treat the general case for W by induction on the dimension. The $\dim W = 0$ case is trivial. The case where W is irreducible is handled in cases 1 and 2. If W is not irreducible, there exists a proper nonzero subrepresentation $U \leq W$. In that case W/U is a subrepresentation of V/U of codimension 1. By the induction hypothesis, there exists a 1-dimensional subspace $L \subset V/U$ such that $V/U = (W/U) \oplus L$. Since subspaces of V/U are all of the form W'/U , where W' is a subspace of V containing U , we can write $L = W'/U$, where $W' \leq V$ is a subrepresentation containing U . We have $V = W + W'$ and $W \cap W' = U$. The codimension of U in W' is 1, so we may again apply the induction hypothesis and find a subrepresentation $W'' \subset W'$ such that $W' = U \oplus W''$. Then $W = W \oplus W''$, so W'' is a complementary subrepresentation, as desired. $\square$

Proof of Theorem 3.11. Let W be a subrepresentation of an $\mathfrak{sl}_2(\mathbb{C})$ -representation V . We will put ourselves in the situation of Proposition 3.19. To this end, consider the Hom space of linear maps $\text{Hom}(V, W)$, which we recall has the structure of a $\mathfrak{g}$ -representation via $(x \cdot \varphi)(v) = x \cdot (\varphi(v)) - \varphi(x \cdot v)$. Let $\mathcal{V}$ be the subspace of

those $\varphi \in \text{Hom}(V, W)$ such that $\varphi|_W = c\text{Id}_W$ for some $c \in \mathbb{C}$. In other words, $\mathcal{V}$ are those maps that restrict to a scalar multiple of the identity on W . Let $\mathcal{W} \subset \mathcal{V}$ be the subspace of maps φ such that $\varphi|_W = 0$, i.e. they restrict to zero on W . We claim that $\mathcal{W}, \mathcal{V}$ are actually subrepresentations of $\text{Hom}(V, W)$, and that $\mathcal{W}$ has codimension 1 in $\mathcal{V}$. Indeed, if $x \in \mathfrak{g}$, $\varphi \in \mathcal{V}$ and $w \in W$, then $(x \cdot \varphi)(w) = \varphi(x \cdot w) - x \cdot \varphi(w)$. Since $\varphi|_W = c\text{Id}_W$, the latter equals $c(x \cdot w) - x \cdot (cw) = 0$. So $x \cdot \mathcal{V} \subset \mathcal{W}$, hence both are indeed subrepresentations. To show that $\mathcal{W}$ has codimension 1 in $\mathcal{V}$, consider the map $F: \mathcal{V} \rightarrow \mathbb{C}$ sending φ to c , where $c \in \mathbb{C}$ is the unique element satisfying $\varphi|_W = c\text{Id}_W$. Then F is linear and surjective, hence the kernel, which is by definition $\mathcal{W}$, has codimension 1.

We can now apply Proposition 3.19, and obtain a trivial subrepresentation $\mathcal{W}' \subset \mathcal{V}$ such that $\mathcal{V} = \mathcal{W} \oplus \mathcal{W}'$. Let $\varphi \in \mathcal{W}'$ be a nonzero element. Since $\mathcal{W}'$ is a trivial $\mathfrak{sl}_2(\mathbb{C})$ -representation, $x \cdot \varphi = 0$ for all $x \in \mathfrak{sl}_2(\mathbb{C})$. In other words, the linear map $\varphi: V \rightarrow W$ is $\mathfrak{g}$ -equivariant! Since it restricts to $c\text{Id}_W$ on W for some nonzero c , $W' = \ker(\varphi)$ is a subrepresentation satisfying $V = W \oplus W'$, as desired. $\square$

The proof of Theorem 3.11 was fairly complicated. Fortunately, the same proof will generalize to arbitrary semisimple Lie algebras, once we have a definition of the Casimir element for a general semisimple Lie algebra.

Remark 3.20 (Non-examinable). *I am not a huge fan of the proof of Theorem 3.11 given above. There is a much better proof, if one is willing to admit some facts about the relation between Lie groups and Lie algebras. Consider the compact group $G = \text{SU}(2) = \{A \in \text{SL}_2(\mathbb{C}) : AA^t = I\}$. If $G \rightarrow \text{GL}_n(\mathbb{C})$ is a representation, then by taking the derivative and complexifying, we get a representation of $\mathfrak{sl}_2(\mathbb{C})$. This association induces an equivalence of categories between the category of representations of $\text{SU}(2)$ and the category of representations of $\mathfrak{sl}_2(\mathbb{C})$. (This is the part that uses some hard inputs from analysis and topology.) Using this equivalence, it suffices to prove that every representation of $\text{SU}(2)$ is completely reducible. This can be proved by constructing an $\text{SU}(2)$ -invariant inner product using the Haar integral, which naturally generalizes the proof of complete reducibility in the finite group case. (Don't worry if this doesn't make sense to you, we will never use it in what follows.)*

4 Structure theory of general Lie algebras: theorems of Engel, Lie and Cartan

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(In class we said a few words about tensor products, see the appendix posted on Moodle.)

The next big goal of the course is the classification of complex semisimple Lie algebras. To this end, we need to make various definitions and prove intermediate results concerning nilpotent/solvable Lie algebras, which will take us a few lectures.

4.1 Ideals

Let $\mathfrak{g}$ be a Lie algebra over a field F .

Definition 4.1. An *ideal* of $\mathfrak{g}$ is a subspace $I \subset \mathfrak{g}$ such that $[x, i] \in I$ for all $x \in \mathfrak{g}$ and $i \in I$.

Obviously, an ideal is also a subalgebra, so is a Lie algebra by restricting the bracket.

Remark 4.2. An equivalent definition of an ideal is that when considering $\mathfrak{g}$ as a $\mathfrak{g}$ -module using the adjoint representation, an ideal is exactly a subrepresentation of $\mathfrak{g}$.

If $\varphi: \mathfrak{g} \rightarrow \mathfrak{h}$ is a homomorphism of Lie algebras, then $\ker(\varphi)$ is an ideal of $\mathfrak{g}$. In fact, every ideal arises in this way. If $I \subset \mathfrak{g}$ is an ideal, then the quotient space $\mathfrak{g}/I$ inherits the structure of a Lie algebra via $[x+I, y+I] = [x, y] + I$. The projection $\mathfrak{g} \rightarrow \mathfrak{g}/I$ is a homomorphism of Lie algebras whose kernel is I .

Remark 4.3. If $F = \mathbb{R}$ and $\mathfrak{g}$ is the Lie algebra of a Lie group G , then a closed subgroup $H \subset G$ gives rise to a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$, while a normal closed subgroup $H \subset G$ gives rise to an ideal $\mathfrak{h} \subset \mathfrak{g}$.

Definition 4.4. We say $\mathfrak{g}$ is *simple* if $\mathfrak{g}$ is not abelian (in other words, $[\mathfrak{g}, \mathfrak{g}] \neq 0$) and the only ideals of $\mathfrak{g}$ are 0 and $\mathfrak{g}$ itself.

Equivalently, $\mathfrak{g}$ is simple if and only if the adjoint representation $\mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is nonzero and irreducible.

Example 4.5. We have seen before that $\mathfrak{sl}_2(\mathbb{C})$ is simple, since the adjoint representation $V(2)$ is irreducible. It turns out that all the Lie algebras $\mathfrak{sl}_n(\mathbb{C})$ ($n \geq 2$), $\mathfrak{sp}_{2\ell}(\mathbb{C})$ ($\ell \geq 1$) and $\mathfrak{so}_n(\mathbb{C})$ ($n \geq 5$) are simple.

We will later classify simple Lie algebras when $F = \mathbb{C}$. It turns out that the above example lists almost all of them, apart from 5 exceptions, called exceptional Lie algebras.

4.2 Nilpotent and solvable Lie algebras

Let $\mathfrak{g}$ be a Lie algebra over a field F .

Definition 4.6. If S_1, S_2 are subsets of $\mathfrak{g}$, write $[S_1, S_2]$ for the span of $\{[s_1, s_2]: s_i \in S_i\}$.

Definition 4.7. The *derived subalgebra* of $\mathfrak{g}$ is by definition the subalgebra $[\mathfrak{g}, \mathfrak{g}]$.

This is indeed a subalgebra, because it is even an ideal.

Example 4.8. We have $[\mathfrak{gl}_n, \mathfrak{gl}_n] = \mathfrak{sl}_n$.

Just like in the case of groups, we can define the derived series and central series. Set $\mathfrak{g}^{(0)} = \mathfrak{g}^0 = \mathfrak{g}$ and inductively define

$$\mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}] \quad (4.2.1)$$

$$\mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}]. \quad (4.2.2)$$

Then $\mathfrak{g}^{(n)}$ and $\mathfrak{g}^n$ are all ideals of $\mathfrak{g}$ (you will prove this on example sheet 2) and the chain of ideals

$$\mathfrak{g} \supset \mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] \supset \mathfrak{g}^{(2)} = [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \supset \cdots \quad (4.2.3)$$

is called the *derived series* and the chain of ideals

$$\mathfrak{g} \supset \mathfrak{g}^1 = [\mathfrak{g}, \mathfrak{g}] \supset \mathfrak{g}^2 = [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supset \cdots \quad (4.2.4)$$

is called the *(lower) central series*.

Definition 4.9. We say $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n . We say $\mathfrak{g}$ is *nilpotent* if $\mathfrak{g}^n = 0$ for some n .

Note that $\mathfrak{g}^{(n)} \subset \mathfrak{g}^n$, so every nilpotent Lie algebra is solvable. The converse is not true, we will see an example when discussing the most important examples of nilpotent and solvable Lie algebras.

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Example 4.10. Let $\mathfrak{g} = \mathfrak{n}_n$ be the subalgebra of strictly upper triangular matrices in $\mathfrak{gl}_n$. We say an element $A \in \mathfrak{n}_n$ has 'level $\geq k$ ' if $a_{ij} = 0$ whenever $j - i < k$. So every element of $\mathfrak{n}_n$ has level ≥ 1 . Let e_{ij} be the elementary matrix with only nonzero entry at the (i, j) -position, whose entry equals 1. Then a computation shows that

$$[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj}. \quad (4.2.5)$$

It follows that if $A, B \in \mathfrak{n}_n$ are such that A has level $\geq k$ and B has level $\geq m$, then $[A, B]$ has level $\geq k + m$. Therefore every element of $\mathfrak{g}^k$ has level $\geq k + 1$. So $\mathfrak{g}^n = 0$ and $\mathfrak{g}$ is nilpotent.

Example 4.11. If $\mathfrak{g} = \mathfrak{b}_n \subset \mathfrak{gl}_n$ is upper triangular matrices, then the language of the previous example applies, and every element of $[\mathfrak{g}, \mathfrak{g}]$ has level ≥ 1 , and inductively every element of $\mathfrak{g}^{(k)}$ has level $\geq 2^{k-1}$, so $\mathfrak{g}$ is solvable. However, $\mathfrak{g}^1 = \mathfrak{g}^{(1)} = \mathfrak{n}_n$ and $\mathfrak{g}^k = \mathfrak{n}_n$ for all $k \geq 1$ so $\mathfrak{g}$ is not nilpotent.

Example 4.12. Every abelian Lie algebra is nilpotent.

Lemma 4.13. If $\mathfrak{g}$ is nilpotent or solvable, then so is any quotient and subalgebra of $\mathfrak{g}$.

Proof. If $\mathfrak{h} \subset \mathfrak{g}$ is a subalgebra, then $\mathfrak{h}^n \subset \mathfrak{g}^n$ and $\mathfrak{h}^{(n)} \subset \mathfrak{g}^{(n)}$ for all $n \geq 0$. If $I \subset \mathfrak{g}$ is an ideal, then $(\mathfrak{g}/I)^n = \mathfrak{g}^n + I$ and $(\mathfrak{g}/I)^{(n)} = \mathfrak{g}^{(n)} + I$. These imply the statements. $\square$

4.3 Engel's theorem

Recall that an endomorphism $\varphi \in \mathfrak{gl}(V)$ is said to be nilpotent if $\varphi^n = 0$ for some $n \geq 1$. Equivalently, all of its eigenvalues are zero, equivalently its characteristic polynomial is $T^{\dim V}$.

Lemma 4.14. If $\mathfrak{g}$ is nilpotent, then $\text{ad}_x = \text{ad}(x) \in \mathfrak{gl}(\mathfrak{g})$ is nilpotent for all $x \in \mathfrak{g}$.

Proof. By a reformulation of the definition of nilpotence, a Lie algebra is nilpotent if and only if there exists an $n \geq 1$ such that $\text{ad}_{x_1} \circ \cdots \circ \text{ad}_{x_n} = 0$ for all $x_i \in \mathfrak{g}$. Taking $x = x_1 = \cdots = x_n$, we get $\text{ad}_x^n = 0$, so ad_x is nilpotent. $\square$

Remark 4.15. If $\mathfrak{g}$ is a nilpotent subalgebra of $\mathfrak{gl}(V)$, then it is not necessarily true that x is nilpotent for all $x \in \mathfrak{g}$. For example, the subalgebra of diagonal matrices in $\mathfrak{gl}_n(V)$ is nilpotent but contains non-nilpotent elements.

However, we do have the following handy criterion for checking whether a Lie algebra is nilpotent:

Theorem 4.16 (Engel). A Lie algebra $\mathfrak{g}$ is nilpotent if and only if $\text{ad}(x)$ is nilpotent for all $x \in \mathfrak{g}$.

This theorem and a few of the next ones are all boring linear algebra. To convince you of this, we will prove Engel's theorem and omit the proofs of the others; you can find them in Humphreys.

We will use the following intermediate results:

Lemma 4.17. If $x \in \mathfrak{gl}(V)$ is nilpotent, then $\text{ad}(x)$ is also nilpotent.

Proof. Let $L_x: \mathfrak{gl}(V) \rightarrow \mathfrak{gl}(V)$ be the linear map $y \mapsto xy$ and $R_x: \mathfrak{gl}(V) \rightarrow \mathfrak{gl}(V)$ the linear map $y \mapsto yx$. Then L_x and R_x are both nilpotent, since x is. Moreover $L_x \circ R_x = R_x \circ L_x$. Therefore $\text{ad}(x) = L_x - R_x$ is also nilpotent, since the sum of two commuting nilpotent endomorphisms is again nilpotent. $\square$

Proposition 4.18. If $\mathfrak{g} \subset \mathfrak{gl}(V)$ is a subalgebra and x is nilpotent for every element of $\mathfrak{g}$, then there exists a nonzero element $v \in V$ such that $x \cdot v = 0$ for all $x \in \mathfrak{g}$.

Proof. We use induction on $\dim \mathfrak{g}$. When $\dim \mathfrak{g} = 1$, $\mathfrak{g}$ is spanned by one nilpotent element $x \in \mathfrak{gl}(V)$. There is an integer $n \geq 1$ such that $x^n \neq 0$ but $x^{n+1} = 0$. Therefore any nonzero element v in the image of x^n satisfies $x \cdot v = 0$. For the induction step, let $\mathfrak{k}$ be a maximal proper subalgebra of $\mathfrak{g}$. If $x \in \mathfrak{k}$, then since x is nilpotent, $\text{ad}(x): \mathfrak{g} \rightarrow \mathfrak{g}$ is again nilpotent by the previous lemma, hence the induced action on $\mathfrak{g}/\mathfrak{k}$ is also nilpotent. By the induction hypothesis and the fact that $\dim \mathfrak{g}/\mathfrak{k} < \dim \mathfrak{g}$, there exists a nonzero element $\bar{y} \in \mathfrak{g}/\mathfrak{k}$ with $[x, \bar{y}] = 0$ in $\mathfrak{g}/\mathfrak{k}$ for all $x \in \mathfrak{k}$. Lifting $\bar{y}$ to an element $y \in \mathfrak{g} \setminus \mathfrak{k}$, we get that $[x, y] \in \mathfrak{k}$ for all $x \in \mathfrak{k}$. Therefore $\mathfrak{k} + F \cdot y$ is a subalgebra of $\mathfrak{g}$ strictly containing $\mathfrak{k}$. By maximality $\mathfrak{g} = \mathfrak{k} + F \cdot y$. In addition, $\mathfrak{k}$ is an ideal of $\mathfrak{g}$, since it is a subalgebra and it is stable under y . By the induction hypothesis, the subspace $W = \{v \in V : k \cdot v = 0 \text{ for all } k \in \mathfrak{k}\}$ is non-empty. It is $\mathfrak{k}$ -stable by definition, but also stable by y : indeed if $w \in W$ and $k \in \mathfrak{k}$ then

$$k \cdot (y \cdot w) = [k, y] \cdot w + y \cdot (k \cdot w)$$

We have $k \cdot w = 0$ and $[k, y] \in \mathfrak{k}$ by the above so $[k, y] \cdot w = 0$ too. So y restricts to a nilpotent endomorphism of W . By the $\dim V = 1$ case of the theorem, there exists a nonzero $w \in W$ such that $y \cdot w = 0$. Then w satisfies the conclusion of the proposition. $\square$

Corollary 4.19. *If $\mathfrak{g} \subset \mathfrak{gl}(V)$ is a subalgebra and x is nilpotent for every element of $\mathfrak{g}$, then there exists a flag (V_i) of V such that $x \cdot V_i \subset V_{i-1}$ for all $x \in \mathfrak{g}$ and i . Consequently, there exists a basis of V such that $\mathfrak{g}$ is contained in $\mathfrak{n}_n$ and $\mathfrak{g}$ is a nilpotent Lie algebra.*

Proof. By Proposition 4.18, there exists a nonzero $v_1 \in V$ such that $\mathfrak{g} \cdot v_1 = 0$. Let $V_1 = \text{span}\{v_1\}$ and $W = V/V_1$. Then the induced action of $\mathfrak{g}$ on W is by nilpotent endomorphisms. By induction on $\dim V$, we get a flag $W_1 \subset \cdots \subset W_{n-1} = W$ such that $x \cdot W_i \subset W_{i-1}$. Write $W_i = V_{i+1}/V_1$, then the flag $V_1 \subset V_2 \subset \cdots \subset V_n = V$ satisfies the conclusions of the corollary. $\square$

Proof of Engel's theorem. The forward direction is the easy Lemma 4.14. Conversely, assume every element of $\mathfrak{g}$ is ad-nilpotent. By Corollary 4.19, the image of $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. The kernel of ad is by definition the center

$$Z(\mathfrak{g}) = \{x \in \mathfrak{g} : [x, y] = 0 \text{ for all } y \in \mathfrak{g}\}. \quad (4.3.1)$$

Therefore $\mathfrak{g}/Z(\mathfrak{g})$ is isomorphic to the image of ad , hence is nilpotent. This implies that $\mathfrak{g}$ is itself nilpotent. Indeed, if $(\mathfrak{g}/Z(\mathfrak{g}))^n = 0$, then $\mathfrak{g}^n \subset Z(\mathfrak{g})$, so $\mathfrak{g}^{n+1} = [\mathfrak{g}, \mathfrak{g}^n] \subset [\mathfrak{g}, Z(\mathfrak{g})] = 0$. $\square$

A **flag** of a vector space V is a chain of subspace $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ such that $\dim V_i = i$ for all i . Given a flag (V_i) , the set of all $\varphi \in \mathfrak{gl}(V)$ such that $\varphi(V_i) \subset V_i$ for all i is a Lie subalgebra isomorphic to $\mathfrak{b}_n$, and the set of all $\varphi \in \mathfrak{gl}(V)$ such that $\varphi(V_i) \subset V_{i-1}$ for all i is a Lie subalgebra isomorphic to $\mathfrak{n}_n$.

Engel's theorem doesn't have any assumptions on the characteristic of F . Almost all of the theorems later in the course do.

4.4 Lie's theorem

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starts here

From now on, assume $F = \mathbb{C}$.

Theorem 4.20 (Lie). *Let V be a vector space and $\mathfrak{g} \subset \mathfrak{gl}(V)$ a solvable subalgebra. Then there exists a flag (V_i) of V such that $x \cdot V_i \subset V_i$ for all $x \in \mathfrak{g}$ and i . Consequently, there exists a basis so that $\mathfrak{g}$ is contained in the subalgebra $\mathfrak{b}_n$ of upper triangular matrices.*

Proof. The proof is similar to Engel's theorem, and is omitted; see Humphreys, §4.1. $\square$

In particular, by looking at V_1 , there exists a nonzero element $v \in V$ such that for all $x \in \mathfrak{g}$, $x \cdot v = \lambda_x v$ for some $\lambda_x \in \mathbb{C}$.

Corollary 4.21. *A Lie algebra $\mathfrak{g}$ is solvable if and only if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.*

Proof. If $\mathfrak{g}$ is solvable, then by Lie's theorem there exists a basis of $\mathfrak{g}$ so that $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}) \simeq \mathfrak{gl}_n$ lands in $\mathfrak{b}_n$. Therefore $\text{ad}([\mathfrak{g}, \mathfrak{g}])$ lands in $[\mathfrak{b}_n, \mathfrak{b}_n] = \mathfrak{n}_n$. It follows that $\text{ad}(x)$ is nilpotent for every $x \in [\mathfrak{g}, \mathfrak{g}]$, so $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent by Engel's theorem.

For the converse, assume that $D(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}]$ is nilpotent. Then a confusing but elementary calculation by induction shows $\mathfrak{g}^{(n+1)} \subset D(\mathfrak{g})^n$ for all $n \geq 1$. This implies $\mathfrak{g}$ is solvable. $\square$

4.5 Cartan's theorem

The following theorem is a handy criterion for when a subalgebra of $\mathfrak{gl}(V)$ is solvable. We will again not prove it, as it involves more boring linear algebra, see Humphreys §4.3.

Theorem 4.22 (Cartan's trace criterion). *Let $\mathfrak{g}$ be a subalgebra of $\mathfrak{gl}(V)$. Then $\mathfrak{g}$ is solvable if and only if $\text{tr}(xy) = 0$ for all $x \in \mathfrak{g}$ and $y \in [\mathfrak{g}, \mathfrak{g}]$.*

5 Semisimple Lie algebras

For the remainder of the course, we will work over $F = \mathbb{C}$.

5.1 Basic definitions

Definition 5.1. A Lie algebra $\mathfrak{g}$ is **semisimple** if $\mathfrak{g} = I_1 \oplus \cdots \oplus I_k$, where each I_j is an ideal that is simple as a Lie algebra.

Remark 5.2. A Lie algebra is semisimple if and only if the adjoint representation is faithful and completely reducible. (Exercise)

Every simple Lie algebra is semisimple. The zero Lie algebra is semisimple, corresponding to the empty direct sum of simple Lie algebras.

Example 5.3. The Lie algebra $\mathfrak{so}_4(\mathbb{C})$ turns out to be isomorphic to $\mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_2(\mathbb{C})$,

5.2 The Killing form

Let $\mathfrak{g}$ be a Lie algebra.

Definition 5.4. Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a representation. The **trace form** of V is the bilinear form $(-, -)_V$ on $\mathfrak{g} \times \mathfrak{g}$ defined by the formula

$$(x, y)_V = \text{tr}_V(\rho(x)\rho(y)). \quad (5.2.1)$$

Recall that the trace of an element $X \in \mathfrak{gl}_n$ is invariant under conjugation, so the trace of an endomorphism $\varphi \in \mathfrak{gl}(V)$ is well defined and independent of the choice of basis of V .

Lemma 5.5. *The form $(-, -)_V : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is symmetric, bilinear and satisfies*

$$([x, y], z)_V = (x, [y, z])_V$$

for all $x, y, z \in \mathfrak{g}$.

Proof. The fact that $(-, -)_V$ is bilinear follows from the linearity of ρ and the trace. The other properties follow from the fact that $\text{tr}(AB) = \text{tr}(BA)$. $\square$

Definition 5.6. *The **Killing form** of $\mathfrak{g}$ is the trace form of the adjoint representation. Explicitly, it is the symmetric bilinear form given by*

$$\kappa(x, y) = \text{tr}(\text{ad}_x \circ \text{ad}_y) \quad (5.2.2)$$

for all $x, y \in \mathfrak{g}$.

By Lemma 5.5, the Killing form satisfies

$$\kappa([x, y], z) = \kappa(x, [y, z]) \quad (5.2.3)$$

for all $x, y, z \in \mathfrak{g}$.

Example 5.7. *We can compute the Killing form of $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ using the calculations from Example 2.16. We have*

$$\text{ad}_e = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{ad}_h = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad \text{ad}_f = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

and the Gram matrix of κ with respect to the basis $\{e, h, f\}$ is

$$\begin{pmatrix} 0 & 0 & 4 \\ 0 & 8 & 0 \\ 4 & 0 & 0 \end{pmatrix}. \quad (5.2.4)$$

Notice that this form is nondegenerate.

5.3 Cartan–Killing criterion for semisimplicity

Lemma 5.8. *1. If I is an ideal of $\mathfrak{g}$ such that I and $\mathfrak{g}/I$ are solvable, then $\mathfrak{g}$ is solvable.*

2. If I, J are two solvable ideals of $\mathfrak{g}$, then so is $I + J$.

Proof. 1. There is an $n \geq 1$ such that $(\mathfrak{g}/I)^{(n)} = 0$, so $\mathfrak{g}^{(n)} \subset I$. There is an $m \geq 1$ such that $I^{(m)} = 0$. Therefore $\mathfrak{g}^{(n+m)} = (\mathfrak{g}^{(n)})^{(m)} \subset I^{(m)} = 0$.

2. Note that $I + J$ is an ideal of $\mathfrak{g}$, J is an ideal of $I + J$, $I \cap J$ is an ideal of I , and there is an isomorphism of Lie algebras $(I + J)/J \simeq I/(I \cap J)$. Since I is solvable, so is the quotient $I/(I \cap J)$. Therefore J and $(I + J)/J$ are solvable. This implies (by Part 1) that $I + J$ is solvable. $\square$

By Lemma 5.8, there is a unique maximal solvable ideal.

Definition 5.9. The *radical* of $\mathfrak{g}$, denoted $\text{Rad}(\mathfrak{g})$, is the unique maximal solvable ideal of $\mathfrak{g}$.

Theorem 5.10 (Cartan–Killing). *The following are equivalent for a Lie algebra $\mathfrak{g}$ over $\mathbb{C}$:*

1. $\mathfrak{g}$ is semisimple;
2. $\text{Rad}(\mathfrak{g}) = 0$, i.e., $\mathfrak{g}$ has no solvable ideals;
3. The Killing form $\kappa: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is nondegenerate.

We first need another lemma.

Lemma 5.11. *If $I \subset \mathfrak{g}$ is an ideal with Killing form $\kappa_I: I \times I \rightarrow \mathbb{C}$, then $\kappa_I(x, y) = \kappa(x, y)$ for all $x, y \in I$.*

Proof. Choose a basis of I and extend it to a basis of $\mathfrak{g}$. Consider the adjoint representations $\text{ad}_{\mathfrak{g}}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ and $\text{ad}_I: I \rightarrow \mathfrak{gl}(I)$. If $x \in I$, then $\text{ad}_{\mathfrak{g}}(x)$ looks like the block matrix

$$\begin{pmatrix} \text{ad}_I(x) & * \\ 0 & 0 \end{pmatrix} \quad (5.3.1)$$

Therefore $\text{tr}(\text{ad}_{\mathfrak{g}}(x) \text{ad}_{\mathfrak{g}}(y)) = \text{tr}(\text{ad}_I(x) \text{ad}_I(y))$ for all $x, y \in I$. $\square$

Proof of Theorem 5.10. (1) $\Rightarrow$ (2): Write $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k$, where $\mathfrak{g}_i$ are simple ideals of $\mathfrak{g}$. Let I_i be the projection of $\text{Rad}(\mathfrak{g})$ onto $\mathfrak{g}_i$. Then I_i is a solvable ideal of $\mathfrak{g}_i$ by Lemma 4.13. Since $\mathfrak{g}_i$ is simple, $[\mathfrak{g}_i, \mathfrak{g}_i] = \mathfrak{g}_i$ and so $\mathfrak{g}_i$ is not solvable. We conclude that $I_i = 0$ for all i . This implies that $\text{Rad}(\mathfrak{g}) = 0$.

(2) $\Rightarrow$ (3): let

$$\mathfrak{g}^{\perp} = \{x \in \mathfrak{g} : \kappa(x, y) = 0 \text{ for all } y \in \mathfrak{g}\}. \quad (5.3.2)$$

The identity $\kappa([x, y], z) = \kappa(x, [y, z])$ of (5.2.3) implies that $\mathfrak{g}^{\perp}$ is an ideal of $\mathfrak{g}$. We now claim that $\mathfrak{g}^{\perp}$ is also solvable. Indeed, consider the adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. It follows from Cartan's trace criterion (Theorem 4.22) that $\text{ad}(\mathfrak{g}^{\perp})$ is solvable. Moreover the kernel of ad equals the centre $Z(\mathfrak{g})$, which is also solvable. It follows from Lemma 5.8(1) that $\mathfrak{g}^{\perp}$ is also solvable. We conclude that $\mathfrak{g}^{\perp} \subset \text{Rad}(\mathfrak{g}) = 0$, so κ is nondegenerate.

(3) $\Rightarrow$ (1): We need to show that $\mathfrak{g}$ is semisimple. Let us first show that $\mathfrak{g}$ has no nonzero abelian ideals. Indeed, if $I \subset \mathfrak{g}$ is such an ideal, then we have (similarly to (5.3.1)) that

$$\text{ad}(x) = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}, \quad \text{ad}(y) = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \quad (5.3.3)$$

if $x \in I$ and $y \in \mathfrak{g}$. Therefore $\text{tr}(\text{ad}(x) \text{ad}(y)) = 0$ and hence $I \subset \mathfrak{g}^{\perp} = 0$.

Therefore, to show that $\mathfrak{g}$ is semisimple, it suffices to show (by the above paragraph and Lemma 3.9) that for every ideal $I \subset \mathfrak{g}$, there is an ideal $J \subset \mathfrak{g}$ such that $\mathfrak{g} = I \oplus J$. Let $I \subset \mathfrak{g}$ be an ideal and let

$$J = \{x \in \mathfrak{g} : \kappa(x, y) = 0 \text{ for all } y \in I\}$$

be the orthogonal complement of I in $\mathfrak{g}$. We claim that $\mathfrak{g} = I \oplus J$. We first show that $K = I \cap J = 0$. Note that K is an ideal of $\mathfrak{g}$. If $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is the adjoint representation, then $\text{ad}(K)$ is solvable by Cartan's trace criterion. Since $K \cap Z(\mathfrak{g}) = \ker(\text{ad}) \cap K$ is solvable too, K is solvable by Lemma 5.8. If $K \neq 0$, there is an $n \geq 0$ such that $K^{(n)} \neq 0$ but $K^{(n+1)} = 0$. Therefore $K^{(n)}$ is a nonzero abelian ideal, contradicting the fact that no such ideal exist. Therefore $K = I \cap J = 0$. Since κ is nondegenerate, $\dim I + \dim J = \dim \mathfrak{g}$. (This is a general fact: if $(-, -): V \times V \rightarrow \mathbb{C}$ is a bilinear form and $W \subset V$ a subspace, then $\dim W + \dim W^{\perp} = \dim V$.) By dimension reasons, $\mathfrak{g} = I \oplus J$. $\square$

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Lemma 5.12. *Every ideal and quotient of a semisimple Lie algebra is semisimple.*

Proof. Let $\mathfrak{g}$ be a semisimple Lie algebra and write $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k$, where each $\mathfrak{g}_i$ is a simple ideal. Let $I \subset \mathfrak{g}$ be an ideal. Since the adjoint representation of $\mathfrak{g}$ is completely reducible and $I \subset \mathfrak{g}$ is a subrepresentation, there exists a complementary subrepresentation to I , in other words an ideal $J \subset \mathfrak{g}$ such that $\mathfrak{g} = I \oplus J$. Since $[I, J] = 0$ (this being contained in $I \cap J = 0$), we have

$$[\mathfrak{g}, \mathfrak{g}] = [I, I] \oplus [J, J]$$

Since $\mathfrak{g}$ is semisimple, $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. Since $[I, I] \subset I$ and $[J, J] \subset J$, we must in fact have $I = [I, I]$ and $J = [J, J]$. Let S be the subset of $i \in \{1, \dots, k\}$ such that $I \cap \mathfrak{g}_i \neq 0$. We claim that $I = \bigoplus_{i \in S} \mathfrak{g}_i$. Indeed, if $i \in S$ then $I \cap \mathfrak{g}_i = \mathfrak{g}_i$ by the simplicity of $\mathfrak{g}_i$. Moreover if $i \notin S$ then $[I, \mathfrak{g}_i] \subset I \cap \mathfrak{g}_i = 0$. So

$$I = [I, I] = [I, \mathfrak{g}] = \bigoplus_{i=1}^k [I, \mathfrak{g}_i] = \bigoplus_{i \in S} [I, \mathfrak{g}_i].$$

If $i \in S$ is such that $[I, \mathfrak{g}_i] = 0$, then the above direct sum decomposition shows that $I \subset \bigoplus_{j \in S \setminus \{i\}} \mathfrak{g}_j$, which has trivial intersection with $\mathfrak{g}_i$, contradiction. Therefore $[I, \mathfrak{g}_i] = \mathfrak{g}_i$ for all $i \in S$ and $I = \bigoplus_{i \in S} \mathfrak{g}_i$ is semisimple. Similarly $J = \bigoplus_{i \notin S} \mathfrak{g}_i$ and so $\mathfrak{g}/I \simeq J$ is also semisimple. $\square$

Remark 5.13. *It is not true that every subalgebra of a semisimple Lie algebra is semisimple. For example, $\mathfrak{sl}_n(\mathbb{C})$ is semisimple, but $\mathfrak{n}_n(\mathbb{C}) \subset \mathfrak{sl}_n(\mathbb{C})$ is nilpotent, hence not semisimple.*

Remark 5.14. *You will prove on Example Sheet 2 that the Lie algebras $\mathfrak{sl}_n(\mathbb{C})$, $\mathfrak{sp}_{2\ell}(\mathbb{C})$, $\mathfrak{so}_n(\mathbb{C})$ are semisimple.*

5.4 Complete reducibility

Let $\mathfrak{g}$ be a semisimple Lie algebra (over $\mathbb{C}$).

Theorem 5.15 (Weyl). *Every representation of $\mathfrak{g}$ is completely reducible.*

This follows from adapting the proof of the case $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$. We give the ingredients here, and leave the details as an exercise on Example Sheet 2.

Lemma 5.16. *Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a faithful representation. Then the trace form $(-, -)_V: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is nondegenerate.*

Proof. Similarly to the proof of Theorem 5.10, it follows from Cartan's trace criterion that the kernel of the trace form is a solvable ideal, so this ideal must be zero, and the form must be nondegenerate. $\square$

Definition 5.17. *Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a faithful representation. Let $x_1, \dots, x_m$ be a basis of $\mathfrak{g}$ and let $y_1, \dots, y_m$ be the dual basis with respect to the trace form: $(x_i, y_j)_V = \delta_{ij}$ for all i, j . Then the linear operator*

$$\Omega_\rho = \sum_{i=1}^m \rho(x_i) \rho(y_i) \in \text{End}(V) \tag{5.4.1}$$

is called the [Casimir element](#) associated to ρ .

It turns out that Ω_ρ does not depend on the choice of basis of $\mathfrak{g}$. We will not need or prove this, but it justifies leaving this choice implicit in the notation.

Example 5.18. If $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ and we choose basis $\{e, h, f\}$ and we consider the trace form of the defining representation ρ , then $\{f, \frac{1}{2}h, e\}$ is the dual basis, and $\Omega_\rho = \rho(e)\rho(f) + \rho(f)\rho(e) + \rho(h)^2/2$ is the Casimir element from Definition 3.15.

Proposition 5.19. In the notation of the definition, $\Omega_\rho: V \rightarrow V$ is $\mathfrak{g}$ -equivariant. We have $\text{tr}(\Omega_\rho) = \dim \mathfrak{g}$.

Proof. The fact that Ω_ρ is $\mathfrak{g}$ -equivariant is a calculation: let $x \in \mathfrak{g}$ and write $[x, x_i] = \sum_j a_{ij}x_j$ and $[x, y_i] = \sum_j b_{ij}y_j$. The relation $([x, x_i], y_k)_V = -(x_i, [x, y_j])_V$ shows that $a_{ik} = -b_{ki}$. The relation $[A, BC] = [A, B]C + B[A, C]$ for $A, B, C \in \text{End}(V)$ shows that

$$[\rho(x), \Omega_\rho] = \sum_{i=1}^m [\rho(x), \rho(x_i)\rho(y_i)] = \sum_i ([\rho(x), \rho(x_i)]\rho(y_i) + \rho(x_i)[\rho(x), \rho(y_i)]).$$

Using the fact that $a_{ij} = -b_{ji}$, we see that this expression is zero! To calculate its trace, we have

$$\text{tr}(\Omega_\rho) = \sum_{i=1}^m \text{tr}(\rho(x_i)\rho(y_i)) = \sum_{i=1}^m (x_i, y_i)_V = m = \dim \mathfrak{g}.$$

□

5.5 Jordan decomposition

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Let V be a vector space. Recall that $x \in \mathfrak{gl}(V) = \text{End}(V)$ is **semisimple** if it is diagonalizable (that is, it has a basis of eigenvectors), and **nilpotent** if $x^n = 0$ for some $n \geq 1$. Note that if x is both semisimple and nilpotent, then x has a basis of eigenvectors and all its eigenvalues are zero, so $x = 0$.

Proposition 5.20 (Concrete Jordan decomposition). Let $x \in \mathfrak{gl}(V)$ be an element.

1. There exists unique elements $x_s, x_n \in \mathfrak{gl}(V)$ with the following properties: x_s is semisimple, x_n is nilpotent,

$$x = x_s + x_n,$$

$$\text{and } [x_s, x_n] = 0.$$

2. There exists polynomials $p_s, p_n \in \mathbb{C}[t]$ without constant term, such that $x_s = p_s(x)$ and $x_n = p_n(x)$.
3. If $y \in \text{End}(V)$ commutes with x , then it commutes with x_s and x_n . If $B \subset A \subset V$ are subspaces such that $x(A) \subset B$, then $x_s(A) \subset B$ and $x_n(A) \subset B$.

Proof. The proof is routine linear algebra, see Humphreys, Section 4.2. Note that (2) immediately implies (3). □

The decomposition $x = x_s + x_n$ is called the **(concrete) Jordan decomposition** of x .

Example 5.21. We illustrate Proposition 5.20 in the case of $V = \mathbb{C}^2$ and $x = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ a single Jordan block.

Then $x_s = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ and $x_n = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Moreover if $\lambda \neq 0$ then $p_s(t) = 2t - \lambda^{-1}t^2$ and $p_n(t) = \lambda^{-1}t^2 - t$. The general case of Proposition 5.20 can be reduced to the case of a single Jordan block, for which a similarly explicit analysis can be given.

Theorem 5.22. Let $\mathfrak{g}$ be a semisimple subalgebra of $\mathfrak{gl}(V)$. Let $x \in \mathfrak{g}$ and let $x = x_s + x_n$ be its Jordan decomposition in $\mathfrak{gl}(V)$. Then $x_s, x_n \in \mathfrak{g}$.

Proof. Omitted, see Humphreys, §6.4. □

Remark 5.23. This certainly fails for general Lie subalgebras of $\mathfrak{gl}(V)$, since any non-semisimple and non-nilpotent element x defines a one-dimensional subalgebra $\text{span}\{x\} \subset \mathfrak{gl}(V)$ not containing x_s and x_n .

Since the adjoint representation of a semisimple Lie algebra is always faithful, $\mathfrak{g}$ can be seen as a subalgebra of $\mathfrak{gl}(\mathfrak{g})$, which allows us to define an ‘abstract’ Jordan decomposition.

Definition 5.24. If $\mathfrak{g}$ is semisimple and $x \in \mathfrak{g}$, let $x_s, x_n \in \mathfrak{g}$ be the unique elements such that $\text{ad}(x) = \text{ad}(x_s) + \text{ad}(x_n)$ is the Jordan decomposition of $\text{ad}(x)$ in $\mathfrak{gl}(\mathfrak{g})$. (Such elements exist by Theorem 5.22.) Then $x = x_s + x_n$ is called the *(abstract) Jordan decomposition* of x .

If $\mathfrak{g}$ is a semisimple subalgebra of $\mathfrak{gl}(V)$ and $x \in \mathfrak{g}$ there are now (a priori) two different ‘Jordan decompositions’: the concrete one and abstract one. We show that these agree using the following lemma:

Lemma 5.25. If $x \in \mathfrak{gl}(V)$ has Jordan decomposition $x = x_s + x_n$, then $\text{ad}(x) \in \mathfrak{gl}(\mathfrak{gl}(V))$ has Jordan decomposition $\text{ad}(x) = \text{ad}(x_s) + \text{ad}(x_n)$.

Proof. Since x_s is semisimple, there is a basis of V in which x_s is diagonal. A calculation then shows that (in this basis) elementary matrices of $\mathfrak{gl}(V)$ form a basis of eigenvectors for $\text{ad}(x_s)$, so $\text{ad}(x_s)$ is semisimple. We have shown in Lemma 4.17 of Lecture 7 that $\text{ad}(x_n)$ is nilpotent. Moreover $[\text{ad}(x_s), \text{ad}(x_n)] = \text{ad}([x_s, x_n]) = 0$. Therefore $\text{ad}(x) = \text{ad}(x_s) + \text{ad}(x_n)$ satisfies all the properties of the Jordan decomposition, hence must equal to it by uniqueness. □

Corollary 5.26. If $\mathfrak{g}$ is a semisimple subalgebra of $\mathfrak{gl}(V)$ and $x \in \mathfrak{g}$, then the concrete and abstract Jordan decompositions of x coincide.

We will therefore simply speak of ‘the’ Jordan decomposition of an element $x \in \mathfrak{g}$, since there can be no ambiguity.

Proof. If $x \in \mathfrak{g}$ has concrete Jordan decomposition $x = x_s + x_n$, then $x_s, x_n \in \mathfrak{g}$ by Theorem 5.22. Moreover $\text{ad}(x_s), \text{ad}(x_n) \in \mathfrak{gl}(\mathfrak{g}(V))$ are semisimple/nilpotent endomorphisms that commute. Hence so are their restrictions $\text{ad}_{\mathfrak{g}}(x_s), \text{ad}_{\mathfrak{g}}(x_n)$. So $\text{ad}_{\mathfrak{g}}(x) \in \mathfrak{gl}(\mathfrak{g})$ has Jordan decomposition $\text{ad}_{\mathfrak{g}}(x) = \text{ad}_{\mathfrak{g}}(x_s) + \text{ad}_{\mathfrak{g}}(x_n)$. Therefore $x = x_s + x_n$ is also the abstract Jordan decomposition of x . □

A very similar argument shows that the Jordan decomposition is compatible with any representation of $\mathfrak{g}$:

Proposition 5.27. Let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra and $x \in \mathfrak{g}$ with Jordan decomposition $x = x_s + x_n$. Then $\rho(x)_s = \rho(x_s)$ and $\rho(x)_n = \rho(x_n)$.

Proof. (Omitted in class, so non-examinable.) Let $\mathfrak{h} \subset \mathfrak{gl}(V)$ be the image of ρ . Since $\mathfrak{h}$ is quotient of the semisimple $\mathfrak{g}$, $\mathfrak{h}$ is semisimple too. Since $\text{ad}_{\mathfrak{g}}(x_s)$ is semisimple, $\mathfrak{g}$ is spanned by eigenvectors $y_1, \dots, y_n$ of $\text{ad}_{\mathfrak{g}}(x_s)$. Since ρ is a Lie algebra homomorphism, $\rho(y_i)$ are eigenvectors for $\text{ad}_{\mathfrak{h}}(x_s)$, so the eigenvectors of $\text{ad}_{\mathfrak{h}}(\rho(x_s))$ span $\mathfrak{h}$. In other words, $\text{ad}_{\mathfrak{h}}(\rho(x_s)) \in \mathfrak{gl}(\mathfrak{h})$ is semisimple. Similarly, since $\rho \circ \text{ad}_{\mathfrak{g}}(x_n)^k = \text{ad}_{\mathfrak{h}}(\rho(x_n))^k \circ \rho$ and ρ surjects onto $\mathfrak{h}$, $\text{ad}_{\mathfrak{h}}(\rho(x_n))$ is nilpotent. Since $[\rho(x_s), \rho(x_n)] = 0$ and $\rho(x) = \rho(x_s) + \rho(x_n)$, $\text{ad}_{\mathfrak{h}}(\rho(x)) = \text{ad}_{\mathfrak{h}}(\rho(x_s)) + \text{ad}_{\mathfrak{h}}(\rho(x_n))$ is the Jordan decomposition of $\text{ad}_{\mathfrak{h}}(\rho(x))$. So $\rho(x) = \rho(x_s) + \rho(x_n)$ is the abstract Jordan decomposition of $\rho(x) \in \mathfrak{h}$. By Corollary 5.26, this is also the concrete Jordan decomposition of $\rho(x) \in \mathfrak{gl}(V)$. □

If $\mathfrak{g}$ is a semisimple Lie algebra, we say $x \in \mathfrak{g}$ is **semisimple** if $x = x_s$ and $x \in \mathfrak{g}$ is **nilpotent** if $x = x_n$.

6 Root space decomposition of a semisimple Lie algebra

Having all our preliminaries at hand, we can now properly start to analyze semisimple Lie algebras in more detail. At the heart of this will be a certain combinatorial structure associated to such a Lie algebra, namely the root space decomposition.

From now on, let $\mathfrak{g}$ be a semisimple Lie algebra.

6.1 Cartan subalgebras

Definition 6.1. A subalgebra $\mathfrak{t} \subset \mathfrak{g}$ is called **toral** if it is abelian every element of $\mathfrak{t}$ is semisimple (seen as an element of $\mathfrak{g}$). If $\mathfrak{t}$ is toral and not contained in a strictly bigger toral subalgebra, then $\mathfrak{t}$ is called a maximal toral subalgebra or a **Cartan subalgebra**, or CSA for short.

Example 6.2. Diagonal matrices (of trace zero) in $\mathfrak{sl}_n$ are a CSA. (You will prove this on sheet 2.) Diagonal matrices in $\mathfrak{so}_n(\mathbb{C})$ and $\mathfrak{sp}_{2\ell}(\mathbb{C})$ are also CSA's. This is the biggest reason why we chose the symmetric/bilinear forms as we did.

Remark 6.3. We will later see that CSA's are essentially unique: if $\mathfrak{t}, \mathfrak{t}'$ are two CSA's, there exists a Lie algebra automorphism $\varphi: \mathfrak{g} \rightarrow \mathfrak{g}$ such that $\varphi(\mathfrak{t}) = \mathfrak{t}'$.

Since the zero subalgebra is toral, it is clear that CSA's exist. In fact, it is also easy to see that a CSA is nonzero when $\mathfrak{g}$ is: if not, then every element x of $\mathfrak{g}$ would have the property that $\text{ad}(x)$ is nilpotent. So then $\mathfrak{g}$ would be nilpotent by Engel's theorem, contradicting semisimplicity.

6.2 Root space decomposition

Let $\mathfrak{g}$ be a semisimple Lie algebra and $\mathfrak{t} \subset \mathfrak{g}$ a choice of Cartan subalgebra. The key tool in everything that follows is to analyze the adjoint action of $\mathfrak{t}$ on $\mathfrak{g}$. To this end, we recall the following lemma:

Lemma 6.4. Let $\mathfrak{h}$ be an abelian Lie algebra and $\rho: \mathfrak{h} \rightarrow \mathfrak{gl}(V)$ a representation such that $\rho(x)$ is semisimple for all $x \in \mathfrak{h}$. Then there exist linear functionals $\lambda_1, \dots, \lambda_k: \mathfrak{h} \rightarrow \mathbb{C}$ such that

$$V = \bigoplus_{i=1}^k V_{\lambda_i}, \quad (6.2.1)$$

where $V_{\lambda} = \{v \in V: x \cdot v = \lambda(x)v \text{ for all } x \in \mathfrak{h}\}$.

Proof. Pick a basis $t_1, \dots, t_r$ of $\mathfrak{h}$. Then $\rho(t_1), \dots, \rho(t_r)$ are semisimple commuting endomorphisms of V . Moreover a linear map $\lambda: \mathfrak{h} \rightarrow \mathbb{C}$ is the same as an r -tuple of complex numbers $\lambda(t_1), \dots, \lambda(t_r)$. Therefore the statement of the lemma is equivalent to saying that the endomorphisms are simultaneously $\rho(t_i)$ diagonalizable. (We phrased the lemma in a basis independent way because we do not always want to pick bases.) This is easily

seen by induction on r . If $r = 1$, then $\rho(t_1)$ is diagonalizable by definition. If $r \geq 2$, let $\mathfrak{h}' = \text{span}\{t_1, \dots, t_{r-1}\}$ and use the induction hypothesis to find linear functionals $\lambda'_1, \dots, \lambda'_m: \mathfrak{h}' \rightarrow \mathbb{C}$ such that

$$V = \bigoplus_{i=1}^m V_{\lambda'_i}.$$

Since $\rho(t_r)$ commutes with $\rho(t_i)$ for all i , we see that $\rho(t_r)$ preserves $V_{\lambda'_i}$ for all i . Consequently, we may decompose $V_{\lambda'_i}$ into a direct sum of eigenspaces for the action of $\rho(t_r)$, and extend λ'_i to some linear functionals on $\mathfrak{h}$ so that (6.2.1) holds. $\square$

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The restriction of $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ to $\mathfrak{t}$ is a representation $\mathfrak{t} \rightarrow \mathfrak{gl}(\mathfrak{g})$ that satisfies the assumptions of Lemma 6.4, by definition of a Cartan subalgebra. Therefore, there exist linear functionals $S \subset \mathfrak{t}^*$ such that

$$\mathfrak{g} = \bigoplus_{\lambda \in S} \mathfrak{g}_\lambda \quad (6.2.2)$$

where $\mathfrak{g}_\lambda = \{x \in \mathfrak{g} : [t, x] = \lambda(t)x \text{ for all } t \in \mathfrak{t}\}$. The zero functional $0 \in \mathfrak{t}^*$ is contained in S , since $\mathfrak{t} \subset \mathfrak{g}_0$. Traditionally, we write $\Phi = S \setminus \{0\}$ and so we can write

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha. \quad (6.2.3)$$

This is called the **root space decomposition**. It is absolutely essential to everything that follows. Elements in Φ are called **roots**.

Example 6.5. Suppose $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$, with its standard basis $\{e, h, f\}$. Then $\mathfrak{t} = \text{span}\{h\}$ is a Cartan subalgebra, so $\mathfrak{t}^*$ has dual basis h^* , the unique linear functional on $\mathfrak{t}$ satisfying $h^*(h) = 1$. Then $\text{ad}(h)(e) = 2e$ and $\text{ad}(h)(f) = -2f$, so $\Phi = \{\pm\alpha\}$, where $\alpha = 2h^*$, and $\mathfrak{g}_\alpha = \text{span}\{e\}$ and $\mathfrak{g}_{-\alpha} = \text{span}\{f\}$. Remember this example!

Example 6.6. Let $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ and $\mathfrak{t}$ the subset of diagonal matrices in $\mathfrak{g}$. On example sheet 2, you will show that $\mathfrak{t}$ is a CSA of $\mathfrak{g}$. For $i = 1, 2, 3$ let $e_i: \mathfrak{t} \rightarrow \mathbb{C}$ be the functional sending $\text{diag}(t_1, t_2, t_3)$ to t_i . Then $\mathfrak{t}^*$ is spanned by e_1, e_2, e_3 and they satisfy the relation $e_1 + e_2 + e_3 = 0$. We may calculate that $\text{ad}(\text{diag}(t_1, t_2, t_3))(E_{12}) = (t_1 - t_2)E_{12}$, and similarly $\text{ad}(\text{diag}(t_1, t_2, t_3))(E_{ij}) = (t_i - t_j)E_{ij}$. It follows that $\Phi = \{\pm(e_i - e_j) : 1 \leq i < j \leq 3\}$ and E_{ij} spans $\mathfrak{g}_{e_i - e_j}$.

6.3 First properties of roots

Lemma 6.7. If $\alpha, \beta \in \mathfrak{t}^*$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$.

Fulton and Harris call this the ‘fundamental calculation’.

Proof. This follows from the Jacobi identity! Indeed, if $t \in \mathfrak{t}$, $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_\beta$, then

$$\text{ad}(t)([x, y]) = [t, [x, y]] = -[y, [t, x]] - [x, [y, t]] = \alpha(t)[x, y] + \beta(t)[x, y] = (\alpha + \beta)(t)[x, y].$$

$\square$

Lemma 6.8. If $x \in \mathfrak{g}_\alpha$ and $\alpha \neq 0$, then $\text{ad}(x)$ is nilpotent.

Proof. (I didn't cover this lemma in class, so repeat when we use it.) By the previous lemma, $\text{ad}(x)$ sends $\mathfrak{g}_\beta$ to $\mathfrak{g}_{\beta+\alpha}$. Since $\alpha \neq 0$ and the set of roots is finite, there is an $n \geq 1$ such that $\mathfrak{g}_{\beta+n\alpha} = 0$ for all $\beta \in \mathfrak{t}^*$ such that $\mathfrak{g}_\beta \neq 0$. $\square$

Lemma 6.9. *If $\alpha, \beta \in \mathfrak{t}^*$ and $\alpha + \beta \neq 0$, then $\mathfrak{g}_\alpha \perp \mathfrak{g}_\beta$ under the Killing form.*

Proof. Let $t \in \mathfrak{t}$ be an element such that $(\alpha + \beta)(t) \neq 0$. Let $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_\beta$. By (5.2.3), $\kappa([t, x], y) = -\kappa(x, [t, y])$. Therefore $(\alpha + \beta)(t)\kappa(x, y) = 0$, hence $\kappa(x, y) = 0$. $\square$

6.4 The zero weight space

Note that in the root space decomposition of $\mathfrak{g}$, the zero weight space

$$\mathfrak{g}_0 = \{x \in \mathfrak{g} : [t, x] = 0 \text{ for all } t \in \mathfrak{t}\}$$

equals the centralizer of $\mathfrak{t}$ in $\mathfrak{g}$. (You can take this as the definition of the centralizer.) Since $\mathfrak{t}$ is abelian, $\mathfrak{t} \subset \mathfrak{g}_0$. In fact, equality holds:

Proposition 6.10. *In the notation of (6.2.3), $\mathfrak{g}_0 = \mathfrak{t}$.*

Proof. Omitted, see Humphreys, §8.2. $\square$

Henceforth we write the root space decomposition (6.2.3) from now on as

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha. \quad (6.4.1)$$

Proposition 6.11. *The restriction of the Killing form $\kappa: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ to $\mathfrak{t}$ is nondegenerate.*

Proof. By Lemma 6.9, $\mathfrak{g}_0 \perp \mathfrak{g}_\alpha$ for every $\alpha \neq 0$. Therefore if $t \in \mathfrak{g}_0$ is perpendicular to every element of $\mathfrak{g}_0$, it is perpendicular to every element of $\mathfrak{g}$. Since the Killing form is nondegenerate, this implies $x = 0$. So the restriction of κ to $\mathfrak{g}_0$ is nondegenerate. Since $\mathfrak{g}_0 = \mathfrak{t}$ by Proposition 6.10, we're done. $\square$

Since $\kappa|_{\mathfrak{t}}: \mathfrak{t} \times \mathfrak{t} \rightarrow \mathbb{C}$ is nondegenerate, the map

$$\begin{aligned} \mathfrak{t} &\rightarrow \mathfrak{t}^* \\ t &\mapsto \kappa(t, \cdot) \end{aligned}$$

is an isomorphism. Consequently, the following definition makes sense:

Definition 6.12. *If $\lambda \in \mathfrak{t}^*$, let $t_\lambda \in \mathfrak{t}$ be the unique element such that $\kappa(t_\lambda, x) = \lambda(x)$ for all $x \in \mathfrak{t}$.*

6.5 Finding $\mathfrak{sl}_2$'s inside $\mathfrak{g}$

We show that every root gives rise to a copy of $\mathfrak{sl}_2(\mathbb{C})$ inside $\mathfrak{g}$.

Proposition 6.13. 1. Φ spans $\mathfrak{t}^*$.

2. If $\alpha \in \Phi$, then $-\alpha \in \Phi$.

Proof. 1. If not, there exists a nonzero $t \in \mathfrak{t}$ such that $\alpha(t) = 0$ for all $\alpha \in \Phi$. Therefore if $x \in \mathfrak{g}_\alpha$ then $[t, x] = \alpha(t)x = 0$. So $[t, \mathfrak{g}_\alpha] = 0$ for all $\alpha \in \Phi$. Also $[t, \mathfrak{t}] = 0$ since $\mathfrak{t}$ is abelian. Therefore t lies in the centre $Z(\mathfrak{g})$ of $\mathfrak{g}$. But the centre of a semisimple Lie algebra is zero.

2. Recall that $\mathfrak{g}_\alpha$ is orthogonal to $\mathfrak{g}_\beta$ under the Killing form whenever $\alpha + \beta \neq 0$. Since the Killing form is nondegenerate, $\mathfrak{g}_\alpha$ cannot be orthogonal to the whole of $\mathfrak{g}$, so $-\alpha \in \Phi$.

□

Proposition 6.14. 1. If $\alpha \in \Phi$, $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$, then $[x, y] = t_\alpha \kappa(x, y)$.

2. If $\alpha \in \Phi$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ is one-dimensional.

3. $\alpha(t_\alpha) \neq 0$.

Proof. 1. If $t \in \mathfrak{t}$, then we calculate that

$$\kappa(t, [x, y]) = \kappa([t, x], y) = -\kappa([t, x], y) = \kappa(\alpha(t)x, y) = \alpha(t)\kappa(x, y) = \kappa(t, t_\alpha \kappa(x, y))$$

therefore $[x, y] = \kappa(x, y)t_\alpha$.

2. By the previous point, it suffices to show that there exist $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ such that $\kappa(x, y) \neq 0$. If not, this would contradict the nondegeneracy of the Killing form, since $\mathfrak{g}_\alpha \perp \mathfrak{g}_\beta$ whenever $\alpha + \beta \neq 0$.

3. Let $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ such that $\kappa(x, y) = 1$. Then $\{x, t_\alpha, y\}$ satisfy the relations

$$[t_\alpha, x] = \alpha(t_\alpha)x, \quad [t_\alpha, y] = -\alpha(t_\alpha)y, \quad [x, y] = t_\alpha.$$

Denote this subalgebra by $\mathfrak{h}$. Suppose $\alpha(t_\alpha) = 0$. Then $\mathfrak{h}$ is a solvable Lie algebra since $\mathfrak{h}^{(2)} = 0$. By Lie's theorem, there exists a basis of $\mathfrak{g}$ such that $\text{ad}: \mathfrak{h} \rightarrow \mathfrak{gl}(\mathfrak{g})$ lands in upper triangular matrices. Therefore $t_\alpha \in [\mathfrak{h}, \mathfrak{h}]$ lands in strictly upper triangular matrices. So $\text{ad}(t_\alpha) \in \mathfrak{gl}(\mathfrak{g})$ is nilpotent. But it is also semisimple, so $\text{ad}(t_\alpha) = 0$, to $t_\alpha = 0$ since ad is injective for a semisimple Lie algebra. This is a contradiction.

□

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We define for every $\alpha \in \Phi$ the element

$$h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}.$$

Note that $\kappa(t_\alpha, t_\alpha) = \alpha(t_\alpha)$ is nonzero by Proposition 6.14.

Corollary 6.15. Let $\alpha \in \Phi$ and $e_\alpha \in \mathfrak{g}_\alpha$ be nonzero. Then there exists an element $f_\alpha \in \mathfrak{g}_{-\alpha}$ such that $(e_\alpha, h_\alpha, f_\alpha)$ satisfy the standard $\mathfrak{sl}_2(\mathbb{C})$ -relations:

$$[h_\alpha, e_\alpha] = 2e_\alpha, \quad [h_\alpha, f_\alpha] = -2f_\alpha, \quad [e_\alpha, f_\alpha] = h_\alpha.$$

Proof. By the nondegeneracy of the Killing form and the fact that $\mathfrak{g}_\alpha \perp \mathfrak{g}_\beta$ whenever $\alpha + \beta \neq 0$, there exists an $f_\alpha \in \mathfrak{g}_{-\alpha}$ such that $\kappa(e_\alpha, f_\alpha) = 2/\kappa(t_\alpha, t_\alpha)$. Proposition 6.14 then implies that $[e_\alpha, f_\alpha] = h_\alpha$. Moreover $[h_\alpha, e_\alpha] = \alpha(h_\alpha)e_\alpha = 2e_\alpha$ since $\alpha(h_\alpha) = 2\alpha(t_\alpha)/\kappa(t_\alpha, t_\alpha) = 2\kappa(t_\alpha, t_\alpha)/\kappa(t_\alpha, t_\alpha) = 2$. Similarly $[h_\alpha, f_\alpha] = -2f_\alpha$. □

Definition 6.16. For $\alpha \in \Phi$, let $\mathfrak{m}_\alpha \simeq \mathfrak{sl}_2(\mathbb{C})$ be the subalgebra spanned by $\{e_\alpha, h_\alpha, f_\alpha\}$, where the latter is any triple satisfying the conclusions of Corollary 6.15.

We will soon see that $\mathfrak{m}_\alpha$ is independent of any choices made, which justifies the omission of these choices from the notation.

The restriction of the adjoint representation of $\mathfrak{g}$ to $\mathfrak{m}_\alpha$ defines a representation $\mathfrak{m}_\alpha \rightarrow \mathfrak{gl}(\mathfrak{g})$, in other words, a representation of $\mathfrak{sl}_2$! Next we examine the consequences that the representation theory of $\mathfrak{sl}_2$ has on the root space decomposition.

6.6 Root strings

Proposition 6.17. *If $\alpha \in \Phi$, then $\mathfrak{g}_\alpha$ is one-dimensional. Moreover, if $c \in \mathbb{C}$, then $c\alpha \in \Phi$ if and only if $c = \pm 1$. (In other words, no multiple of a root is a root, unless that multiple is ± 1 .)*

Proof. Let $(e_\alpha, h_\alpha, f_\alpha)$ be a triple satisfying the conclusions of Corollary 6.15, spanning the subalgebra $\mathfrak{m}_\alpha$. Let $V = \mathfrak{t} \oplus \bigoplus_{c \in \mathbb{C}} \mathfrak{g}_{c\alpha}$. By Lemma 6.7, V is stable under the adjoint action of $\mathfrak{m}_\alpha$, so defines a representation of $\mathfrak{m}_\alpha$. What is this representation? Let S be the set of $c \in \mathbb{C}$ such that $c\alpha \in \Phi$. Then the weights of V are the eigenvalues of h_α on V , in other words 0 (coming from $\mathfrak{t}$) and $c\alpha(h_\alpha) = 2c$ for $c \in S$. Since the weights of an $\mathfrak{sl}_2(\mathbb{C})$ -representation are known to be integers, $S \subset \frac{1}{2}\mathbb{Z}$.

Consider the subspace $U = \mathfrak{t} + \mathfrak{m}_\alpha \subset V$. Since $[\mathfrak{t}, \mathfrak{m}_\alpha] \subset \mathfrak{m}_\alpha$ and $\mathfrak{m}_\alpha$ is a subalgebra, U is a $\mathfrak{m}_\alpha$ -subrepresentation of V . Consequently, by complete reducibility there exists a subrepresentation $W \subset V$ such that $V = U \oplus W$.

What are the weights of W ? Since 0 is not a weight of W (since the 0-weight space of V is $\mathfrak{t} \subset U$), W has no even weights. In particular, $2\alpha \notin \Phi$, since otherwise $2\alpha(h_\alpha) = 4$ would be a weight of W . We have shown at this point that $2\alpha \notin \Phi$ for every $\alpha \in \Phi$. But this implies that $\frac{1}{2}\alpha \notin \Phi$ for every $\alpha \in \Phi$. Indeed, if $\beta = \frac{1}{2}\alpha$ were a root, then $2\beta = \alpha$ cannot be root, contradiction. So $\frac{1}{2}\alpha \notin \Phi$, hence 1 is not a weight of W .

The previous paragraph shows that both 0 and 1 are not weights of W . Using the explicit description of representations of $\mathfrak{sl}_2(\mathbb{C})$, this implies that W is trivial! So $V = \mathfrak{t} + \mathfrak{m}_\alpha$, which implies the claims of the proposition. $\square$

Corollary 6.18. *If $\alpha \in \Phi$, then*

$$\mathfrak{m}_\alpha = \mathfrak{g}_\alpha \oplus [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \oplus \mathfrak{g}_{-\alpha}.$$

Note that if $(e_\alpha, h_\alpha, f_\alpha)$ is a triple satisfying the conclusions of Corollary 6.15, then every other such triple is of the form $(ce_\alpha, h_\alpha, c^{-1}f_\alpha)$ for some $c \in \mathbb{C}^\times$. So a choice of e_α uniquely determines f_α .

We now examine root strings. Let $\beta \in \Phi$ and consider

$$V = \bigoplus_{\lambda \in S} \mathfrak{g}_\lambda = \bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_{\beta + k\alpha}. \quad (6.6.1)$$

This is stable under the adjoint action of $\mathfrak{m}_\alpha$ by Lemma 6.7, hence defines a representation of $\mathfrak{m}_\alpha$. Then S is called the **α -root string through β** . If $\beta = \pm\alpha$, then we have analyzed this representation in Proposition 6.17, so assume $\beta \neq \pm\alpha$. We use this representation to show:

Proposition 6.19. *Let $\alpha, \beta \in \Phi$ be such that $\beta \neq \pm\alpha$.*

1. *Let r be the largest integer such that $\beta - r\alpha \in \Phi$, and let q be the largest integer such that $\beta + q\alpha \in \Phi$. Then*

$$\{\beta + k\alpha : k \in \mathbb{Z}\} \cap \Phi = \{\beta + k\alpha : -r \leq k \leq q\}$$

$$\text{and } \beta(h_\alpha) = r - q.$$

2. $\beta - \beta(h_\alpha)\alpha \in \Phi$.

3. If $\alpha + \beta \in \Phi$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$. (Note we have previously shown that the left hand side is a subspace of the right hand side. We are now asserting equality when $\alpha + \beta \in \Phi$.)

Proof. Consider the set $S = \{k \in \mathbb{Z}: \beta + k\alpha \in \Phi\}$ and the root string representation V of $\mathfrak{m}_\alpha$ described in (6.6.1). Then $\text{ad}(h_\alpha)$ acts on $\mathfrak{g}_{\beta+k\alpha}$ via the scalar $\beta(h_\alpha) + 2k$. Since each $\mathfrak{g}_{\beta+k\alpha}$ is one-dimensional, all the weights of V have multiplicity 1 and the same parity. By $\mathfrak{sl}_2(\mathbb{C})$ -theory, V is irreducible! So the weights are of the form $\{n, n-2, \dots, -n\}$ for some integer $n \geq 0$. This implies all three claims of the proposition.

In more detail, we must have $n = (\beta + q\alpha)(h_\alpha)$ and $-n = (\beta - r\alpha)(h_\alpha)$. Equating these gives $\beta(h_\alpha) = r - q$, proving part 1. Since $-r \leq -\beta(h_\alpha) \leq q$, $\beta - \beta(h_\alpha)\alpha \in \Phi$, proving part 2. Part 3 follows from the fact that if V is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ and λ and $\lambda + 2$ are weights of V , then $\text{ad}(e_\alpha): V_\lambda \rightarrow V_{\lambda+2}$ is surjective. $\square$

6.7 Φ is a root system

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starts here

Recall from §6.4 that the Killing form restricted to $\mathfrak{t}$ defines an isomorphism $\mathfrak{t} \rightarrow \mathfrak{t}^*$ whose inverse we denote by $\lambda \mapsto t_\lambda$. We can use this isomorphism to transport the Killing form to $\mathfrak{t}^*$: if $\lambda, \mu \in \mathfrak{t}^*$, define

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu).$$

Then $(-, -): \mathfrak{t}^* \times \mathfrak{t}^* \rightarrow \mathbb{C}$ is a nondegenerate bilinear form. Note that $(\lambda, \mu) = \lambda(t_\mu) = \mu(t_\lambda)$.

Lemma 6.20. If $\alpha, \beta \in \Phi$, then $\frac{2(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}$.

Proof. We have $\frac{2(\beta, \alpha)}{(\alpha, \alpha)} = \beta(h_\alpha)$. The latter is in $\mathbb{Z}$ by Proposition 6.19 if $\beta \neq \pm\alpha$. If $\beta = \pm\alpha$, then $\beta(h_\alpha) = \pm 2 \in \mathbb{Z}$ as well. $\square$

Since Φ spans $\mathfrak{t}^*$, there exists a subset $\{\alpha_1, \dots, \alpha_r\} \subset \Phi$ that is a basis of $\mathfrak{t}^*$.

Lemma 6.21. Every root β is a $\mathbb{Q}$ -linear combination of the elements α_i .

Proof. Write $\beta = \sum_{i=1}^r c_i \alpha_i$ with $c_i \in \mathbb{C}$. We need to show that $c_i \in \mathbb{Q}$. Pairing the above identity with α_j and multiplying by $2/(\alpha_j, \alpha_j)$, we get

$$\frac{2(\beta, \alpha_j)}{(\alpha_j, \alpha_j)} = \sum_{i=1}^r c_i \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}$$

for each $j = 1, \dots, r$. We view the above identities as a collection of linear equations in the variables $c_1, \dots, c_r$. Since $(-, -)$ is nondegenerate and $\alpha_1, \dots, \alpha_r$ is a basis, the matrix $((\alpha_i, \alpha_j))_{1 \leq i, j \leq r}$ is invertible. Therefore the matrix $(2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j))_{1 \leq i, j \leq r}$ is invertible too. Moreover, all the coefficients of this system of equations are integers by the previous lemma. By linear algebra over the field $\mathbb{Q}$, there is a unique solution to this equation and $c_i \in \mathbb{Q}$ for all i . $\square$

Let $E = \mathbb{R}\Phi \subset \mathfrak{t}^*$ be the $\mathbb{R}$ -span of Φ , i.e. the subset of $\mathfrak{t}^*$ that can be written as $\mathbb{R}$ -linear combinations of roots. By the previous lemma, this is also the $\mathbb{R}$ -span of a basis of $\mathfrak{t}^*$ consisting of roots, so $\dim_{\mathbb{R}} E = \dim_{\mathbb{C}} \mathfrak{t}^*$.

Theorem 6.22. In the above notation:

1. The restriction of the form $(-, -)$ to $E \times E$ is $\mathbb{R}$ -valued and positive definite.
2. If $\alpha \in \Phi$ and $c \in \mathbb{R}$, then $c\alpha \in \Phi$ if and only if $c = \pm 1$.
3. If $\alpha, \beta \in \Phi$, then

$$\beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha \in \Phi.$$

4. If $\alpha, \beta \in \Phi$, then $\frac{2(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}$.

Proof. Part 2 follows from Proposition 6.17. Parts 3 and 4 follow from Proposition 6.19 if $\beta \neq \pm\alpha$, and are easily checked to be also true when $\beta = \pm\alpha$. So it remains to prove Part 1.

We first prove that $(\alpha, \beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \Phi$. Indeed, if $\lambda, \mu \in \mathfrak{t}^*$, then $(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \text{tr}(\text{ad}_{t_\lambda} \circ \text{ad}_{t_\mu})$ equals the sum of the eigenvalues of $\text{ad}_{t_\lambda} \circ \text{ad}_{t_\mu}$ on $\mathfrak{g}$. Using the root space decomposition (6.4.1), this sum is exactly

$$(\lambda, \mu) = \sum_{\alpha \in \Phi} \lambda(\alpha)\mu(\alpha) = \sum_{\alpha} (\alpha, \lambda)(\alpha, \mu). \quad (6.7.1)$$

Taking $\lambda = \mu = \beta \in \Phi$ and dividing by $(\beta, \beta)^2$, we get

$$\frac{1}{(\beta, \beta)} = \sum_{\alpha \in \Phi} \frac{(\alpha, \beta)^2}{(\beta, \beta)^2}$$

By Part 4, all the summands on the right hand side are rational numbers. So $(\beta, \beta) \in \mathbb{Q}$ for all $\beta \in \Phi$. Again by Part 4, we conclude that $(\alpha, \beta) \in \mathbb{Q}$. Therefore $(-, -)$ is real valued on $E \times E$.

If $\lambda \in E$, then (6.7.1) shows that $(\lambda, \lambda) = \sum_{\alpha} (\alpha, \lambda)^2$. Since $(\alpha, \lambda) \in \mathbb{R}$, $(\lambda, \lambda) \geq 0$. If $(\lambda, \lambda) = 0$, then $(\alpha, \lambda) = 0$ for all $\alpha \in \Phi$. Since $(-, -)$ is nondegenerate and Φ spans $\mathfrak{t}^*$ (as a $\mathbb{C}$ -vector space), this implies $\lambda \neq 0$. So $(-, -)$ is indeed positive definite on $E \times E$. $\square$

This theorem says that the triple $(\Phi, E, (-, -))$ is a ‘root system’. It turns out that this data (up to a suitable isomorphism) exactly classifies the semisimple Lie algebra $\mathfrak{g}$! To this end, we will study root systems abstractly in the next section.

7 Root systems

We forget for a moment that we are studying Lie algebras, and study a combinatorial object that is purely within the realm of Euclidean geometry, called a root system.

We slightly deviate from our standing assumption that all vector spaces are over $\mathbb{C}$; we will now consider many (finite-dimensional) vector spaces over $\mathbb{R}$.

7.1 Definitions

Definition 7.1. A *Euclidean space* is a pair $(E, (-, -))$, where E is a (finite-dimensional) vector space over $\mathbb{R}$ and $(-, -): E \times E \rightarrow \mathbb{R}$ is a positive definite bilinear form (sometimes called an inner product).

We will often write a Euclidean space simply as E , suppressing $(-, -)$ from the notation. For every Euclidean space $(E, (-, -))$, there is a basis of E such that the form $(-, -)$ is the ‘standard’ inner product whose Gram matrix is the identity matrix.

If $\lambda, \alpha \in E$ and $\alpha \neq 0$, define the number

$$\langle \lambda, \check{\alpha} \rangle = 2 \frac{(\lambda, \alpha)}{(\alpha, \alpha)} \quad (7.1.1)$$

and the linear map

$$w_\alpha: E \rightarrow E \quad (7.1.2)$$

$$v \mapsto v - \langle v, \check{\alpha} \rangle \alpha \quad (7.1.3)$$

It is easy to check that $w_\alpha^2 = 1$. Geometrically, w_α is the reflection around the hyperplane perpendicular to α . So for example, $w_\alpha(\alpha) = -\alpha$.

Warning 7.2. The pairing $\langle \lambda, \check{\alpha} \rangle$ is linear in the first variable, but not necessarily in the second.

Definition 7.3. Let $(E, (-, -))$ be a Euclidean space. A **root system** in E is a finite subset $\Phi \subset E$ with the following properties:

1. Φ spans E and $0 \notin \Phi$
2. If $\alpha \in \Phi$ and $c \in \mathbb{R}$, then $c\alpha \in \Phi$ if and only if $c = \pm 1$.
3. If $\alpha \in \Phi$, then $w_\alpha(\Phi) = \Phi$.
4. If $\alpha, \beta \in \Phi$, then $\langle \beta, \check{\alpha} \rangle = 2(\beta, \alpha)/(\alpha, \alpha) \in \mathbb{Z}$.

Technically, a root system is a triple $(\Phi, E, (-, -))$, but we typically abbreviate this by simply writing (Φ, E) or even simply Φ .

Definition 7.4. The **rank** of a root system (Φ, E) is $\dim_{\mathbb{R}} E$.

Definition 7.5. Let (Φ, E) and (Φ', E') be root systems. An **isomorphism** $f: (\Phi, E) \rightarrow (\Phi', E')$ between root systems (Φ, E) and (Φ', E') is an $\mathbb{R}$ -linear isomorphism $f: E \rightarrow E'$ such that $f(\Phi) = \Phi'$ and such that $\langle f(\alpha), f(\beta) \rangle = \langle \alpha, \check{\beta} \rangle$ for all $\alpha, \beta \in \Phi$.

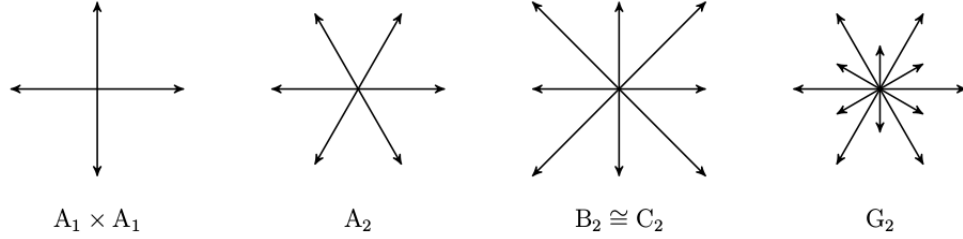
Note that we are not asserting that f must preserve the inner products; simply that it preserves the expressions $\langle \alpha, \check{\beta} \rangle$ for $\alpha, \beta \in \Phi$.

7.2 Examples

Example 7.6. If $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{C}$ and $\mathfrak{t} \subset \mathfrak{g}$ a Cartan subalgebra, then $\Phi \subset E = \mathbb{R}\Phi \subset \mathfrak{t}^*$ (equipped with the pairing induced by the Killing form) is a root system, by Theorem 6.22.

Example 7.7 (Rank 1). The subset $\Phi = \{\pm 1\} \subset \mathbb{R}$, where $\mathbb{R}$ has the standard inner product, is a root system. Up to isomorphism, it is the unique root system of rank 1. We note that $\mathfrak{sl}_2(\mathbb{C})$, with Cartan subalgebra $\mathfrak{t} = \text{span}\{h\}$, gives rise to this rank 1 root system. It is called the A_1 -root system.

The next picture displays (up to isomorphism) all root systems of rank 2.



There is a nice geometric interpretation of Axiom 3 in Definition 7.3. If $\alpha, \beta \in \Phi$, let $\bar{\beta}$ be the projection of β onto the line spanned by α ; in a formula, $\bar{\beta} = \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha$. Then Axiom 3 is equivalent to the statement that $\bar{\beta} = c\alpha$ for some $c \in \frac{1}{2}\mathbb{Z}$. In other words: the projection of a root onto the line spanned by another root must be an integer or half-integer multiple of that root.

Example 7.8. Let $n \geq 1$ be an integer and consider the vector space $\mathbb{R}^{n+1}$ with standard basis $e_0, \dots, e_n$. Let $E \subset \mathbb{R}^{n+1}$ be the subspace of all $\sum x_i e_i$ with $\sum x_i = 0$, equipped with the restriction of the standard inner product on $\mathbb{R}^{n+1}$. Consider $\Phi = \{e_i - e_j : 0 \leq i \neq j \leq n\} \subset E$. Then one can check that (Φ, E) is a root system, called the root system of type A_n . Indeed, reflection around the root $e_i - e_j$ corresponds to transposing the i th and j th coordinate of E , which clearly preserves Φ . Moreover $\langle \alpha, \alpha \rangle = 2$ for all $\alpha \in \Phi$ so $\langle \beta, \check{\alpha} \rangle = \langle \alpha, \beta \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$.

7.3 Weyl group of a root system

Recall that for each $\alpha \in \Phi$, there is an involution $w_\alpha : E \rightarrow E$ given by $\lambda \mapsto \lambda - \langle \lambda, \check{\alpha} \rangle \alpha$, which is reflection around the hyperplane perpendicular to α .

Definition 7.9. The **Weyl group** of (Φ, E) , typically denoted by $W(\Phi)$ or just W , is the subgroup of $GL(E)$ generated by the reflections $\{w_\alpha : \alpha \in \Phi\}$.

Lemma 7.10. The Weyl group $W(\Phi)$ is finite.

Proof. Note that $W(\Phi)$ preserves Φ by the axioms of a root system, so $W(\Phi)$ acts on Φ . This action is faithful since Φ spans E . Therefore $W(\Phi)$ is a subgroup of the group of permutations of the set Φ , which is a finite group. $\square$

Example 7.11. We have $W(A_1) = \{1, w_\alpha\} \simeq C_2$ is cyclic of order 2. Moreover $W(A_1 \times A_1) = C_2 \times C_2$. To determine $W(A_2)$, let α be the horizontal right-pointing root and β the left upwards diagonal root. Then $r = w_\alpha w_\beta$ is rotation by $2\pi/3 = 120^\circ$ degrees, so $r^3 = 1$. Therefore $W(A_2) \simeq D_3 \simeq S_3$, where we write D_n for the dihedral group of order $2n$. Similarly we can check (exercise) that $W(B_2) \simeq D_4$ and $W(G_2) \simeq D_6$. We leave the verifications of these isomorphisms as an exercise.

Example 7.12. The Weyl group of the A_n root system is S_{n+1} . (Exercise)

7.4 Irreducible root systems

Definition 7.13. We say a root system (Φ, E) is **reducible** if there exists a partition $\Phi = \Phi_1 \sqcup \Phi_2$ with $\Phi_1 \neq \emptyset, \Phi$ such that $\Phi_1 \perp \Phi_2$. Otherwise, we say (Φ, E) is irreducible.

$\langle \beta, \check{\alpha} \rangle$	$\langle \alpha, \check{\beta} \rangle$	$(\beta, \beta)/(\alpha, \alpha)$	θ
0	0	?	$\pi/2 = 90^\circ$
1	1	1	$\pi/3 = 60^\circ$
-1	-1	1	$2\pi/3 = 120^\circ$
2	1	2	$\pi/4 = 45^\circ$
-2	-1	2	$3\pi/4 = 135^\circ$
3	1	3	$\pi/6 = 30^\circ$
-3	-1	3	$5\pi/6 = 150^\circ$

Table 1: Angles and length ratios of non-proportional roots

If (Φ, E) is reducible and $\Phi = \Phi_1 \sqcup \Phi_2$ is a partition such that $\Phi_1 \perp \Phi_2$, let $E_i = \mathbb{R}\Phi_i$ for $i = 1, 2$. Then $E = E_1 \oplus E_2$ and Φ_i is a root system in E_i . Therefore (E, Φ) is ‘built’ out of smaller root systems in this case. It follows that to classify all root systems, it suffices to classify irreducible ones.

Remark 7.14. *It would be better to use the terminology ‘decomposable’ and ‘indecomposable’, however reducible/irreducible is more standard.*

In the examples of rank 2, $A_1 \times A_1$ is reducible, while A_2, B_2, G_2 are all irreducible.

7.5 Angles in root systems

Let (Φ, E) be a root system. We severely constrain the values of $\langle \alpha, \check{\beta} \rangle$, hence the possible angles between roots.

Lemma 7.15. *If $\alpha, \beta \in \Phi$ with $\alpha \neq \pm\beta$, then*

$$\langle \alpha, \check{\beta} \rangle \langle \beta, \check{\alpha} \rangle \in \{0, 1, 2, 3\}$$

Proof. Write $\|v\| = (v, v)^{1/2}$ for $v \in E$. Let $0 \leq \theta < \pi$ be the angle between α and β . Then $(\alpha, \beta) = \|\alpha\| \|\beta\| \cos(\theta)$, hence

$$\langle \alpha, \check{\beta} \rangle \langle \beta, \check{\alpha} \rangle = 4 \frac{(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} = 4(\cos(\theta))^2.$$

By Axiom 3 of a root system, this must be an integer. On the other hand, $0 \leq (\cos(\theta))^2 \leq 1$, and $\cos(\theta)^2 \neq 1$ since we assumed $\beta \neq \pm\alpha$. So $4(\cos(\theta))^2 \in \{0, 1, 2, 3\}$. $\square$

This easy lemma has a great number of consequences. First of all, since both terms $\langle \alpha, \check{\beta} \rangle$ and $\langle \beta, \check{\alpha} \rangle$ are integers and their product is $\{0, \pm 1, \pm 2, \pm 3\}$, they both lie in $\{0, \pm 1, \pm 2, \pm 3\}$. After possibly switching the roles of α and β , we may assume $(\beta, \beta) \geq (\alpha, \alpha)$. Then Table 1 lists of all the possibilities of the angles and length ratios. Note that all these options appear in the rank 2 root systems.

Remark 7.16. *It is now a good exercise (see example sheet 3) to check that $A_1 \times A_1, A_2, B_2$ and G_2 are the only rank 2 root systems up to isomorphism.*

Corollary 7.17. *If $\alpha, \beta \in \Phi$ are such that $(\alpha, \beta) < 0$ and $\beta \neq \pm\alpha$, then $\alpha + \beta \in \Phi$.*

Proof. Examining Table 1 shows that after possibly swapping α and β so that $(\beta, \beta) \geq (\alpha, \alpha)$, we have $\langle \alpha, \check{\beta} \rangle = -1$. But then $w_\beta(\alpha) = \alpha - \langle \alpha, \check{\beta} \rangle \beta = \beta + \alpha \in \Phi$, as desired. $\square$

Corollary 7.18. If Φ is irreducible, there can be at most two distinct lengths of roots, i.e., the set $\{(\alpha, \alpha) : \alpha \in \Phi\}$ has cardinality ≤ 2 .

Proof. Exercise, using the constraints on the length ratios and angles of roots. $\square$

Corollary 7.19. Root strings are of length at most 4.

Proof. In fact, write down all the root strings in each case of Table 1 as an exercise to see that this holds. $\square$

Definition 7.20. An irreducible root system is *simply laced* if all the roots are of the same length.

In that case $\langle \alpha, \beta \rangle \in \{0, \pm 1\}$ for all $\alpha, \beta \in \Phi$ with $\beta \neq \pm \alpha$.

7.6 Root bases

Definition 7.21. A subset $\Delta = \{\alpha_1, \dots, \alpha_\ell\} \subset \Phi$ is called a *root basis* if

1. Δ is an $\mathbb{R}$ -basis of E ; and
2. Every $\alpha \in \Phi$ is of the form

$$\alpha = \sum_{i=1}^{\ell} c_i \alpha_i, \quad (7.6.1)$$

where c_i are integers, and either $c_i \geq 0$ for all i or $c_i \leq 0$ for all i . (Briefly: c_i are integers of the same sign.)

We say elements of Δ are *simple roots*. A root $\alpha \in \Phi$ is *positive* (with respect to Δ) if $c_i \geq 0$ for all i in the expression (7.6.1). The set of positive roots is denoted by Φ^+ .

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Our next task is to show that root bases exist.

Example 7.22. Convince yourself that root bases exist in rank 2. Note that a root basis is not unique, since the Weyl group sends one root bases to another one.

Example 7.23. If $\Phi = \{e_i - e_j : 0 \leq i \neq j \leq n\}$ is of type A_n , then $\Delta = \{e_0 - e_1, e_1 - e_2, \dots, e_{n-1} - e_n\}$ is a root basis, with set of positive roots $\{e_i - e_j : i < j\}$.

We will give a practical way to construct root bases. For $\alpha \in \Phi$, recall that $H_\alpha = \{x \in E : (x, \alpha) = 0\}$ denotes the hyperplane orthogonal to α .

It is an elementary fact from Euclidean geometry that $E \neq \bigcup_{\alpha \in \Phi} H_\alpha$. (Find your own proof if you're not convinced!) Therefore we may pick a vector

$$\gamma \in E \setminus \left(\bigcup_{\alpha \in \Phi} H_\alpha \right).$$

Let

$$\begin{aligned} \Phi_\gamma^+ &= \{\alpha \in \Phi : (\gamma, \alpha) > 0\} \\ \Phi_\gamma^- &= -\Phi_\gamma^+ = \{\alpha \in \Phi : (\gamma, \alpha) < 0\}. \end{aligned}$$

We say an element $\alpha \in \Phi_\gamma^+$ is decomposable if there exist $\alpha_1, \alpha_2 \in \Phi_\gamma^+$ such that $\alpha = \alpha_1 + \alpha_2$, and indecomposable otherwise. Let Δ_γ be the set of indecomposable elements of Φ_γ^+ .

Theorem 7.24. *The set Δ_γ is a root basis of Φ , and a root is positive with respect to Δ_γ if and only if it lies in Φ_γ^+ . Moreover, every root basis is of the form Δ_γ for some $\gamma \in E \setminus \cup_\alpha H_\alpha$.*

Proof. Note that we have $\Phi = \Phi_\gamma^+ \sqcup (-\Phi_\gamma^+)$ since $(\gamma, \alpha) \neq 0$ for all $\alpha \in \Phi$. We prove three claims.

Claim 1: if $\alpha \in \Phi_\gamma^+$, then $\alpha = \sum_{i=1}^\ell c_i \alpha_i$ with $c_i \in \mathbb{Z}_{\geq 0}$ and $\alpha_i \in \Delta_\gamma$. Indeed, if α is indecomposable, this is clear. Otherwise, write $\alpha = \beta_1 + \delta_1$ with $\beta_1, \delta_1 \in \Phi_\gamma^+$. If both β_1, δ_1 are indecomposable, we are done again. Otherwise, replace α by β_1 and δ_1 and repeat this process for both. If this process would not terminate, we would obtain a sequence of elements $\beta_1, \beta_2, \dots$ in Φ_γ^+ such that $(\gamma, \beta_1) > (\gamma, \beta_2) > \dots$, contradicting the finiteness of Φ . (If you do not like this argument, you can also argue by contradiction, and take a counterexample α for which (γ, α) is minimal.)

Claim 2: if $\alpha, \beta \in \Delta_\gamma$ and $\beta \neq \pm\alpha$, then $(\alpha, \beta) \leq 0$. Indeed, if $(\alpha, \beta) > 0$, then applying Corollary 7.17 to the pair $(-\alpha, \beta)$ shows that $\pm(\beta - \alpha) \in \Phi$. After possibly reordering α, β we may assume $\beta - \alpha \in \Phi_\gamma^+$, so $\beta = (\beta - \alpha) + \alpha$, contradicting indecomposability of β .

Claim 3: the elements of Δ_γ are linearly independent. In fact, by Claim 2 it suffices to show the more general statement that if $S = \{\lambda_1, \dots, \lambda_n\} \subset E$ is a subset of vectors such that $(\lambda_i, \lambda_j) \leq 0$ for all $i \neq j$ and $(\gamma, \lambda_i) > 0$ for all i , then S consists of linearly independent elements. Indeed, suppose that $\sum c_i \lambda_i = 0$ is a nontrivial relation. After reordering and relabeling the coefficients, we may write this relation as

$$e = \sum_{i=1}^m c_i \lambda_i = \sum_{j=m+1}^n c_j \lambda_j,$$

where $c_i \geq 0$, $c_j \geq 0$, and not all of them are zero. We then compute that

$$(e, e) = \sum_{1 \leq i \leq m, m+1 \leq j \leq n} c_i c_j (\lambda_i, \lambda_j) \leq 0.$$

Since $(-, -)$ is positive definite, $e = 0$. Therefore $(e, \gamma) = 0$. On the other hand, $(e, \gamma) = \sum \lambda_i (\gamma, \lambda_i) \geq 0$, so $\lambda_i = 0$ for all $1 \leq i \leq m$, and similarly $\lambda_j = 0$ for all $m+1 \leq j \leq n$.

By assumption, Φ spans E . Therefore Φ_γ^+ spans E . By Claim 1, Δ_γ spans E . By Claim 3, Δ_γ is a basis of E . By Claim 1 again, Δ_γ is a root basis.

To show every root basis is of this form, let $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ be a root basis with positive roots Φ^+ . Let γ be an element such that $(\alpha, \gamma) > 0$ for all $\alpha \in \Delta$; as a concrete example of such a γ , let $v_1, \dots, v_\ell$ be the dual basis of Δ with respect to $(-, -)$, so $(\alpha_i, v_j) = \delta_{ij}$, and take $\gamma = v_1 + \dots + v_\ell$. Then $(\alpha, \gamma) > 0$ for all $\alpha \in \Phi^+$ so $\Phi^+ \subset \Phi_\gamma^+$. Since these finite sets have the same cardinality, $\Phi^+ = \Phi_\gamma^+$. Since every element of Φ^+ is a $\mathbb{Z}_{\geq 0}$ -linear combination of elements of Δ , Δ consists of indecomposable elements of Φ^+ , so $\Delta \subset \Delta_\gamma$. Again by cardinality reasons, $\Delta = \Delta_\gamma$. $\square$

We extract one very useful corollary of the proof (claim 2) of the above theorem:

Corollary 7.25. *If Δ is a root basis and $\alpha, \beta \in \Delta$ are distinct, then $(\alpha, \beta) \leq 0$.*

Connected components of $E \setminus \cup_\alpha H_\alpha$ are called [Weyl chambers](#).

Lemma 7.26. *If $\gamma, \gamma' \in E \setminus \cup_\alpha H_\alpha$, then $\Delta_\gamma = \Delta_{\gamma'}$ if and only if γ, γ' lie in the same Weyl chamber.*

Proof. The elements γ, γ' lie in the same Weyl chamber if and only if (α, γ) and (α, γ') have the same sign for all $\alpha \in \Phi$. The latter holds if and only if $\Phi_\gamma^+ = \Phi_{\gamma'}^+$, if and only if $\Delta_\gamma = \Delta_{\gamma'}$. $\square$

This lemma shows that the map $\gamma \mapsto \Delta_\gamma$ induces a bijection

$$\{\text{Weyl chambers}\} \xrightarrow{1:1} \{\text{Root bases of } \Phi\}. \quad (7.6.2)$$

We denote the inverse of this bijection by $\Delta \mapsto \mathcal{C}(\Delta)$. If Δ is a fixed root basis of Φ , we call the Weyl chamber $\mathcal{C}(\Delta)$ corresponding to it the **fundamental Weyl chamber**. It is explicitly given by

$$\mathcal{C}(\Delta) = \{x \in E : (x, \alpha) > 0 \text{ for all } \alpha \in \Delta\}.$$

Example 7.27. Draw the fundamental Weyl chamber for all the rank 2 root systems.

Let Φ be a root system with root basis $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$.

Definition 7.28. Let $\alpha = \sum_{i=1}^\ell c_i \alpha_i$. We define the **root height** of α (with respect to Δ) to be $\sum_i c_i$.

Recall that Φ^+ denotes the set of positive roots of Φ with respect to Δ .

Lemma 7.29. If $\alpha \in \Phi$ is positive and not simple, there is a simple root $\alpha_i \in \Delta$ such that $\alpha - \alpha_i \in \Phi^+$.

Proof. Suppose $(\alpha, \alpha_i) \leq 0$ for all i . Writing $\alpha = \sum c_i \alpha_i$ with $c_i \geq 0$, we see that $(\alpha, \alpha) = \sum c_i (\alpha, \alpha_i) \leq 0$, implying that $\alpha = 0$, contradiction. So there is an i such that $(\alpha, \alpha_i) > 0$. Therefore $\alpha - \alpha_i \in \Phi$ by Lemma 7.17. This root must be positive, since one of the coordinates of α distinct from α_i must be nonzero and positive. $\square$

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Corollary 7.30. If $\beta \in \Phi^+$, then there is a sequence $\beta_1, \dots, \beta_n$ of (not necessarily distinct) simple roots such that $\beta_1 + \dots + \beta_n = \beta$ and $\beta_1 + \dots + \beta_k$ is a root for all $1 \leq k \leq n$.

Proof. Use the previous lemma and induction on the root height. $\square$

7.7 Facts about the Weyl group

Let (Φ, E) be a root system with root basis $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ and Weyl group W . It is easy to see that if $w \in W$, then $w(\Delta)$ is again a root bases of Φ , with positive roots $w(\Phi^+)$. Moreover since W acts via isometries on E , it preserves the root hyperplanes H_α , hence it acts on the set of Weyl chambers. These actions are compatible under the bijection (7.6.2). Explicitly: $\Delta_{w(\gamma)} = w(\Delta_\gamma)$ for an element $\gamma \in E \setminus \cup_\alpha H_\alpha$.

We will now prove two fundamental properties about the Weyl group W : it permutes the root bases *simply transitively* and it is generated by reflections of simple roots.

To prove this, we need two lemmas.

Lemma 7.31. If $\alpha_i \in \Delta$ is a simple root, then w_{α_i} preserves $\Phi^+ \setminus \{\alpha_i\}$.

Proof. See example sheet 3. (The proof is easy.) $\square$

Lemma 7.32. Let $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$. Then $w_{\alpha_i}(\rho) = \rho - \alpha_i$ for all $\alpha_i \in \Delta$.

Proof. Follows immediately from previous lemma. □

Theorem 7.33. *In the above notation:*

1. W acts simply transitively on root bases.
2. For every $\alpha \in \Phi$, there is a $w \in W$ with $w(\alpha) \in \Delta$. (Such a w might not be unique.)
3. W is generated by the simple reflections $\{w_{\alpha_i} : 1 \leq i \leq \ell\}$.

Proof. Let $w_i = w_{\alpha_i}$ and let W' be the subgroup of W generated by the simple reflections $\{w_i : 1 \leq i \leq \ell\}$. We will prove that W' acts transitively on root bases and use this to show that $W' = W$.

Claim 1: W' acts transitively on root bases. Indeed, using the bijection (7.6.2), it suffices to show that W' acts transitively on Weyl chambers. Let $\mathcal{C}$ be a Weyl chamber and $\gamma \in \mathcal{C}$ arbitrary. Let $w \in W'$ be an element such that the expression $(w(\gamma), \rho)$ is maximal among elements of W' , where ρ is the element of Lemma 7.32. By assumption, for all i we have $(w_i w(\gamma), \rho) \leq (w(\gamma), \rho)$, and moreover $(w_i w(\gamma), \rho) = (w(\gamma), w_i(\rho)) = (w(\gamma), \rho - \alpha_i)$, so $(w(\gamma), \alpha_i) \geq 0$. Since γ does not lie on any root hyperplane, neither does $w(\gamma)$, so $(w(\gamma), \alpha_i) > 0$ for all i . In other words, $w(\gamma)$ lies in the fundamental Weyl chamber $\mathcal{C}(\Delta)$. So $w \cdot \mathcal{C} = \mathcal{C}(\Delta)$.

Claim 2: for every $\alpha \in \Phi$, there is an $w \in W'$ such that $w(\alpha) \in \Delta$. Indeed, by the previous claim, it suffices to prove that α is contained in some root basis of Φ . Let $\gamma \in E \setminus \cup_{\alpha} H_{\alpha}$ be an element such that $(\gamma, \alpha) > 0$ and $|(\gamma, \beta)| > (\gamma, \alpha)$ for all $\beta \in \Phi \setminus \{\pm\alpha\}$. (Such a γ can be found by taking an element of $H_{\alpha} \setminus \cup_{\beta \neq \pm\alpha} H_{\beta}$ and slightly perturbing it.) Then it is easy to see that α is an indecomposable element of Φ_{γ}^+ , so $\alpha \in \Delta_{\gamma}$.

We can now prove that $W = W'$. Let $\alpha \in \Phi$ and let $w \in W'$ be an element such that $w(\alpha) \in \Delta$, which exists by Claim 2. We compute that $w_{w(\alpha)} = w w_{\alpha} w^{-1}$. Therefore $w_{\alpha} = w^{-1} w_{w(\alpha)} w \in W'$. Since W is generated by $\{w_{\alpha} : \alpha \in \Phi\}$, we conclude that $W = W'$.

It only remains to prove that W acts simply transitively on root bases. In other words, the stabilizer of Δ in W is trivial. We leave this for Example Sheet 3. □

Remark 7.34. The fact that W acts transitively on Weyl chambers can be visualised nicely in the rank 2 case: when going from one Weyl chamber $\mathcal{C}$ to another one $\mathcal{C}'$, one can make a sequence $\mathcal{C} = \mathcal{C}_0, \mathcal{C}_1, \dots, \mathcal{C}_n = \mathcal{C}'$, where $\mathcal{C}_i$ shares a root hyperplane H_{α_i} with $\mathcal{C}_{i-1}$ as facet. Then the elements $w_{\alpha_n} \cdots w_{\alpha_1}$ will send $\mathcal{C}$ to $\mathcal{C}'$.

7.8 Cartan matrix and Dynkin diagram of a root system

Let (Φ, E) be a root system and let $\Delta = \{\alpha_1, \dots, \alpha_{\ell}\}$ be a root basis of Φ .

Definition 7.35. The matrix $C = (\langle \alpha_i, \check{\alpha}_j \rangle)_{1 \leq i, j \leq \ell}$ is called the **Cartan matrix** of Φ . The entries of this matrix are called **Cartan integers**.

The Cartan matrix of Φ is well defined up to conjugation by a permutation matrix, i.e. up to reordering the indices. This follows from the fact that any two root bases are W -conjugate.

Example 7.36. Here are the Cartan matrices of the rank 1 and 2 root systems:

- $A_1 : (2)$
- $A_1 \times A_1 : \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

- $A_2 : \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$
- $B_2 : \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$
- $G_2 : \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$

It turns out that the Cartan matrix contains all the information about a root system:

Proposition 7.37. *Let (Φ', E') be another root system with root basis $\Delta' = \{\alpha'_1, \dots, \alpha'_\ell\}$. Suppose that $\langle \alpha_i, \check{\alpha}_j \rangle = \langle \alpha'_i, \check{\alpha}'_j \rangle$ for all $1 \leq i, j \leq n$. Then there is a unique isomorphism of root systems $f: (\Phi, E) \rightarrow (\Phi', E')$ such that $f(\alpha_i) = \alpha'_i$.*

Proof. Since Δ and Δ' are bases of E and E' respectively, there exists a unique $\mathbb{R}$ -linear isomorphism $f: E \rightarrow E'$ sending α_i to α'_i for all i . We need to show that f maps Φ to Φ' and has the property that $\langle f(\alpha), f(\check{\beta}) \rangle = \langle \alpha, \check{\beta} \rangle$ for all $\alpha, \beta \in \Phi$. We will show both these statements using the Weyl groups W and W' of Φ and Φ' respectively. By assumption and by linearity of $\langle -, \check{\cdot} \rangle$ in the first variable, we have $\langle f(\lambda), f(\check{\alpha}_i) \rangle = \langle \lambda, \check{\alpha}_i \rangle$ for all $\lambda \in E$. Therefore $w_{f(\alpha_i)}(f(\lambda)) = f(w_{\alpha_i}(\lambda))$. In other words, $w_{\alpha'_i} = f \circ w_{\alpha_i} \circ f^{-1}$. It follows that the group homomorphism $w \mapsto f \circ w \circ f^{-1}$ from $GL(E)$ to $GL(E')$ maps simple reflections in W to simple reflections in W' . Since the simple reflections generate the Weyl group, it follows that this assignment maps W onto W' .

We can use this to prove the claims above. Indeed, if $\alpha \in \Phi$ is a root, let $w \in W$ be an element such that $w(\alpha) = \alpha_i \in \Delta$. Let $w' = fwf^{-1} \in W'$. Then $w'(f(\alpha)) = f(w(\alpha)) = f(\alpha_i) = \alpha'_i$. So $f(\alpha) = w'^{-1}(\alpha'_i) \in \Phi'$ since W' preserves the roots. So $f(\Phi) = \Phi'$. Similarly, using the fact that $\langle f(\lambda), f(\check{\alpha}_i) \rangle = \langle \lambda, \check{\alpha}_i \rangle$ for all i and $\lambda \in E$ and using the Weyl group action, we have $\langle f(\lambda), f(\check{\alpha}) \rangle = \langle \lambda, \check{\alpha} \rangle$ for all $\alpha \in \Phi$. $\square$

Remark 7.38. *In fact, it is possible to construct Φ from the Cartan matrix, by iteratively constructing root strings and using Corollary 7.30. See the end of Section 11.1 of Humphreys for details.*

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It turns out that a Cartan matrix is typically quite sparse (i.e., contains a lot of zeroes), so it is convenient to display its information pictorially using a certain graph.

Definition 7.39. *The **Dynkin diagram** associated to the data (Φ, Δ) , sometimes denoted $D(\Phi)$, is a graph defined by the following properties:*

1. *Its vertices are indexed by elements of Δ .*
2. *Two vertices α, β are connected by an edge if and only if $(\alpha, \beta) \neq 0$. In that case, we draw exactly $\langle \alpha, \check{\beta} \rangle \langle \beta, \check{\alpha} \rangle$ edges between them. (This number is 1, 2 or 3 by Lemma 7.15.)*
3. *Suppose α, β are connected by an edge. If $(\alpha, \alpha) = (\beta, \beta)$, we declare the edge to be undirected. If $(\alpha, \alpha) < (\beta, \beta)$, we draw a ' $<$ ' on the edge from α to β , like $\alpha \rightleftarrows \beta$.*

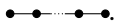
Note that the Dynkin diagram does not depend on the choice of root basis since they're all W -conjugate.

Example 7.40. *Here are the Dynkin diagrams of the rank 1 and 2 root systems:*

- $A_1 : \bullet$
- $A_1 \times A_1 : \bullet \quad \bullet$

- A_2 : 
- B_2 : 
- G_2 : 

In the cases B_2, G_2 , the leftmost vertex corresponds to the long root in the root basis.

Example 7.41. Consider the A_n root system $\Phi = \{e_i - e_j : 1 \leq i, j \leq n+1\}$. It has root basis $\Delta = \{e_1 - e_2, \dots, e_n - e_{n+1}\}$. The Dynkin diagram is .

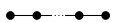
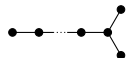
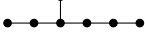
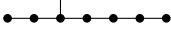
Remark 7.42. It turns out the Cartan matrix determines the group structure of the Weyl group W . Indeed, if $C = (c_{ij})$ is the Cartan matrix, then W has generators $w_1, \dots, w_\ell$ and relations w_i^2 for all i and $(w_i w_j)^{c_{ij}+2}$ for all $i \neq j$. We will not need or prove this.

Remark 7.43. The unoriented graph underlying the Dynkin diagram (that is, the graph obtained by forgetting the orientations of some of the edges) is called the Coxeter graph of Φ .

On Example Sheet 3, you will prove that Φ is irreducible if and only if its Dynkin diagram is connected.

7.9 Classification of irreducible root systems

Theorem 7.44. If Φ is an irreducible root system, its Dynkin diagram is one of the following graphs:

- (A_n) : 
- (B_n) : 
- (C_n) : 
- (D_n) : 
- (E_6) : 
- (E_7) : 
- (E_8) : 
- (F_4) : 
- (G_2) : 

Proof. This involves some combinatorics and Euclidean geometry, which for reasons of time we will not do here, see Humphreys §11.4. □

If we want to avoid duplications in the above list, we should assume $n \geq 2$ in the B_n case, $n \geq 3$ in the C_n case, and $n \geq 4$ in the D_n case.

The four infinite families are called the [classical root systems/Dynkin diagrams](#), and the five remaining cases are called the [exceptional root systems](#).

We will soon discuss the classical root systems in detail and say some things about the exceptional cases. First, we will finish the classification of simple Lie algebras.

8 The classification of semisimple Lie algebras

Summarizing our work so far, we have constructed maps between (isomorphism classes of) the following objects:

$$\left\{ \begin{array}{l} \text{Pairs } (\mathfrak{g}, \mathfrak{t}), \text{ where} \\ \mathfrak{g} \text{ semisimple, } \mathfrak{t} \text{ a CSA} \end{array} \right\} \rightarrow \{\text{Root systems}\} \hookrightarrow \{\text{Dynkin diagrams}\}. \quad (8.0.1)$$

We will now indicate why both of these maps are bijections and the first one does not depend on the choice of $\mathfrak{t}$.

8.1 Independence of $\mathfrak{t}$

Theorem 8.1 (Conjugacy theorem). *Let $\mathfrak{g}$ be a semisimple Lie algebra and $\mathfrak{t}, \mathfrak{t}'$ be Cartan subalgebras. Then there exists a Lie algebra automorphism $f: \mathfrak{g} \rightarrow \mathfrak{g}$ such that $f(\mathfrak{t}) = \mathfrak{t}'$.*

Proof. Omitted, see Humphreys Section 16.4. Warning: unlike the other results whose proofs we omitted, this one actually requires some serious work. $\square$

Corollary 8.2. *In the notation of the theorem, suppose that $\mathfrak{t}, \mathfrak{t}'$ give rise to the root systems (Φ, E) and (Φ', E') respectively. Then (Φ, E) and (Φ', E') are isomorphic.*

Proof. Indeed, if $f: \mathfrak{g} \rightarrow \mathfrak{g}$ is an isomorphism sending $\mathfrak{t}$ to $\mathfrak{t}'$, then f induces an isomorphism $\tilde{f}: \mathfrak{t}^\vee \rightarrow \mathfrak{t}'^\vee$ of vector spaces that preserves the Killing form and the subset of roots, so induces an isomorphism of root systems. $\square$

It follows that in the maps of (8.0.1), the first one is independent of the choice of $\mathfrak{t}$.

Remark 8.3. *In fact, the automorphism f can be chosen to be inner, which is a notion we will not define here, but intuitively means that it comes from conjugation by an element of a (complex Lie group) G with Lie algebra $\mathfrak{g}$. For example, the inner automorphisms of $\mathfrak{sl}_n(\mathbb{C})$ are those induced by conjugating by an element of $\text{SL}_n(\mathbb{C})$.*

Lemma 8.4. *Let $\mathfrak{g}$ be a semisimple Lie algebra with Cartan subalgebra $\mathfrak{t}$, giving rise to the root system Φ . Then $\mathfrak{g}$ is simple if and only if Φ is irreducible, if and only if its Dynkin diagram is connected. A semisimple Lie algebra $\mathfrak{g}$ is simple if and only if its associated root system Φ is irreducible, if and only if its Dynkin diagram is connected.*

Proof. (Leave this as an exercise.) If $\mathfrak{g}$ is not simple, then we can write $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ with $\mathfrak{g}_i$ nonzero ideals of $\mathfrak{g}$ which are themselves semisimple. Choose a Cartan subalgebra $\mathfrak{t}_i$ of $\mathfrak{g}_i$, then $\mathfrak{t}_1 \oplus \mathfrak{t}_2 \subset \mathfrak{g}$ is a Cartan subalgebra of $\mathfrak{g}$, which we may assume (after applying an automorphism of $\mathfrak{g}$) equals $\mathfrak{t}$ by Theorem 8.1. Then $\Phi = \Phi_1 \sqcup \Phi_2$ in $\mathfrak{t}^\vee = \mathfrak{t}_1^\vee \oplus \mathfrak{t}_2^\vee$, where Φ_i is the set of roots of $\mathfrak{g}_i$ in $\mathfrak{t}_i^\vee$. Conversely, suppose $\Phi = \Phi_1 \sqcup \Phi_2$ is reducible. For a root $\alpha \in \Phi$, recall that $h_\alpha \in \mathfrak{t}$ denotes the element in the $\mathfrak{sl}_2$ -triple constructed from α . Let $\mathfrak{t}_i = \text{span}\{h_\alpha: \alpha \in \Phi_i\}$ and $\mathfrak{g}_i = \mathfrak{t}_i + \sum_{\alpha \in \Phi_i} \mathfrak{g}_\alpha$. Then $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ and $\mathfrak{g}_i$ are nonzero ideals of $\mathfrak{g}$. This shows that $\mathfrak{g}$ is simple if and only if Φ is irreducible. On example sheet 3, you will show that this is equivalent to the Dynkin diagram being connected. $\square$

8.2 Constructing a Lie algebra from a Cartan matrix

Theorem 8.5. 1. Given a Dynkin diagram D of type $A - G$, there is a simple Lie algebra whose root system has Dynkin diagram D .

2. Let $\mathfrak{g}, \mathfrak{g}'$ be semisimple Lie algebras with Cartan subalgebras $\mathfrak{t}, \mathfrak{t}'$ respectively, giving rise to root systems Φ, Φ' . Let Δ, Δ' be root bases of Φ, Φ' and let $e_\alpha \in \mathfrak{g}_\alpha, e_{\alpha'} \in \mathfrak{g}_{\alpha'}$ be choices of nonzero elements for each $\alpha \in \Delta$ and $\alpha' \in \Delta'$. Let $f: \Phi \rightarrow \Phi'$ be an isomorphism of root systems sending Δ to Δ' .

Then there is a unique isomorphism of Lie algebras $\tilde{f}: \mathfrak{g} \rightarrow \mathfrak{g}'$ with the following property: $\tilde{f}$ sends $\mathfrak{t}$ to $\mathfrak{t}'$ and maps e_α to $e_{f(\alpha)}$ for all $\alpha \in \Delta$.

Proof. There are multiple ways of proving these theorems, the most elegant due to Serre, using generators and relations, see Humphreys §18.3. Below, we will give an explicit construction for $\mathfrak{g}$ for the classical Lie algebras. $\square$

We conclude that there are bijections between isomorphism classes of the following objects:

$$\{\text{Complex simple Lie algebras } \mathfrak{g}\} \xleftrightarrow{1:1} \{\text{Irreducible root systems}\} \xleftrightarrow{1:1} \{\text{Connected Dynkin diagrams}\}.$$

These bijections are commonly referred to as the [Cartan–Killing classification](#).

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8.3 The classical root systems and Lie algebras

We make the Cartan–Killing classification completely explicit for the Dynkin diagrams of type A, B, C, D . It turns out that these Dynkin diagrams exactly correspond to the classical Lie algebras $\mathfrak{sl}, \mathfrak{sp}, \mathfrak{so}$ studied before. We call these Lie algebras and their root systems and Dynkin diagrams [classical](#).

For example, let $\mathfrak{g} = \mathfrak{sl}_{n+1}$ and $\mathfrak{t} \subset \mathfrak{g}$ the Cartan subalgebra consisting of diagonal matrices. Then $\mathfrak{t}^\vee$ is the quotient of $\mathbb{C}e_1 \oplus \cdots \oplus \mathbb{C}e_{n+1}$ by the one-dimensional subspace $e_1 + \cdots + e_{n+1}$. (Here e_i the linear functional on $\mathfrak{t}$ sending an element $\text{diag}(t_1, \dots, t_{n+1})$ to t_i .) Then $\Phi = \{e_i - e_j : 1 \leq i \neq j \leq n+1\}$. Moreover $\Delta = \{e_1 - e_2, \dots, e_n - e_{n+1}\} = \{\alpha_1, \dots, \alpha_n\}$ is a basis of simple roots, and using (for example) root strings we may compute that if $i \neq j$ then $\langle \alpha_i, \tilde{\alpha}_j \rangle = -1$ if $|i - j| = 1$ and zero otherwise. Therefore the Cartan matrix of Φ is the same as that of the A_n root system, so Φ has type A_n . The Weyl group of Φ is generated by reflections corresponding to the roots $e_i - e_j$, which have the effect of swapping the i and j coordinate. It follows that $W \simeq S_{n+1}$. (Note: we do not need to compute the Killing form explicitly for this.)

For orthogonal and symplectic Lie algebras we can do an analogous computation and we arrive at Table 2. We leave the verification of this as an essential exercise. Explain Weyl groups.

In the B_n and C_n case the C_2^n -factor in W acts by sign changes on the coordinates. In the D_n case, we view C_2^{n-1} as the subgroup of elements of C_2^n whose product is the identity, and then C_2^{n-1} again acts by sign changes.

Since B_2 and C_2 are isomorphic, the Cartan–Killing classification immediately implies that $\mathfrak{so}_5$ and $\mathfrak{sp}_4$ are isomorphic. Similarly, since D_3 and A_3 are isomorphic, $\mathfrak{so}_4$ and $\mathfrak{sp}_6$ are isomorphic. It is possible but extremely tedious to check these facts ‘by hand’.

Type	$\Phi \subset \mathbb{R}^n$	Δ	W	$\mathfrak{g}$	Dimension
B_n	$\{e_i\} \cup \{\pm e_i \pm e_j : i \neq j\}$	$\{e_1 - e_2, \dots, e_{n-1} - e_n, e_n\}$	$S_n \ltimes C_2^n$	$\mathfrak{so}_{2n+1}$	$2n^2 + n$
C_n	$\{2e_i\} \cup \{\pm e_i \pm e_j : i \neq j\}$	$\{e_1 - e_2, \dots, e_{n-1} - e_n, 2e_n\}$	$S_n \ltimes C_2^n$	$\mathfrak{sp}_{2n}$	$2n^2 + n$
D_n	$\{\pm e_i \pm e_j : i \neq j\}$	$\{e_1 - e_2, \dots, e_{n-1} - e_n, e_{n-1} + e_n\}$	$S_n \ltimes C_2^{n-1}$	$\mathfrak{so}_{2n}$	$2n^2 - n$

Table 2: Classical Lie algebras and their root systems

8.4 The exceptional root systems and Lie algebras

This section is non-examinable. If a simple Lie algebra $\mathfrak{g}$ (or irreducible root system Φ) has Dynkin diagram of type E, F or G , we call it **exceptional**. By the abstract Cartan–Killing classification, we know that there are five exceptional Lie algebras $\mathfrak{g}_2, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7$ and $\mathfrak{e}_8$. However, actually constructing them is much harder than for the classical cases. We have summarized some facts about these Lie algebras and their root systems in Table 3. Each next entry in the table contains the previous one, that is, there is a chain of inclusions

$$\mathfrak{g}_2 \subset \mathfrak{f}_4 \subset \mathfrak{e}_6 \subset \mathfrak{e}_7 \subset \mathfrak{e}_8.$$

We now make some comments about this, focusing on the smallest one $\mathfrak{g}_2$ and the biggest one $\mathfrak{e}_8$.

We have already seen the root system of type G_2 : it has twelve roots, and its Weyl group is dihedral of order 12. To construct the Lie algebra $\mathfrak{g}_2$, consider the octonions $\mathbb{O}$, which is a certain 8-dimensional non-associative algebra. Then the set of derivations of $\mathbb{O}$, just as in example sheet 2, for a Lie algebra, which turns out to be a Lie algebra of type G_2 . There is a trace map on octonions, and restricting to trace zero elements defines a 7-dimensional subspace of $\mathbb{O}$ on which $\mathfrak{g}_2$ acts faithfully. This defines a representation $\mathfrak{g}_2 \rightarrow \mathfrak{gl}_7$. It turns out $\mathfrak{g}_2$ lands in $\mathfrak{so}_7$ (with respect to a certain choice of basis).

What about E_8 ? We start by describing the root system. Consider the following lattice in $\mathbb{R}^8$:

$$\Lambda = \left\{ (x_i) \in \mathbb{Z}^8 \cup \left(\frac{1}{2} + \mathbb{Z} \right)^8 : \sum x_i \in 2\mathbb{Z} \right\}.$$

That is, elements of Λ are 8-tuples which are either all integers or all half-integers, and whose sum is an even integer. The restriction of the inner product on $\mathbb{R}^8$ defines a positive definite, even unimodular lattice $(-, -): \Lambda \times \Lambda \rightarrow \mathbb{Z}$. (Even means that $(\lambda, \lambda) \in 2\mathbb{Z}$ for all $\lambda \in \Lambda$. Unimodular means that the determinant of the matrix $((b_i, b_j))$ is ± 1 for any $\mathbb{Z}$ -basis of Λ .) This is up to isomorphism the unique lattice of rank 8 with these properties. Let

$$\Phi = \{ \alpha \in \Lambda : (\alpha, \alpha) = 2 \}.$$

Then it is not hard to see that Φ is a root system. It has basis of simple roots

$$\{e_1 - e_2, \dots, e_6 - e_7, e_6 + e_7, -\frac{1}{2}(e_1 + \dots + e_8)\}$$

whose Cartan matrix has Dynkin diagram of type E_8 . The Weyl group W of Φ is extraordinarily large, of order 696729600. Constructing the Lie algebra $\mathfrak{e}_8$ is harder. I highly recommend Skip Garibaldi's survey article <https://arxiv.org/pdf/1605.01721> if you're interested in reading more about E_8 .

There is a website where all root systems are explicitly described which is handy for calculations: <http://www.liegroups.org/dissemination/spherical/explorer/cgi-bin/rootSystem.cgi>.

Type	$\#\Phi^+$	$\#W$	$\dim \mathfrak{g}$	Smallest faithful rep
G_2	6	12	14	7
F_4	24	$2^7 \cdot 3^2$	52	26
E_6	36	$2^7 \cdot 3^4 \cdot 5$	78	27
E_7	63	$2^{10} \cdot 3^4 \cdot 5 \cdot 7$	133	56
E_8	120	$2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$	248	248

Table 3: Some facts about exceptional Lie algebras

9 Representations of semisimple Lie algebras

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The final goal of the course is the classification of finite-dimensional representations of semisimple Lie algebras using their root systems.

9.1 Root and weight lattice

Let (Φ, E) be a root system. Recall that a subset $L \subset E$ is called a lattice if it is the $\mathbb{Z}$ -span of an $\mathbb{R}$ -basis of E . (Equivalently, L is a discrete subgroup of E and the quotient E/L is compact.)

Definition 9.1. The **root lattice** $\mathbb{Z}\Phi$ is by definition the $\mathbb{Z}$ -span of Φ in E . The **weight lattice**, typically denoted by X , is the set $\{\lambda \in E : \langle \lambda, \check{\alpha} \rangle \in \mathbb{Z} \text{ for all } \alpha \in \Phi\}$.

These are both lattices. Indeed, fix a root basis $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$ of Φ . Then $\mathbb{Z}\Phi = \mathbb{Z}\alpha_1 \oplus \dots \oplus \mathbb{Z}\alpha_\ell$. In the case of X , let $\omega_1, \dots, \omega_\ell$ be the unique elements of E with the property that

$$\langle \omega_i, \check{\alpha}_j \rangle = \delta_{ij}. \quad (9.1.1)$$

These are called the **fundamental weights**. On example sheet 3, you have shown that $X = \{\lambda \in E : \langle \lambda, \check{\alpha}_i \rangle \in \mathbb{Z} \text{ for all } i\}$. Therefore $X = \mathbb{Z}\omega_1 \oplus \dots \oplus \mathbb{Z}\omega_\ell$.

Example 9.2. If $\Phi = \{\pm\alpha\}$ is of type A_1 , then $\mathbb{Z}\Phi = \mathbb{Z}\alpha$ and $X = \mathbb{Z}(\alpha/2)$.

Example 9.3. We can do some examples in the rank 2 cases. For example, in the A_2 case, if $\Delta = \{\alpha, \beta\}$ is a root basis, then $\omega_1 = (2\alpha + \beta)/3$ and $\omega_2 = (2\beta + \alpha)/3$.

Since $\langle \alpha, \check{\beta} \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$, we have $\mathbb{Z}\Phi \subset X$, but this inclusion is strict in general. The quotient $X/\mathbb{Z}\Phi$ is a finite group, called the fundamental group of Φ . (This is a general fact: if $L_1 \subset L_2$ are lattices of E then L_2/L_1 is finite.) The order of $X/\mathbb{Z}\Phi$ equals the determinant of the Cartan matrix of Φ . Note that W permutes X .

Example 9.4. In the A_n case, $\mathbb{Z}\Phi \subset X$ has index $n + 1$.

Remark 9.5 (Can be safely ignored). If G is a compact Lie group with trivial center and Lie algebra $\mathfrak{g}$ such that $\mathfrak{g} \otimes \mathbb{C}$ is semisimple with root system Φ , then the (topological) fundamental group of G is isomorphic to the fundamental group of Φ , motivating why we call $X/\mathbb{Z}\Phi$ the fundamental group of the root system.

Definition 9.6. An element $\lambda \in X$ is called **dominant** if $\langle \lambda, \check{\alpha} \rangle \geq 0$ for all $\alpha \in \Phi$ (equivalently, for all $\alpha \in \Delta$).

An element $\lambda \in X$ is dominant if and only if $\lambda = \sum c_i \omega_i$ with $c_i \in \mathbb{Z}_{\geq 0}$, if and only if λ lies in the closure of the fundamental Weyl chamber

9.2 Weights of a representation

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{t} \subset \mathfrak{g}$, giving rise to a set of roots $\Phi \subset \mathfrak{t}^\vee$. Choose a root basis $\Delta = \{\alpha_1, \dots, \alpha_\ell\}$, giving rise to a set of positive roots $\Phi^+ \subset \Phi$.

The root and weight lattices satisfy $\mathbb{Z}\Phi \subset X \subset \mathfrak{t}^\vee$.

Lemma 9.7. *An element $\lambda \in \mathfrak{t}^\vee$ lies in X if and only if $\lambda(h_\alpha) \in \mathbb{Z}$ for all $\alpha \in \Phi$ (equivalently, all $\alpha \in \Delta$).*

Proof. This follows from the identity $\lambda(h_\alpha) = \langle \lambda, \check{\alpha} \rangle$ and the definition of X . $\square$

Let V be a finite-dimensional representation of $\mathfrak{g}$. Since $\mathfrak{t}$ is abelian and consists of semisimple elements of $\mathfrak{g}$, V breaks up into simultaneous eigenspaces under the action of $\mathfrak{t}$ by Lemma 6.4:

$$V = \bigoplus_{\lambda \in \mathfrak{t}^\vee} V_\lambda, \quad (9.2.1)$$

where $V_\lambda = \{v \in V : t \cdot v = \lambda(t)v \text{ for all } t \in \mathfrak{t}\}$. We make some basic observations about this decomposition, which mimic the basic observations made in the $\mathfrak{sl}_2$ case. Note that the Weyl group W acts on X .

Proposition 9.8. 1. *Let $\alpha \in \Phi$ and $e_\alpha \in \mathfrak{g}_\alpha$ be nonzero. Then $e_\alpha V_\lambda \subset V_{\lambda+\alpha}$.*

2. *If $V_\lambda \neq 0$, then $\lambda \in X$, in other words λ lies in the weight lattice.*

3. *$\dim(V_\lambda) = \dim(V_{w(\lambda)})$ for all $w \in W$.*

Proof. Part 1 is a simple computation: if $t \in \mathfrak{t}$ and $v \in V_\lambda$, then

$$t \cdot (e_\alpha \cdot v) = e_\alpha \cdot (t \cdot v) + [t, e_\alpha] \cdot v = e_\alpha(\lambda(t)v) + \alpha(t)e_\alpha \cdot v = (\lambda + \alpha)(t)(e_\alpha \cdot v).$$

This computation is similar to the computation that $e \cdot V_\lambda \subset V_{\lambda+2}$ in the $\mathfrak{sl}_2$ case.

The other two parts will follow from $\mathfrak{sl}_2$ -theory. Indeed, let $\alpha \in \Phi$ and choose $e_\alpha \in \mathfrak{g}_\alpha$ and $e_{-\alpha} \in \mathfrak{g}_{-\alpha}$ such that $\{e_\alpha, h_\alpha, e_{-\alpha}\}$ form an $\mathfrak{sl}_2$ -triple, spanning a subalgebra $\mathfrak{m}_\alpha \subset \mathfrak{g}$ isomorphic to $\mathfrak{sl}_2$. View V as an $\mathfrak{m}_\alpha$ -representation by restriction. Then the weights of this representation must be integers. In other words, $\lambda(h_\alpha) \in \mathbb{Z}$ for all λ such that $V_\lambda \neq 0$. Since this is true for every root, the previous lemma shows that $\lambda \in X$, proving Part 2.

To prove Part 3, we will show that the equality of dimensions for $w = w_\alpha$ (this is enough since the Weyl group is generated by reflections). Decompose V as a direct sum of irreducible $\mathfrak{m}_\alpha$ -representations:

$$V|_{\mathfrak{m}_\alpha} = \bigoplus V^{(i)},$$

where $V^{(i)}$ is an irreducible $\mathfrak{m}_\alpha$ -representation. There exists a basis $v_1, \dots, v_n$ of V_λ such that each v_i lies in a distinct $V^{(j)}$. Then apply e_α or $e_{-\alpha}$ repeatedly and use the fact that $w_\alpha(\lambda)(h_\alpha) = -\lambda(h_\alpha)$. $\square$

Therefore, the weights of V lie in the weight lattice X and are preserved by the Weyl group.

Definition 9.9. *In the above notation, suppose that $v \in V_\lambda$ is nonzero and satisfies $e_\alpha \cdot v = 0$ for all $\alpha \in \Phi^+$. Then λ is called a highest weight and v is called a highest weight vector.*

(In lectures, I only required that $e_\alpha \cdot v = 0$ for all $\alpha \in \Delta$. This is equivalent to requiring it for all positive roots, by induction on the root height, Lemma 7.29 and Part 3 of Proposition 6.19.)

Lemma 9.10. *In the above notation:*

1. V has a highest weight vector.
2. A highest weight of V is dominant.

Proof. 1. Follows from the finite-dimensionality of V and Part 1 of Proposition 9.8.

2. Follows from the fact that the highest weight of an $\mathfrak{sl}_2$ -representation is a nonnegative integer.

□

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In the $\mathfrak{sl}_2$ case, it was easy to explicitly see that an irreducible representation has a unique highest weight, and that this weight uniquely determines the representation. The same statements will be true for general semisimple $\mathfrak{g}$, but require some preparations to prove. Ultimately, the conclusion will be that for every dominant $\lambda \in X$, there is a unique irreducible $\mathfrak{g}$ -representation $V(\lambda)$ with highest weight λ .

9.3 The universal enveloping algebra of a Lie algebra

In this section, we drop our convention that vector spaces are always assumed to be finite-dimensional. So we will explicitly state when we assume this. Everything we say works over a general field of characteristic zero but we will keep our assumption that our base field is $\mathbb{C}$.

Definition 9.11. An *algebra* A is a vector space together with a bilinear map $A \times A \rightarrow A$, typically denoted by $(x, y) \mapsto x \cdot y$. It is *unital* if there exists a (necessarily unique) element $e \in A$ such that $x \cdot e = e \cdot x = x$ for all $x \in A$. It is *associative* if the product is associative: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in A$.

Given an associative algebra A , we may define a bracket $[x, y] = (x \cdot y) - (y \cdot x)$ on A , and the pair $\text{Lie}(A) := (A, [-, -])$ is then a Lie algebra.

Example 9.12. If V is a vector space and $A = \text{End}(V)$ (with algebra structure given by composing linear maps), then $\text{Lie}(A) = \mathfrak{gl}(V)$.

A *representation* of an associative algebra is a homomorphism of algebras $A \rightarrow \text{End}(V)$ for some vector space V .

Recall that for a finite group G , there is an algebra $\mathbb{C}[G]$, the group algebra, such that representations for G correspond bijectively to representations of $\mathbb{C}[G]$. For Lie algebras, the role of $\mathbb{C}[G]$ is played by the universal enveloping algebra $\mathcal{U}(\mathfrak{g})$. The study of this algebra will therefore be useful in the study of the representations of $\mathfrak{g}$.

Given a Lie algebra $\mathfrak{g}$ over $\mathbb{C}$ (not necessarily semisimple or finite-dimensional), we will define a unital associative algebra $\mathcal{U}(\mathfrak{g})$ together with a map $\iota: \mathfrak{g} \rightarrow \mathcal{U}(\mathfrak{g})$ satisfying the following property:

Theorem 9.13 (Universal property). *If A is a unital associative algebra and $f: \mathfrak{g} \rightarrow (A, [-, -])$ is a homomorphism of Lie algebras, there is a unique algebra homomorphism $\mathcal{U}(f): \mathcal{U}(\mathfrak{g}) \rightarrow A$ satisfying $f \circ \iota = f$.*

(In category theory lingo, $\mathcal{U}(\mathfrak{g})$ is left-adjoint to the functor $A \mapsto (A, [-, -])$.)

We will now construct $\mathcal{U}(\mathfrak{g})$ explicitly, but you should keep in mind that the universal property is the key property that we are aiming for.

Definition 9.14. Let V be a vector space. Let

$$\mathcal{T}(V) = \bigoplus_{i \geq 0} V^{\otimes i},$$

where $V^{\otimes 0} = \mathbb{C}$ by convention. Composition of tensors defines bilinear maps $V^{\otimes m} \times V^{\otimes n} \rightarrow V^{\otimes(m+n)}$, which combine to define an algebra structure on $\mathcal{T}(V)$; this is called *the tensor algebra* of V . It is a unital associative algebra.

A *two-sided ideal* of an algebra A is a subspace I satisfying $x \cdot I \subset I$ and $I \cdot x \subset I$ for all $x \in A$. In that case, the quotient A/I , with the product $(x + I) \cdot (y + I) = x \cdot y + I$, is again an algebra.

Definition 9.15. The *symmetric algebra* of a vector space V is the quotient of $\mathcal{T}(V)$ by the two-sided ideal generated by $\{x \otimes y - y \otimes x : x, y \in V\}$.

Both algebras $\mathcal{T}(V)$ and $\text{Sym}(V)$ are naturally graded, and the graded pieces are exactly the tensor powers $V^{\otimes n}$ and symmetric powers $\text{Sym}^n V$ respectively.

Definition 9.16. Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. The *universal enveloping algebra* $\mathcal{U}(\mathfrak{g})$ of $\mathfrak{g}$ is the quotient $\mathcal{T}(\mathfrak{g})/J$ of $\mathcal{T}(\mathfrak{g})$ by the two-sided ideal J generated by $\{x \otimes y - y \otimes x - [x, y] : x, y \in \mathfrak{g}\}$.

So $\mathcal{U}(\mathfrak{g})$ is a unital associative algebra, which is not necessarily commutative.

We often simply write $x_1 \cdots x_n$ instead of $x_1 \otimes \cdots \otimes x_n$. Note that mapping x to itself in $\mathfrak{g}^{\otimes 1} = \mathfrak{g}$ defines a linear map $\iota : \mathfrak{g} \rightarrow \mathcal{U}(\mathfrak{g})$, which can be viewed as a map of Lie algebra $\iota : \mathfrak{g} \rightarrow \mathcal{U}(\mathfrak{g})$.

Proof of universal property. Let $f : \mathfrak{g} \rightarrow \text{Lie}(A)$ be a homomorphism of Lie algebras. Then the assignment $x_1 \otimes \cdots \otimes x_n \mapsto f(x_1) \cdots f(x_n)$ defines a homomorphism of algebras $f' : \mathcal{T}(\mathfrak{g}) \rightarrow A$. Since f respects the Lie brackets, J is contained in the kernel of f' . So f' factors through a map of algebras $\tilde{f} : \mathcal{U}(\mathfrak{g}) \rightarrow A$ and satisfies $\tilde{f} \circ \iota = f$ by construction. Since the algebra $\mathcal{U}(\mathfrak{g})$ is generated by the image of ι , $\tilde{f}$ is unique with this property. $\square$

Applying the universal property to $A = \text{End}(V)$, we see that Lie algebra homomorphisms $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ extend uniquely to algebra homomorphisms $\tilde{\rho} : \mathcal{U}(\mathfrak{g}) \rightarrow \text{End}(V)$. We obtain a bijection:

$$\{\text{Representations of } \mathfrak{g}\} = \{\text{Representations of } \mathcal{U}(\mathfrak{g})\}. \quad (9.3.1)$$

Moreover, this bijection preserves all the natural notions on both sides such as being a subrepresentation/irreducible/...

In particular, every element of $\mathcal{U}(\mathfrak{g})$ defines an element of $\text{End}(V)$, and relations between these elements are dictated by relations in $\mathcal{U}(\mathfrak{g})$. We secretly knew this already! Indeed, if $\mathfrak{g} = \mathfrak{sl}_2$ for example and $v \in V$ an element of a representation then $e \cdot (f \cdot v) = f \cdot (e \cdot v) + [e, f] \cdot v$. But this is expressing that $ef = fe + [e, f]$ in $\mathcal{U}(\mathfrak{g})$.

Example 9.17. Recall that Casimir element $\Omega_\rho = \rho(e)\rho(f) + \rho(f)\rho(e) + \frac{1}{2}\rho(h)^2 \in \text{End}(V)$ of a representation ρ of $\mathfrak{sl}_2$. Writing $\Omega = ef + fe + \frac{1}{2}h^2 \in \mathcal{U}(\mathfrak{g})$, then Ω_ρ is the image of Ω under (the extension to $\mathcal{U}(\mathfrak{g})$ of) ρ . Moreover, Ω lies in the center of $\mathcal{U}(\mathfrak{g})$, and in fact every element of the center is a linear combination of powers of Ω ; this is a consequence of the Harish-Chandra isomorphism.

Example 9.18. If $\mathfrak{g}$ is abelian, then $\mathcal{U}(\mathfrak{g}) = \text{Sym}(\mathfrak{g})$.

The above example already shows that $\mathcal{U}(\mathfrak{g})$ is infinite dimensional even in simple cases. The Poincaré–Birkhoff–Witt theorem (‘PBW theorem’) describes an explicit basis of $\mathcal{U}(\mathfrak{g})$.

To state it, note that $\mathcal{U}(\mathfrak{g})$ is naturally filtered: for $n \geq 0$ let F_n be the image of $\bigoplus_{k \leq n} V^{\otimes k}$ in $\mathcal{U}(\mathfrak{g})$. Then $F_0 \subset F_1 \subset \dots$ and $F_i \cdot F_j \subset F_{i+j}$. Whenever we have a filtered algebra, we can take the associated graded algebra, which is defined to be

$$\text{gr}(\mathcal{U}(\mathfrak{g})) = \bigoplus_{n \geq 0} F_n / F_{n-1} \quad (9.3.2)$$

(let $F_{-1} = 0$ by convention) and with the algebra structure induced componentwise by the multiplication $F_i / F_{i-1} \times F_j / F_{j-1} \rightarrow F_{i+j} / F_{i+j-1}$ induced by $F_i \times F_j \rightarrow F_{i+j}$. Then $\text{gr}(\mathcal{U}(\mathfrak{g}))$ is a unital associative algebra. If $x, y \in \mathfrak{g}$, then $xy - yx = [x, y]$ in $\mathcal{U}(\mathfrak{g})$. Therefore $xy - yx = 0$ in $\text{gr}(\mathcal{U}(\mathfrak{g}))$. In fact, we have:

Lemma 9.19. The algebra $\text{gr}(\mathcal{U}(\mathfrak{g}))$ is commutative.

Proof. Let $\text{gr}^n = F_n / F_{n-1}$. Since $xy - yx = [x, y]$ for all $x, y \in \mathfrak{g}$, we have $[F_1, F_1] \subset F_1$ and hence any two elements in gr^1 commute with each other. Therefore the same is true for products and linear combinations of the elements of gr^1 . Since $\text{gr}(\mathcal{U}(\mathfrak{g}))$ is generated as an algebra by gr^1 , the lemma follows. $\square$

Consider the composition $\mathfrak{g} \rightarrow F_1 \rightarrow F_1 / F_0 \subset \text{gr}(\mathcal{U}(\mathfrak{g}))$ of the natural quotient maps. This is a linear map, hence extends to an algebra homomorphism $\mathcal{T}(\mathfrak{g}) \rightarrow \text{gr}(\mathcal{U}(\mathfrak{g}))$. Since the target is commutative, it contains $\{x \otimes y - y \otimes x : x, y \in \mathfrak{g}\}$ in its kernel. This map therefore factors through an algebra homomorphism $\text{Sym } \mathfrak{g} \rightarrow \mathcal{U}(\mathfrak{g})$, all fitting in a commutative diagram:

$$\begin{array}{ccc} \mathcal{T}(\mathfrak{g}) & & \\ \downarrow & \searrow & \\ \text{Sym}(\mathfrak{g}) & \xrightarrow{\varphi} & \text{gr}(\mathcal{U}(\mathfrak{g})) \end{array}$$

Theorem 9.20 (PBW theorem). The map $\varphi: \text{Sym}(\mathfrak{g}) \rightarrow \text{gr}(\mathcal{U}(\mathfrak{g}))$ is an isomorphism.

Proof. We already showed that φ is surjective, which in many applications is the only thing one needs. The proof that φ is injective is quite gnarly; see Humphreys §17.4. $\square$

Corollary 9.21. Let $x_1, \dots, x_n$ be a basis of $\mathfrak{g}$. Then

$$\{x_1^{k_1} \dots x_n^{k_n} : k_i \in \mathbb{Z}_{\geq 0}\}$$

is a basis of $\mathcal{U}(\mathfrak{g})$.

Proof. This follows from the fact that a basis of $\text{gr}(\mathcal{U})$ lifts to a basis of $\mathcal{U}(\mathfrak{g})$. $\square$

Note that if we change the ordering of the basis of $\mathfrak{g}$, we obtain a new basis of $\mathcal{U}(\mathfrak{g})$, which is quite different from the original one.

What does $\mathcal{U}(\mathfrak{g})$ do for us? For example, we have the following lemma:

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Lemma 9.22. *Let V be a representation of a Lie algebra $\mathfrak{g}$ and $v \in V$ an element. Then $\mathcal{U}(\mathfrak{g}) \cdot v = \{x \cdot v : x \in \mathcal{U}(\mathfrak{g})\}$ is the smallest subrepresentation of V containing v .*

Proof. To be precise, the representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ extends to a homomorphism of algebras $\tilde{\rho}: \mathcal{U}(\mathfrak{g}) \rightarrow \text{End}(V)$, and $x \cdot v$ means by definition $\tilde{\rho}(x)(v)$ if $x \in \mathcal{U}(\mathfrak{g})$. This follows from the general fact that if V is a representation of an algebra A and $v \in V$, then $A \cdot v$ is the smallest A -subrepresentation containing v . $\square$

This implies that if V is an irreducible $\mathfrak{g}$ -representation, then V is always a quotient of $\mathcal{U}(\mathfrak{g})$ by a left ideal. So it makes sense to construct representations using such quotients (more on this later).

9.4 Highest weight modules

Let $\mathfrak{g}$ be a (finite-dimensional) semisimple Lie algebra with Cartan subalgebra $\mathfrak{t}$, corresponding root system Φ , and let $\Delta = \{\alpha_1, \dots, \alpha_\ell\} \subset \Phi$ be a choice of simple roots. In this section, we will consider a class of representations of $\mathfrak{g}$ called highest weight modules and prove some properties about them.

Let V be a (not necessarily finite-dimensional) representation of $\mathfrak{g}$. For such a representation V , we can still define the simultaneous eigenspaces V_λ ($\lambda \in \mathfrak{t}^*$). Analogously to the finite-dimensional case, we say $v \in V$ is a highest weight vector if $v \in V_\lambda \setminus \{0\}$ for some $\lambda \in \mathfrak{t}^*$ and $e_\alpha \cdot v = 0$ for all $\alpha \in \Phi^+$, and in that case λ is a highest weight.

Definition 9.23. *A representation V of $\mathfrak{g}$ is a **highest weight module** if V contains a highest weight vector v that generates V , in the sense that $V = \mathcal{U}(\mathfrak{g}) \cdot v$.*

Lemma 9.24. *If V is finite-dimensional and irreducible, then V is a highest weight module.*

Proof. We have seen that every finite-dimensional $\mathfrak{g}$ -representation contains a highest weight vector v . Since V is irreducible it must equal $\mathcal{U}(\mathfrak{g}) \cdot v$ by Lemma 9.22. $\square$

We are ultimately only interested in finite-dimensional irreducible representations, which are highest weight modules by the previous lemma, but it will be convenient to study this more general class.

Example 9.25. *If $\mathfrak{g} = \mathfrak{sl}_2$ then $V(n)$ is a highest weight module generated by a vector v with $e \cdot v = 0$ and $h \cdot v = nv$. But we can also write down infinite-dimensional highest weight modules. Let V be the vector space with basis $\{v_0, v_1, \dots\}$, and define a $\mathfrak{g}$ -action on V by the rules*

$$e \cdot v_0 = 0, \quad h \cdot v_0 = 0, \quad f \cdot v_i = v_{i+1}$$

and extending linearly and using the bracket relations. (Exercise: check that the previous sentence makes sense.) Then we can calculate that $h \cdot v_n = -2nv_n$ and $e \cdot v_n = n(1-n)v_{n-1}$. We see that v_0 is a highest weight vector generating V , so V is a highest weight module. However v_1 is also a highest weight vector of V , and $\mathcal{U}(\mathfrak{g})v_1 = \text{span}\{v_1, v_2, \dots\}$ is a subrepresentation of V . The above example shows that highest weight modules can be reducible and contain multiple highest weight vectors.

Define $\mathfrak{n}^+ = \sum_{\alpha \in \Phi^+} \mathfrak{g}_\alpha$ and $\mathfrak{n}^- = \sum_{\alpha \in \Phi^-} \mathfrak{g}_\alpha$. Then $\mathfrak{n}, \mathfrak{n}^-$ are subalgebras of $\mathfrak{g}$, and as vector spaces we have $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{t} \oplus \mathfrak{n}^+$.

Lemma 9.26. *If V is a highest weight module generated by highest weight vector $v \in V_\lambda$, then $V = \mathcal{U}(\mathfrak{n}^-) \cdot v$.*

Proof. Choose, for each root $\beta \in \Phi$, a nonzero element $e_\beta \in \mathfrak{g}_\beta$. Let $t_1, \dots, t_\ell$ be a basis of $\mathfrak{t}$. By the (easy part of the) PBW theorem, $V = \mathcal{U}(\mathfrak{g}) \cdot v$ is spanned by elements of the form

$$e_{-\beta_1}^{a_1} \cdots e_{-\beta_n}^{a_n} t_1^{b_1} \cdots t_\ell^{b_\ell} e_{\beta_1}^{c_1} \cdots e_{\beta_n}^{c_n} \cdot v,$$

where $\beta_1, \dots, \beta_n \in \Phi^+$ are the positive roots and $a_i, b_i, c_i \in \mathbb{Z}_{\geq 0}$. Since v is a highest weight vector, $\mathfrak{n}^+ \cdot v = 0$ and $t \cdot v \in \text{span}\{v\}$ for all $t \in \mathfrak{t}$, so it suffices to consider such expressions with $b_i = c_i = 0$ for all i . $\square$

We introduce a convenient partial order on $\mathfrak{t}^*$. If $\lambda, \mu \in \mathfrak{t}^*$, we say $\mu \preceq \lambda$ if $\lambda - \mu = \sum_{i=1}^\ell k_i \alpha_i$, where $k_i \in \mathbb{Z}_{\geq 0}$ and α_i are the simple roots. For example, $0 \preceq \lambda$ exactly if λ is a nonnegative linear combination of simple roots.

Proposition 9.27. *Let V be a highest weight module generated by highest weight vector $v \in V_\lambda$.*

1. *Let $D(\lambda) = \{\mu \in \mathfrak{t}^* : \mu \preceq \lambda\} = \{\lambda - \sum k_i \alpha_i : k_i \in \mathbb{Z}_{\geq 0}\}$. Then*

$$V = \bigoplus_{\mu \in D(\lambda)} V_\mu.$$

2. *If $W \subset V$ is a subrepresentation, then $W = \bigoplus_\mu W_\mu$. (In other words, W is a direct sum of its weight spaces.)*
3. *$\dim(V_\lambda) = 1$ and $\dim(V_\mu)$ is finite-dimensional for all μ .*
4. *V is irreducible if and only if every highest weight vector lies in V_λ .*
5. *V contains a maximal proper subrepresentation.*

Proof. These are all consequences of Lemma 9.26.

1. By the previous lemma and its proof, V is spanned by elements of the form $e_{-\beta_1}^{k_1} \cdots e_{-\beta_n}^{k_n} \cdot v$. By Proposition 9.8, Part 1 (which continues to hold when V is infinite-dimensional), $e_\alpha \cdot V_\lambda \subset V_{\lambda+\alpha}$, so the above element has weight $\lambda - \sum_{i=1}^n k_i \beta_i$. This weight μ satisfies $\mu \preceq \lambda$. Therefore V is indeed generated by the weight spaces V_μ with $\mu \preceq \lambda$.
2. We need to show that if $w \in W$ is of the form $w = \sum_{i=1}^k v_i$ with $v_i \in V_{\lambda_i}$, then $v_i \in W$ for all i . This is a general linear algebra fact, but we will be careful since we're not assuming V is finite dimensional. It is clear when $k = 1$. Let $k > 1$ be the minimal integer for which this property fails for some $w \in W$, so $w = \sum v_i$ and not all v_i are in W . By minimality, all v_i are not in W . Let $t \in \mathfrak{t}$ be such that $\lambda_1(t) \neq \lambda_2(t)$. Then $t \cdot w - \lambda_1(t)w$ lies in W and is a linear combination of $v_2, \dots, v_k$, with the v_2 -coordinate nonzero. By minimality, $v_2 \in W$, contradiction.
3. Since V is generated by the vectors $e_{-\beta_1}^{k_1} \cdots e_{-\beta_n}^{k_n} \cdot v$, we have that $\dim(V_\mu) \leq$ the number of tuples $(k_1, \dots, k_n)$ for which $\mu = \lambda - \sum k_i \beta_i$. The number of such tuples is finite. (Convince yourself of this!) If $\mu = \lambda$, the only such tuple is $(0, \dots, 0)$, so $\dim(V_\lambda) \leq 1$. Since $v \in V_\lambda$ is nonzero, $\dim(V_\lambda) = 1$.
4. If V has a highest weight vector $v_\mu \in V_\mu$ with $\mu \neq \lambda$, then $\mu \preceq \lambda$ and $\mathcal{U}(\mathfrak{g}) \cdot v_\mu = \mathcal{U}(\mathfrak{n}^-) \cdot v_\mu$ is a proper subrepresentation of V . Conversely, suppose that W is a proper nonzero subrepresentation of V and let μ be an element satisfying $W_\mu \neq 0$ and that is maximal with respect to the partial ordering $\preceq$. Such an element exists since $W = \bigoplus W_\mu$ by Part 2. It is different from λ since otherwise W contains v , which generates V , so $W = V$. We claim every nonzero element of W_μ is a highest weight vector. Indeed, if w is such an element and if $\beta \in \Phi^+$, then $e_\beta \cdot w \in W_{\mu+\beta} = 0$ by the maximality of μ .

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5. By Part 2, a subrepresentation $W \subset V$ is proper if and only if $W_\lambda = 0$. This implies (again by Part 2) that a sum of proper subrepresentations is again proper. Therefore the sum of all proper subrepresentations is the unique maximal one.

□

Note that every quotient representation of a highest weight module is a highest weight module. Moreover a highest weight module is always indecomposable as a representation. Therefore if it is finite-dimensional, it must be irreducible by Weyl's complete reducibility theorem.

9.5 Verma modules

At this point, it is not clear that highest weight modules with highest weight $\lambda \in \mathfrak{t}^*$ actually exist. We now construct one, in fact the 'biggest' one, using the universal enveloping algebra.

In general, if A is an associative algebra and $I \subset A$ is a left ideal (in other words, $x \cdot I \subset I$ for all $x \in A$), then the quotient vector space A/I has the structure of an A -representation via $x \cdot (y + I) = x \cdot y + I$.

Definition 9.28. Let $\lambda \in \mathfrak{t}^*$. Let $J(\lambda)$ be the left ideal of $\mathcal{U}(\mathfrak{g})$ generated by the elements $\{e_\alpha : \alpha \in \Phi^+\} \cup \{h - \lambda(h) \cdot 1 : h \in \mathfrak{t}\}$. The *Verma module* for λ is the quotient $M(\lambda) = \mathcal{U}(\mathfrak{g})/J(\lambda)$.

The quotient $M(\lambda)$ has the structure of a $\mathcal{U}(\mathfrak{g})$ -representation; equivalently, it is a $\mathfrak{g}$ -representation. Elements of the ideal $J(\lambda)$ are concretely given by

$$\sum_{i=1}^n x_i e_{\beta_i} + \sum_{j=1}^{\ell} y_j t_j \quad (9.5.1)$$

where $\Phi^+ = \{\beta_1, \dots, \beta_n\}$, $t_1, \dots, t_\ell$ is a basis of $\mathfrak{t}$ and $x_i, y_j \in \mathcal{U}(\mathfrak{g})$.

Example 9.29. If $\mathfrak{g} = \mathfrak{sl}_2$, then $M(0)$ is exactly the infinite-dimensional highest weight module given in Example 9.25.

Let m_λ be the image of $1 \in \mathcal{U}(\mathfrak{g})$ in $M(\lambda)$.

Lemma 9.30. Let $\lambda \in \mathfrak{t}^*$. Then $M(\lambda)$ is a highest weight module generated by highest weight $m_\lambda \in M(\lambda)_\lambda$. If V is another highest weight module generated by highest weight vector $v \in V_\lambda$, then there is a unique surjective map of $\mathfrak{g}$ -representations $M(\lambda) \rightarrow V$ sending m_λ to v .

Proof. The relations of $J(\lambda)$ exactly ensure that $t \cdot m_\lambda = \lambda(t)m_\lambda$ and $e_\beta \cdot m_\lambda = 0$ for all $t \in \mathfrak{t}$ and $\beta \in \Phi^+$. We still need to show $m_\lambda \neq 0$. If m_λ were zero, then $1 \in J(\lambda)$, so 1 would be of the form (9.5.1). This contradicts the PBW theorem. so $M(\lambda)$ is indeed a highest weight module.

If V is generated by highest weight $v \in V_\lambda$, consider the linear map $\varphi: \mathcal{U}(\mathfrak{g}) \rightarrow V$ defined by sending x to $x \cdot v$. This is a surjective map of $\mathfrak{g}$ -representations. The ideal $J(\lambda)$ is contained in the kernel of φ , so φ factors through $M(\lambda)$, as desired. □

So every highest weight module with highest weight λ is a quotient of $M(\lambda)$. By Part 5 of Proposition 9.27, $M(\lambda)$ has a unique maximal proper subrepresentation $W(\lambda) \leq M(\lambda)$, hence a unique irreducible quotient.

Definition 9.31. Let $V(\lambda)$ be the unique irreducible quotient of $M(\lambda)$. In other words, $V(\lambda) = M(\lambda)/W(\lambda)$.

9.6 The classification of irreducible representations

Using fairly abstract considerations we have constructed, for each $\lambda \in \mathfrak{t}^*$, an irreducible highest weight module $V(\lambda)$ with highest weight vector $v_\lambda \in V(\lambda)_\lambda$. Rephrasing what we did so far, we have:

Proposition 9.32. *The map $\lambda \mapsto V(\lambda)$ induces a bijection between $\mathfrak{t}^*$ and isomorphism classes of irreducible highest weight modules of $\mathfrak{g}$.*

Proof. Let V be an irreducible highest weight module generated by highest weight vector $v_\lambda \in V_\lambda$. By Lemma 9.27, V is a quotient of $M(\lambda)$ for some λ . Since $M(\lambda)$ has a unique irreducible quotient, we must have $M(\lambda) \simeq V(\lambda)$. We need to show λ is unique. But this follows from Part 4 of Proposition 9.27, which says that V has a unique highest weight vector up to nonzero multiple. $\square$

Corollary 9.33. *Every finite-dimensional irreducible representation of $\mathfrak{g}$ is of the form $V(\lambda)$ for some unique $\lambda \in \mathfrak{t}^*$.*

Proof. Follows immediately from the previous proposition and the fact that a finite-dimensional irreducible representation is automatically a highest weight module (Lemma 9.10). $\square$

We are almost done with the classification, but one important question remains: which $V(\lambda)$ are finite-dimensional?

Theorem 9.34. *Let $\lambda \in \mathfrak{t}^*$. Then $V(\lambda)$ is finite-dimensional if and only if λ lies in the weight lattice X and is dominant, in other words satisfies $\langle \lambda, \beta \rangle \in \mathbb{Z}_{\geq 0}$ for all $\beta \in \Phi^+$.*

This theorem, together with Corollary 9.33, shows that each finite dimensional irreducible representation has a unique highest weight, and this association defines a bijection:

$$\left\{ \begin{array}{c} \text{Isomorphism classes of finite-} \\ \text{dimensional irreducible representations of } \mathfrak{g} \end{array} \right\} \xleftrightarrow{1:1} \{ \text{Dominant weights } \lambda \in X \} \quad (9.6.1)$$

Example 9.35. *Let $\mathfrak{g} = \mathfrak{sl}_2$. Then $\mathfrak{t}^*$ can be identified with $\mathbb{C}$, using the dual basis h^* . Then $M(\lambda)$ is irreducible and infinite-dimensional if $\lambda \notin \mathbb{Z}$.*

We will use the following calculations in the proof of Theorem 9.34:

Lemma 9.36. *For each simple root $\alpha_i \in \Delta$, let $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$ and $h_i \in \mathfrak{t}$ such that (e_i, h_i, f_i) satisfy the $\mathfrak{sl}_2$ -relations. Then we have the following identities in $\mathcal{U}(\mathfrak{g})$ for all $k \geq 0$:*

$$[e_i, f_j^{k+1}] = 0 \text{ if } i \neq j, \quad (9.6.2)$$

$$[e_i, f_i^{k+1}] = -(k+1)f_i^k(k+1-h_i). \quad (9.6.3)$$

Proof. Indeed, the base case $k=0$ is clear: $[e_i, f_j] \in \mathfrak{g}_{\alpha_i-\alpha_j} = \{0\}$ and $[e_i, f_i] = h_i$. The general case follows from induction and the identity $[A, BC] = [A, B]C + B[A, C]$. $\square$

Proof of Theorem 9.34. If $V(\lambda)$ is finite-dimensional, then λ lies in X and is dominant by considering the $\mathfrak{sl}_2$ -actions of the roots; see Proposition 9.8 and Lemma 9.10. Proving the converse is much harder.

Conversely, suppose that λ is dominant and lies in X . Let $\mathfrak{m}_{\alpha_i}$ be the subalgebra spanned by these elements e_i, h_i, f_i . Let $\Pi(\lambda) = \{\mu \in \mathfrak{t}^* : V(\lambda)_\mu \neq 0\}$. Since each weight space is finite-dimensional, it suffices to prove

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that $\Pi(\lambda)$ is finite. We will prove this by showing that the Weyl group permutes $\Pi(\lambda)$ and has finitely many orbits. Let $v \in V = V(\lambda)$ be a highest weight vector.

Claim 1: for all i , if $n = \lambda(h_i) = \langle \lambda, \check{\alpha}_i \rangle$, then $f_i^{n+1} \cdot v = 0$. Indeed, write $u = f_i^{n+1} \cdot v$. For all j , $e_j \cdot v = 0$ since v is a highest weight vector. Therefore by the first identity in Claim 1, if $j \neq i$ then

$$e_j \cdot u = f_i^{n+1} e_j v = 0.$$

Moreover by the second identity in Claim 1,

$$e_i \cdot u = [e_i, f_i^{n+1}] \cdot v = -(n+1)y_i^n(nv - nv) = 0.$$

Moreover $u \in V_{\lambda - (n+1)\alpha_i}$. If u were nonzero, then u were a highest weight vector of $V(\lambda)$. Since $V(\lambda)$ is irreducible, there are no other highest weight vectors besides those in $V(\lambda)_\lambda$. So $u = 0$.

Claim 2: for each i , $V(\lambda)$ contains a non-zero finite dimensional $\mathfrak{m}_{\alpha_i}$ -representation.

Indeed, we claim that $W = \{v, f_i v, \dots, f_i^n \cdot v\}$ is an $\mathfrak{m}_{\alpha_i}$ -subrepresentation of V . We show that it is stable under f_i, h_i and e_i . By Claim 1, W is stable under f_i . Since each $f_i^k \cdot v$ lies in the weight space $V_{\lambda - k\alpha_i}$, W is stable under h_i . Finally, stability under e_i follows from the second identity in Claim 1.

Claim 3: For each i , $V(\lambda)$ is a (possibly infinite) direct sum of finite dimensional $\mathfrak{m}_{\alpha_i}$ -representations.

Indeed, let V' be the sum of all finite-dimensional $\mathfrak{m}_{\alpha_i}$ -representations contained in V . By Claim 2, V' is nonzero. It therefore suffices to prove that V' is a $\mathfrak{g}$ -subrepresentation by the irreducibility of V . To show this, let $w \in V'$ be arbitrary. Then w is contained in a finite-dimensional $\mathfrak{m}_{\alpha_i}$ -subrepresentation $W \leq V$. Let

$$W' = \mathfrak{g} \cdot W = \sum_{i=1}^{\ell} t_i \cdot W + \sum_{\beta \in \Phi} e_\beta \cdot W,$$

where $t_1, \dots, t_\ell$ is a basis of $\mathfrak{t}$ and $e_\beta \in \mathfrak{g}_\beta$ is nonzero. Then $x \cdot w \in W'$ for all $x \in \mathfrak{g}$ and W' is finite-dimensional, being a sum of finite-dimensional subspaces. Moreover, W' is $\mathfrak{m}_{\alpha_i}$ -invariant, by checking explicitly that it is invariant under the action of e_i, h_i and f_i (exercise). So $W' \subset V'$, and $x \cdot V' \subset V'$ for all $x \in \mathfrak{g}$, so V' is indeed a $\mathfrak{g}$ -subrepresentation.

Claim 4: The set $\Pi(\lambda)$ (with multiplicity) is preserved by the Weyl group: $\dim(V_{w(\mu)}) = \dim(V_\mu)$ for all μ and $w \in W$.

Indeed, using Claim 3, the proof of this fact is identical to the analogous claim when V was assumed to be finite-dimensional in Proposition 9.8; we omit the details.

Claim 5: The set of W -orbits on $\Pi(\lambda)$ is finite.

Indeed, let $\mu \in \Pi(\lambda)$ be an element. Then μ is in the closure of some Weyl chamber $\mathcal{C}$. Since W acts transitively on the Weyl chambers, we can take a representative in $W \cdot \mu$ that lies in the closure of the fundamental Weyl chamber. This means that μ is dominant. Since $\lambda \in X$ and $\mu \preceq \lambda$, $\mu \in X$. So every Weyl group orbit has a representative in the set

$$S = \{\mu \in X : \mu \text{ dominant and } \mu \preceq \lambda\}.$$

We claim that this set is finite. It is clearly a discrete subset of $\mathfrak{t}^*$ since X is discrete. We now prove that it is bounded. If $\mu \in S$, then μ, λ are dominant hence $\lambda + \mu$ is dominant. Since $\lambda - \mu$ is a nonnegative linear combination of simple roots, $(\lambda + \mu, \lambda - \mu) \geq 0$. Therefore $(\mu, \mu) \leq (\lambda, \lambda)$. Since $(-, -)$ is positive definite, S is bounded. Therefore S is finite.

We conclude that $\Pi(\lambda)$ is finite, since the Weyl group W is finite, preserves this set, and has finitely many orbits. Since $\dim(V(\lambda)_\mu)$ is finite for each $\mu \in \Pi(\lambda)$, $V(\lambda)$ is finite-dimensional. $\square$

9.7 The character of $V(\lambda)$

We end by analyzing the weights of $V(\lambda)$. So let $\lambda \in X$ be dominant and set

$$\Pi(\lambda) = \{\mu \in \mathfrak{t}^* : V(\lambda)_\mu \neq 0\}.$$

We view $\Pi(\lambda)$ as a multiset, where μ has weight $\dim V(\lambda)_\mu$. We would like to say as much as possible about $V(\lambda)$. This is what we know so far for $\mu \in \Pi(\lambda)$:

- $\mu \preceq \lambda$.
- W preserves $\Pi(\lambda)$ and the multiplicities.
- $\text{mult}(\lambda) = 1$.
- $\text{mult}(\mu) \leq$ the number of ways of writing $\lambda - \mu$ as a sum of positive roots.

We will add two additional things to this list. Firstly, we can exactly determine which dominant elements lie in $\Pi(\lambda)$.

Proposition 9.37. *Suppose $\lambda, \mu \in X$ are dominant. Then $\mu \in \Pi(\lambda)$ if and only if $\mu \preceq \lambda$.*

Proof. We have already seen that $\mu \in \Pi(\lambda)$ implies $\mu \preceq \lambda$, so consider the converse and assume $\mu \preceq \lambda$. We say an element $\mu' \in X$ is 'good' if $\mu' = \mu + \sum_{i=1}^{\ell} k_i \alpha_i$ lies in $\Pi(\lambda)$ and $k_i \in \mathbb{Z}_{\geq 0}$ for all i . Then we claim that if $k_i > 0$ for some i , there is a simple root α_j such that $\mu' - \alpha_j$ is good. If this is true, then starting with $\mu' = \lambda$ (which is good) and applying this claim iteratively, we find that μ is good, hence lies in $\Pi(\lambda)$.

To prove this, note that $(\sum k_i \alpha_i, \sum k_i \alpha_i) > 0$ since $(-, -)$ is positive definite. So there is a j with $k_j > 0$ such that $(\sum k_i \alpha_i, \alpha_j) > 0$. Since μ is dominant $(\mu, \alpha_j) \geq 0$, so $(\mu', \alpha_j) > 0$. The subspace $\oplus_{n \in \mathbb{Z}} V(\lambda)_{\mu' + n\alpha_j}$ is a representation of $\mathfrak{m}_{\alpha_j} \simeq \mathfrak{sl}_2$, and the μ' -weight space has h_j -weight $\langle \mu', \check{\alpha}_j \rangle > 0$. By $\mathfrak{sl}_2$ -theory, $f_i \in \mathfrak{g}_{-\alpha_i}$ maps $V_{\mu'}$ injectively into $V_{\mu' - \alpha_j}$. So $\mu' - \alpha_j \in \Pi(\lambda)$. Since $k_j \geq 1$, $\mu' - \alpha_j$ is indeed good, as required. $\square$

Corollary 9.38. *If $\lambda, \mu \in X$ and λ is dominant, then $\mu \in \Pi(\lambda)$ if and only if $w(\mu) \preceq \lambda$ for all $w \in W$.*

Proof. We leave this as an exercise. The forward direction is clear since W permutes $\Pi(\lambda)$ and every element of $\Pi(\lambda)$ is $\preceq \lambda$. Conversely, assume $w(\mu) \preceq \lambda$ for all $w \in W$. There exists a $w \in W$ such that $w(\mu)$ is dominant. By the previous proposition, $w(\mu) \in \Pi(\lambda)$, so $\mu \in \Pi(\lambda)$. $\square$

Example 9.39. *Consider $\mathfrak{g}$ of type G_2 and let $\Delta = \{\alpha, \beta\}$ be such that α is the short root. Then $\omega_1 = 2\alpha + \beta$ and we can calculate the weights of $\Pi(\omega_1)$ without multiplicity using Proposition 9.37. It is best to draw a picture here.*

We now know exactly what the set of weights is of $V(\lambda)$. But this does not tell us anything about the multiplicity of a weight μ , i.e., the quantity $\dim(V(\lambda)_\mu)$. We know that $\dim(V(\lambda)_\lambda) = 1$ so λ and all its W -conjugates have multiplicity 1. But in general multiplicities can be higher. Take $\mathfrak{g}$ simple and V the adjoint representation. Then the weights are exactly $\Phi \cup \{0\}$, where each root has multiplicity 1 and 0 has multiplicity $\ell = \dim \mathfrak{t}$. The Weyl character formula tells us about the multiplicities of all the weights at once.

Definition 9.40. *Let $\mathbb{Z}[X]$ be the free abelian group generated by the formal symbols $\{e^\lambda : \lambda \in X\}$. An element of $\mathbb{Z}[X]$ is given by*

$$\sum_{\lambda \in S} c_\lambda e^\lambda,$$

*where $S \subset X$ is a finite subset and $c_\lambda \in \mathbb{Z}$. The assignment $e^\lambda \cdot e^\mu = e^{\lambda+\mu}$ extends by bilinearity to a product on $\mathbb{Z}[X]$, and this product turns $\mathbb{Z}[X]$ into a commutative ring, called the **character ring**.*

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Example 9.41. If $\Phi = \{\pm\alpha\}$, then $X = \mathbb{Z}\{\alpha/2\}$, and an element of $\mathbb{Z}[X]$ is a $\mathbb{Z}$ -linear combination of the elements $e^{n\alpha/2}$. If $t = e^{\alpha/2}$, then $\mathbb{Z}[X] = \mathbb{Z}[t, t^{-1}]$.

In general $\mathbb{Z}[X]$ is isomorphic to $\mathbb{Z}[t_1, t_1^{-1}, \dots, t_\ell, t_\ell^{-1}]$, where ℓ is the rank of Φ . The notation e^λ is purely formal, but has its origins in Lie groups.

Definition 9.42. If V is a finite-dimensional representation of $\mathfrak{g}$, the *character of V* is the element

$$\text{ch}(V) = \sum_{\lambda \in X} \dim(V_\lambda) e^\lambda$$

of $\mathbb{Z}[X]$. An arbitrary element of $\mathbb{Z}[X]$ is called a *formal character*.

Example 9.43. If again $\mathfrak{g} = \mathfrak{sl}_2$ and $t = e^{\alpha/2}$, then $\text{ch}(V(n)) = t^n + t^{n-2} + \dots + t^{-n}$.

The Weyl character formula gives a magic formula for $\text{ch}(V(\lambda))$. To state it, recall the element

$$\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha.$$

We have shown in Lemma 7.32 that $w_{\alpha_i}(\rho) = \rho - \alpha_i$ for every simple root $\alpha_i \in \Delta$. This implies:

Lemma 9.44. If $\omega_1, \dots, \omega_\ell$ are the fundamental weights of Φ with respect to Δ , then $\rho = \omega_1 + \dots + \omega_\ell$.

Recall that if $\mathfrak{g} = \mathfrak{sl}_n$, then the Weyl group is isomorphic to S_n . We have the sign homomorphism $\text{sgn}: S_n \rightarrow \{\pm 1\}$ from group theory. The next definition generalizes this to an arbitrary Weyl group.

Definition 9.45. If $w \in W$ is a product of simple reflections $w_1 \cdots w_n$, define $\text{sgn}(w) = (-1)^n$.

On example sheet 4, you will prove that $\text{sgn}(w)$ is well defined (that is, does not depend on the choice of product decomposition) and is a homomorphism $\text{sgn}: W \rightarrow \{\pm 1\}$. We leave it as an additional exercise that it coincides with the usual sign homomorphism when $W = S_n$. (This follows from the fact that a simple reflection corresponds to a transposition.)

Theorem 9.46 (Weyl character formula). If $\lambda \in X$ is dominant, we have the equality in $\mathbb{Z}[X]$:

$$\text{ch}(V(\lambda)) = \frac{\sum_{w \in W} \text{sgn}(w) e^{w(\lambda + \rho)}}{e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})}. \quad (9.7.1)$$

Proof. To be completely precise, one can move the numerator to the left hand side so that the formula only involves products of terms in $\mathbb{Z}[X]$, no quotients. (Alternatively, $\mathbb{Z}[X]$ is an integral domain, so the identity can be interpreted in the fraction field of $\mathbb{Z}[X]$.) There are many proofs, all difficult. See Humphreys §24.3 for one of them. $\square$

This formula has two immediate corollaries:

Corollary 9.47 (Weyl denominator formula). We have the equality in $\mathbb{Z}[X]$:

$$\sum_{w \in W} \text{sgn}(w) e^{w(\rho)} = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha}) \quad (9.7.2)$$

Proof. Plug in $\lambda = 0$ in the Weyl character formula, and use that $V(0)$ is the trivial representation, so $\text{ch}(V(0)) = e^0 = 1$. $\square$

Corollary 9.48 (Weyl dimension formula). *If $\lambda \in X$ is dominant, then*

$$\dim(V(\lambda)) = \frac{\prod_{\alpha \in \Phi^+} \langle \lambda + \rho, \check{\alpha} \rangle}{\prod_{\alpha \in \Phi^+} \langle \rho, \check{\alpha} \rangle} = \prod_{\alpha \in \Phi^+} \frac{(\lambda + \rho, \alpha)}{(\rho, \alpha)}. \quad (9.7.3)$$

Proof. (non-examinable) Although not quite apparent, this follows from the Weyl character formula using formal manipulations in the ring $\mathbb{Z}[X]$. We would simply like to set $e^\lambda = 1$ for every $\lambda \in X$ in the character formula, but then the quotient becomes an undetermined $0/0$.

To resolve this, we apply L'Hopital's rule by cleverly differentiating in a certain direction. More precisely, for each $\mu \in X$ define the homomorphism

$$\begin{aligned} F_\mu: \mathbb{Z}[X] &\rightarrow \mathbb{Z}[t, t^{-1}] \\ e^\lambda &\mapsto t^{-(\mu, \lambda)} \end{aligned}$$

by extending linearly. Since $F_\mu(e^{\lambda+\nu}) = F_\mu(e^\lambda)F_\mu(e^\nu)$, F_μ is a ring homomorphism. We have

$$F_\mu(\text{ch}(V(\lambda))) = \sum_{\nu \in \Pi(\lambda)} \dim(V(\lambda)_\nu) t^{-(\mu, \nu)}.$$

Applying F_μ to the Weyl denominator formula, we get

$$\begin{aligned} t^{-(\mu, \rho)} \prod_{\alpha \in \Phi^+} (1 - t^{(\alpha, \mu)}) &= \sum_{w \in W} \text{sgn}(w) t^{-(\mu, w(\rho))} \\ &= \sum_{w \in W} \text{sgn}(w^{-1}) t^{-(\mu, w^{-1}(\rho))} \\ &= \sum_{w \in W} \text{sgn}(w) t^{-(w(\mu), \rho)}. \end{aligned}$$

(This follows from replacing w by w^{-1} in the sum and using the identities $\text{sgn}(w^{-1}) = \text{sgn}(w)$ and $(\mu, w^{-1}(\rho)) = (w(\mu), \rho)$.) Plugging in $\mu = \lambda + \rho$, we get

$$\sum_{w \in W} \text{sgn}(w) t^{-(w(\lambda+\rho), \rho)} = t^{-(\lambda+\rho, \rho)} \prod_{\alpha \in \Phi^+} (1 - t^{(\alpha, \lambda+\rho)}). \quad (9.7.4)$$

If we apply F_ρ to the Weyl character formula and use (9.7.4) to replace the numerator, we get

$$F_\rho(\text{ch}(V(\lambda))) = \frac{t^{-(\lambda+\rho, \rho)} \prod_{\alpha \in \Phi^+} (1 - t^{(\alpha, \lambda+\rho)})}{t^{-(\rho, \rho)} \prod_{\alpha \in \Phi^+} (1 - t^{(\alpha, \rho)})}. \quad (9.7.5)$$

Note that the denominator on the right hand side is nonzero: since $\langle \rho, \check{\alpha}_i \rangle = 1$ for all $\alpha_i \in \Delta$, $(\rho, \alpha) > 0$ for every $\alpha \in \Phi^+$.

Now comes the punchline. View equation (9.7.5) as an identity between two functions in the variable $t \in \mathbb{R}$. We want to evaluate the function at $t = 1$, but this is not possible since the quotient is indetermined $0/0$ when $t = 1$. By L'Hopital's rule, we may differentiate the numerator and the denominator and then plug in $t = 1$ instead. So for each $\alpha \in \Phi^+$ we have

$$\lim_{t \rightarrow 1} \frac{(1 - t^{(\alpha, \lambda+\rho)})}{(1 - t^{(\alpha, \rho)})} = \lim_{t \rightarrow 1} \frac{-(\alpha, \lambda + \rho) t^{(\alpha, \lambda+\rho)-1}}{-(\alpha, \rho) t^{(\alpha, \rho)-1}} = \frac{(\alpha, \lambda + \rho)}{(\alpha, \rho)}.$$

Taking the product over all $\alpha \in \Phi^+$ proves the corollary. $\square$

The Weyl character formula tends to be hard to calculate in practice since the Weyl group is quite big so there are quite a few terms in it. There are other formulae to determine the multiplicity of a weight, under the name of Kostant's multiplicity formula and Freudenthal's formula, which we won't state or need but which you can look up if you're interested.

Example 9.49. Let $\mathfrak{g} = \mathfrak{sl}_2$, and $t = e^{\alpha/2}$. We have $\rho = \alpha/2$. So the Weyl character formula reads

$$\text{ch}(V(n)) = \frac{t^{n+1} - t^{-(n+1)}}{t(1 - t^{-2})} = \frac{t^{n+1} - t^{-(n+1)}}{t - t^{-1}} = t^n + t^{n-2} + \cdots + t^{-n}.$$

Example 9.50. Let $\mathfrak{g} = \mathfrak{sl}_3$. So Φ is of type A_2 , hence $\Phi = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)\}$, $\Delta = \{\alpha_1, \alpha_2\}$. We work out the Weyl dimension formula in detail. Let ω_1, ω_2 be the fundamental weights. Let $\lambda \in X$ be dominant. Then $\lambda = n_1\omega_1 + n_2\omega_2$ and $\rho = \omega_1 + \omega_2$. By definition $\langle \omega_i, \check{\alpha}_j \rangle = \delta_{ij}$. So the only nontrivial pairings to calculate are $\langle \omega_i, (\alpha_1 + \alpha_2)^\vee \rangle$. Since Φ is simply laced, $(\alpha_1 + \alpha_2)^\vee = \alpha_1^\vee + \alpha_2^\vee$, so $\langle \omega_i, (\alpha_1 + \alpha_2)^\vee \rangle = 1$ for $i = 1, 2$. The $\langle -, (-) \rangle$ is linear in the first variable, we can easily calculate $\langle \lambda, \check{\alpha} \rangle$ now for every positive root, and the Weyl dimension formula reads

$$\dim(V(\lambda)) = \frac{(n_1 + 1)(n_2 + 1)(n_1 + n_2 + 2)}{2}.$$

It follows that $V(\omega_1)$ and $V(\omega_2)$ are the two smallest nontrivial representations of $\mathfrak{g}$, which are both 3-dimensional. In fact, an explicit calculation shows that (using the standard CSA of $\mathfrak{g}$ and simple root basis) $V(\omega_1)$ is the defining representation and $V(\omega_2)$ is its dual. We can now ask all kinds of questions, like what is the decomposition of $\text{Sym}^2 V(\omega_1)$? It is easy to see that $2\omega_1$ is a highest weight of the latter representation, and the above formula shows that $\dim V(2\omega_1) = 6 = \dim \text{Sym}^2 V(\omega_1)$. So $\text{Sym}^2 V(\omega_1) \simeq V(2\omega_1)$.

Example 9.51. We do a final example for $\mathfrak{g} = \mathfrak{sp}_4$. In that case Φ is of type B_2 , and we let $\Delta = \{\alpha, \beta\}$ be such that α is short. Then we may calculate $\langle \omega_i, \check{\gamma} \rangle$ for every $\gamma \in \Phi^+$, and we get that if $\lambda = n_1\omega_1 + n_2\omega_2$, then

$$\dim V(\lambda) = \frac{(n_1 + 1)(n_2 + 1)(n_1 + n_2 + 2)(n_1 + 2n_2 + 3)}{6}.$$

Since this formula is strictly increasing in both variables n_1, n_2 , and since $\dim V(\omega_1) = 4$ and $\dim V(\omega_2) = 5$, the only nontrivial irreducible 4-dimensional representation of $\mathfrak{sp}_4$ is $V(\omega_1)$, hence this must be the defining representation. We can draw a weight diagram to display the weights of $V = V(\omega_1)$. Using this diagram and properties of the weights, we can decompose $V \otimes V$ into irreducible representations. First of all, note that $V \otimes V = \text{Sym}^2(V) \oplus \bigwedge^2(V)$. Moreover, if V has weights $\{\lambda_1, \dots, \lambda_4\}$, the weights of $\text{Sym}^2(V)$ are $\{\lambda_i + \lambda_j : i \leq j\}$, while the weights of $\bigwedge^2(V)$ are $\{\lambda_i - \lambda_j : i < j\}$. We can now use the weight diagram to show $\text{Sym}^2(V) \simeq V(2\alpha + \beta)$. Since $2\alpha + \beta$ is the highest root, $V(2\alpha + \beta)$ is isomorphic to the adjoint representation.

You will do tons of these examples on example sheet 4.